

Unit 2

KINEMATICS

52 items: 20 from official papers (95 marks), 32 from the booklet

Syllabus sections in this unit

2.1 Equations of motion

Unit 2: Kinematics

1. M/J 26/P22/Q1/b

- (b) A car moves in a straight line along a road. Figure 1.1 shows the variation with time t of the velocity v of the car.

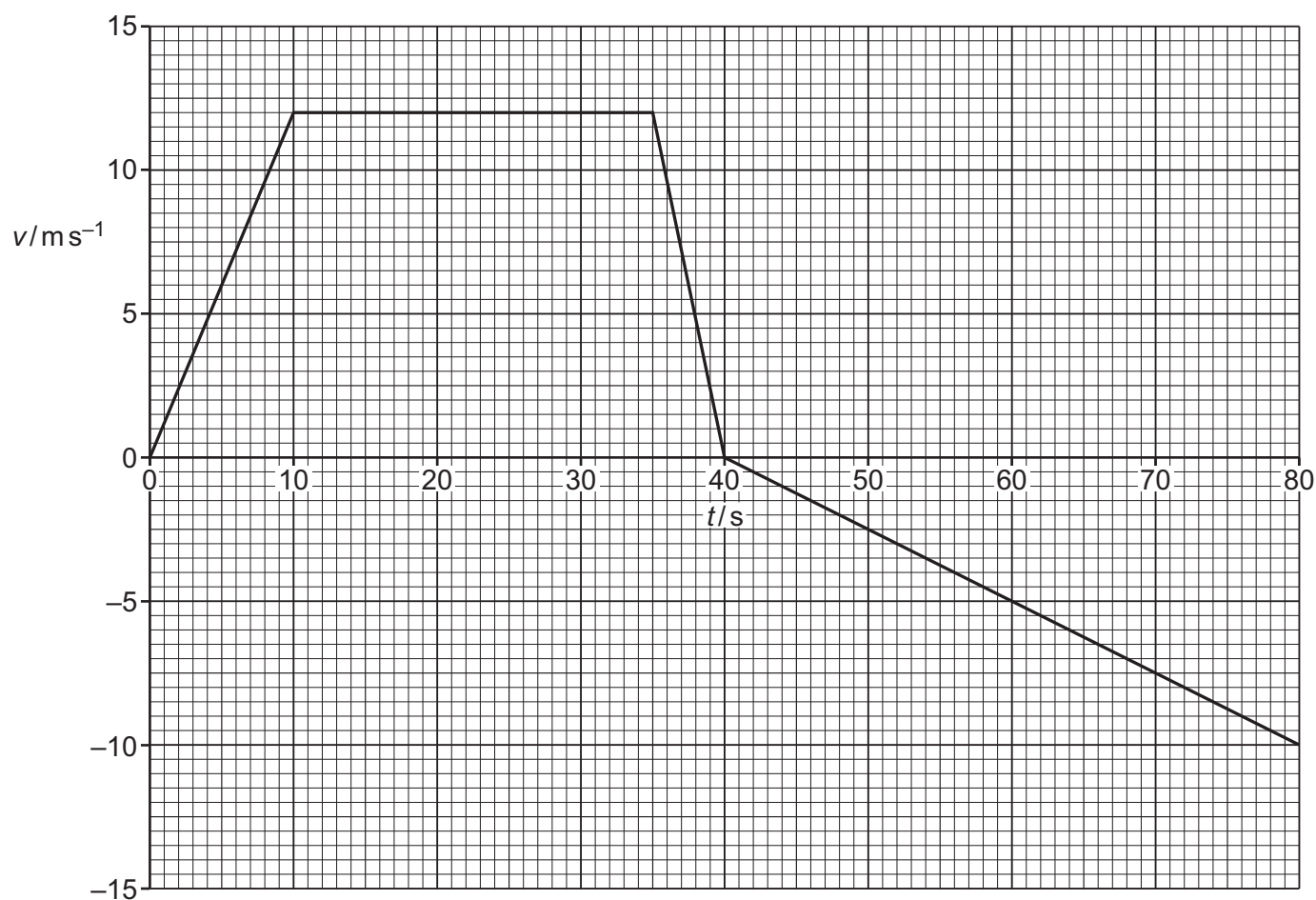


Figure 1.1

- (i) State the feature of the graph that represents the distance travelled by the car between $t = 0$ and $t = 40$ s.

[1]

- (ii) Calculate the distance travelled by the car between $t = 0$ and $t = 40$ s.

distance =

m [1]

- (iii) Determine the maximum magnitude of the acceleration of the car.

acceleration =

ms^{-2} [2]

- (iv) At $t = 25$ s, the car is at a point P.

At $t = 40$ s, the car is at a distance of 150 m from P.

After $t = 40$ s, the car travels with a constant acceleration of -0.25 ms^{-2} until the car is again at point P.

Calculate the speed of the car when it is at point P for the second time.

speed =

ms^{-1} [2]

- (v) At $t = 0$, the displacement of the car is zero.

On Figure 1.2, sketch the variation of the displacement of the car with t from $t = 0$ to $t = 80$ s. Values for displacement are **not** required.

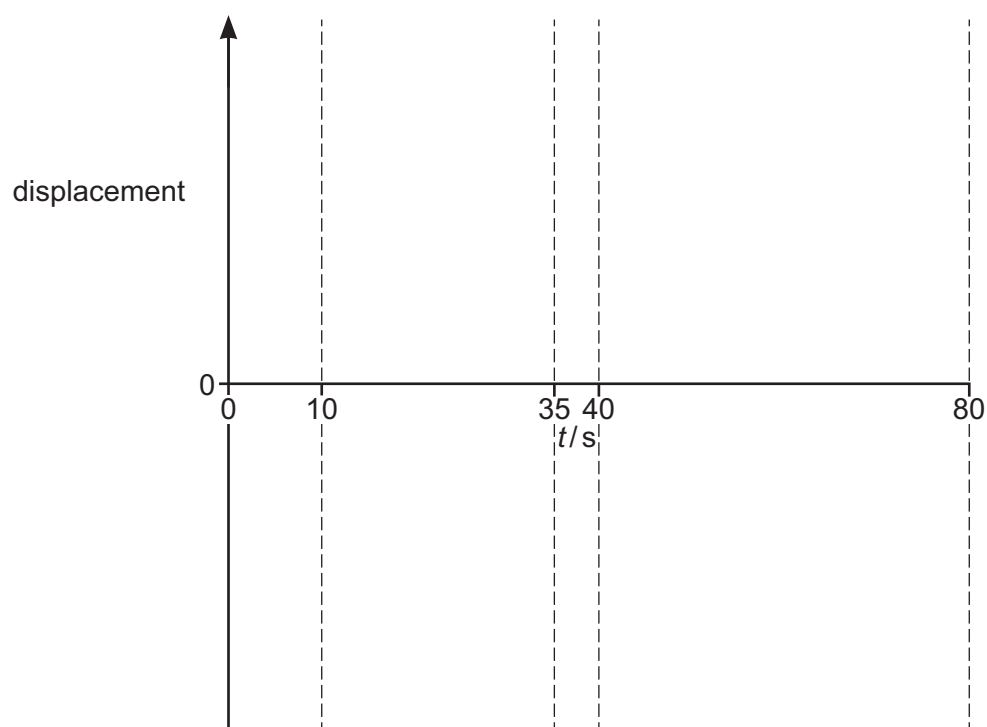


Figure 1.2

[3]

2. M/J 26/P23/Q1/b(i,iii),c(ii)

- (b) A ball X of mass 0.15 kg moves along a horizontal frictionless surface. The ball collides elastically with ball Y that is initially stationary. Balls X and Y have equal diameters but different masses.

Figure 1.1 shows the variation with time of the velocity of ball X before, during and after the collision.

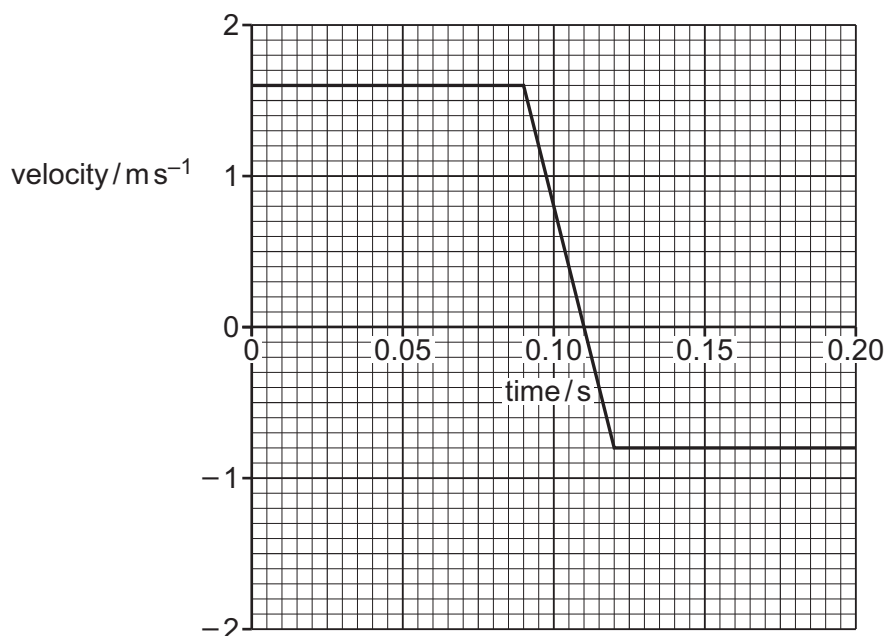


Figure 1.1

- (i) Determine the magnitude of the acceleration of ball X during the collision.

acceleration = m s^{-2} [2]

- (iii) Determine the total distance travelled by ball X during the collision.

distance = m [2]

(c)

- (ii) On Figure 1.1, sketch the variation of the velocity of ball Y with time between time = 0 and time = 0.20 s. [2]

3. M/J 26/P24/Q1/b(ii),c(ii)

- (b)** A car of mass 1400 kg is initially travelling horizontally in a straight line at a constant speed of 15 m s^{-1} . The brakes are applied and the car decelerates uniformly to rest in a distance of 35 m.

Calculate:

- (ii)** the acceleration of the car

acceleration = m s^{-2} [2]

- (iii)** the resultant force on the decelerating car.

force = N [2]

- (c)** The initial speed of the car in **1(b)** is now doubled to 30 m s^{-1} . The brakes of the car are then applied and the car comes to rest. The resultant force on the car remains the same as in **1(b)(iii)**.

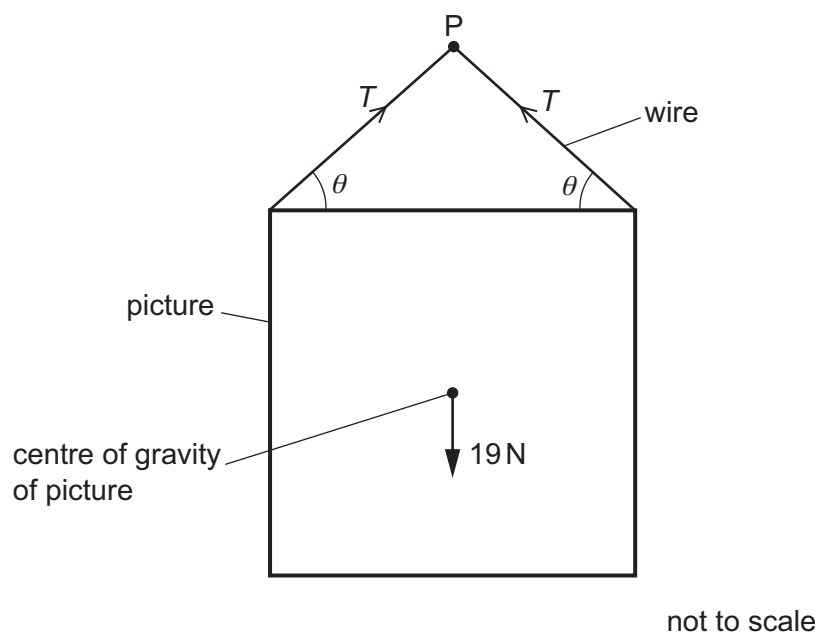
Describe and explain the effect, if any, of doubling the initial speed on:

- (ii)** the distance travelled by the car in coming to rest.

[2]

4. MAR 26/P22/Q4/b

- 4** A uniform square picture is attached to a wire and hangs in equilibrium from a pin at point P, as shown in Figure 4.1.

**Figure 4.1**

The picture has weight 19 N and its centre of gravity is directly below P.

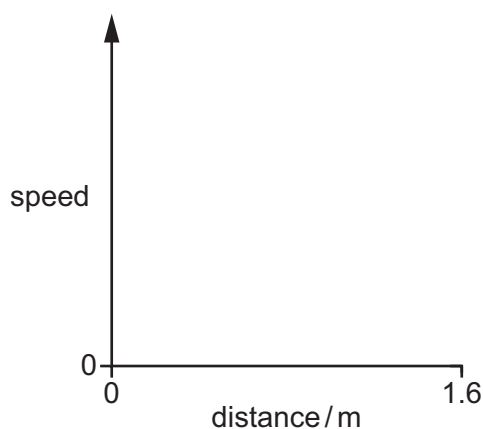
The top side of the picture is horizontal. The wire forms an angle θ with the horizontal. The tension T in the wire is 13 N.

- (b)** The bottom side of the picture shown in Figure 4.1 is 1.6 m above the ground. The pin at P breaks, and the picture falls vertically downwards. Resistive forces acting on the picture are negligible.

- (i)** Calculate the time taken for the bottom side of the picture to hit the ground.

time = _____ s [2]

- (ii)** On Figure 4.2, sketch the variation of the speed of the picture with the distance fallen from its initial position.

**Figure 4.2**

[2]

5. O/N 25/P21/Q1/a,b

- 1** (a) Define acceleration.

[1]

- (b) A rocket is launched vertically from the surface of the Earth.

Fig. 1.1 shows the variation of the velocity of the rocket with time for the first 20 s after its launch.

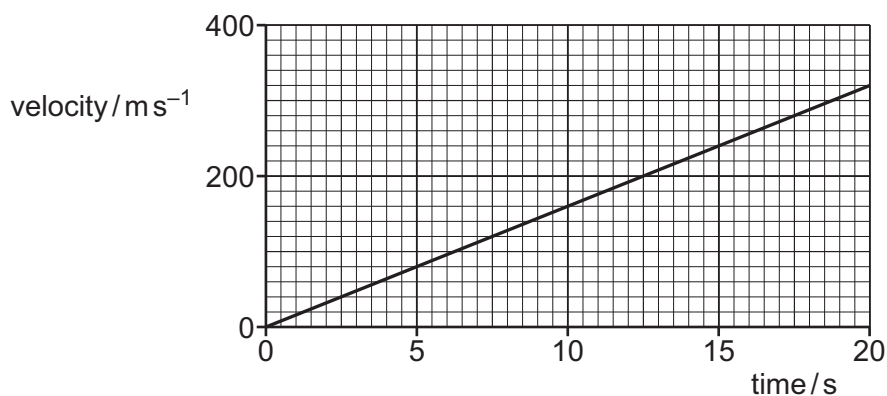


Fig. 1.1

- (i) Determine the acceleration of the rocket.

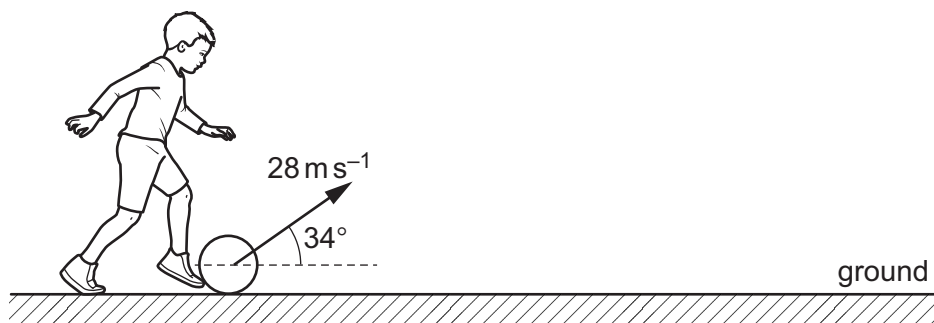
acceleration = ms⁻² [1]

- (ii) Show that the height of the rocket above the surface of the Earth at a time of 20 s after launch is 3.2 km.

[2]

6. O/N 25/P24/Q1/a(ii,iii,iv)

- 1 A child kicks a ball so that it leaves horizontal ground with a velocity of 28 m s^{-1} at an angle of 34° to the horizontal, as shown in Fig. 1.1.

**Fig. 1.1**

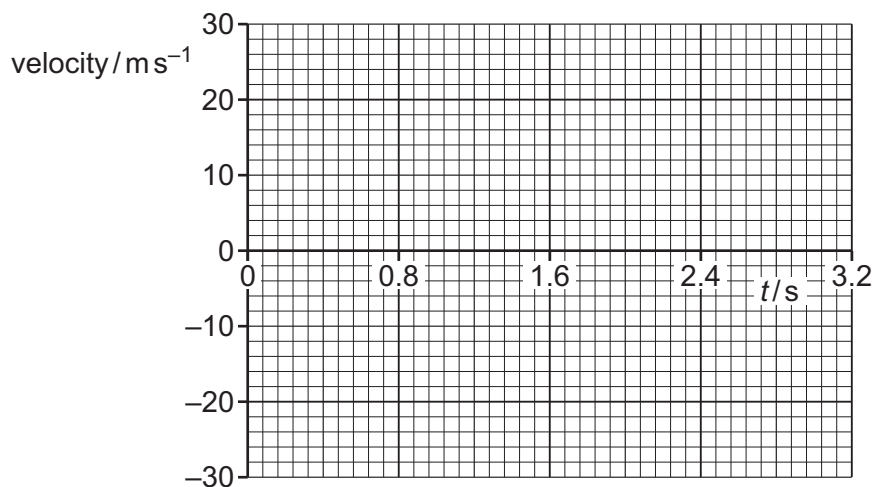
Air resistance is negligible. The ball leaves the ground at time $t = 0$.

(a)

- (ii) Show that the ball reaches its maximum height at time $t = 1.6 \text{ s}$.

[1]

- (iii) On Fig. 1.2, sketch the variation of v_H with time t between $t = 0$ and $t = 3.2 \text{ s}$. Label your line H.

**Fig. 1.2**

[1]

- (iv) On Fig. 1.2, sketch the variation of v_V with time t between $t = 0$ and $t = 3.2 \text{ s}$. Assume that velocity in the upward direction is positive. Label your line V.

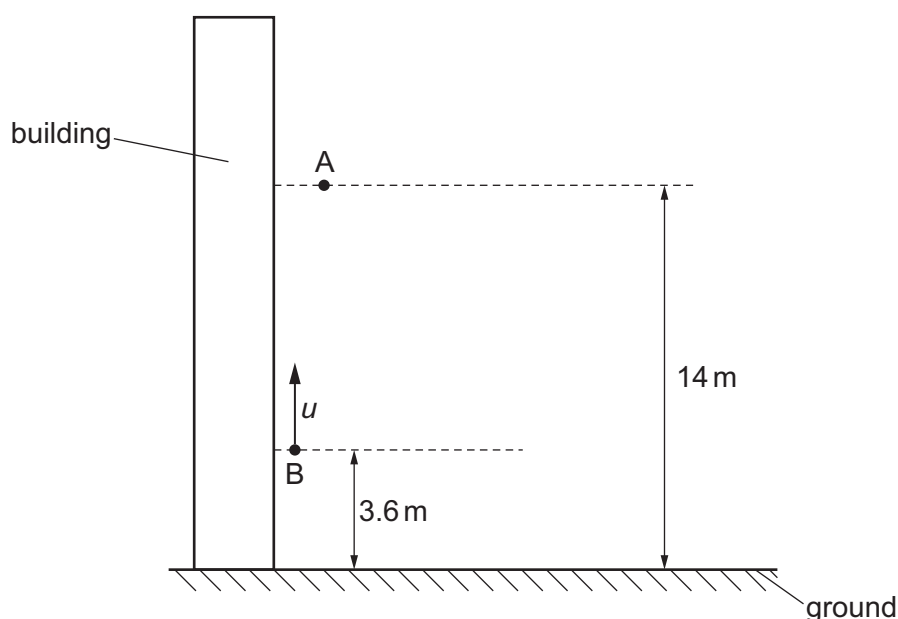
[3]

7. M/J 25/P21/Q1/a,b,c(i)

- 1 (a) Define acceleration.

[1]

- (b) In an experiment, two objects A and B are released from the side of a building, as shown in Fig. 1.1.

**Fig. 1.1** (not to scale)

Object A is released from rest at a height of 14 m above the horizontal ground.
 Object B is released with an initial upwards vertical velocity u at a height of 3.6 m above the ground.
 Both objects take the same time to reach the ground and they do not collide with each other.
 Air resistance is negligible.

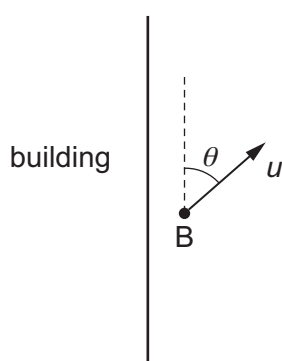
- (i) Calculate the time taken for object A to reach the ground.

time = _____ s [2]

- (ii) Use your answer in **(b)(i)** to calculate u .

u = _____ ms^{-1} [2]

- (c) In a second experiment, object B is released from the same height and given the same initial speed as in **(b)** but at a release angle θ to the vertical, as shown in Fig. 1.2.

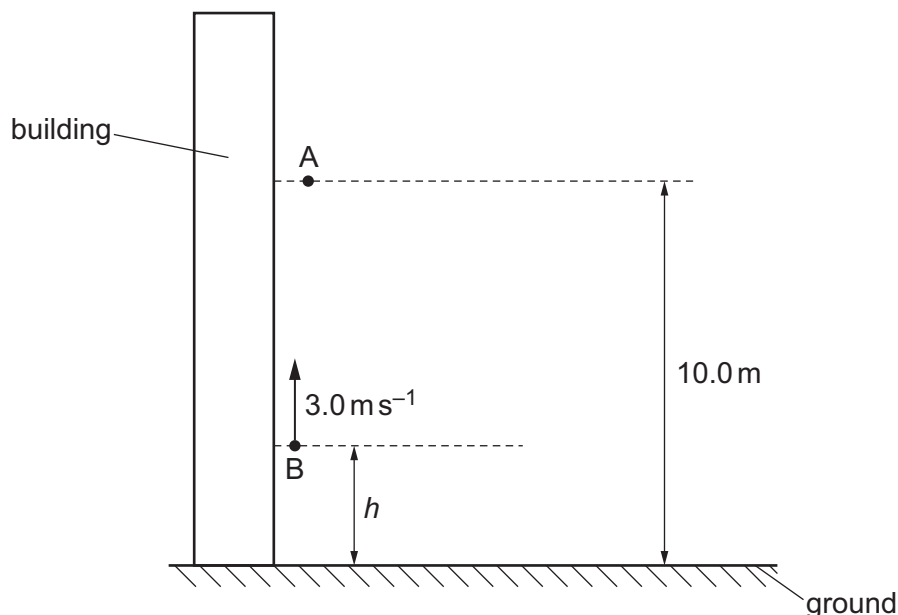
**Fig. 1.2**

- (i) State and explain whether the time taken for object B to reach the ground is less than, the same as or greater than the time in **(b)(i)**.

[2]

8. M/J 25/P23/Q1/a,b,c(i)**1 (a)** Define velocity.

[1]

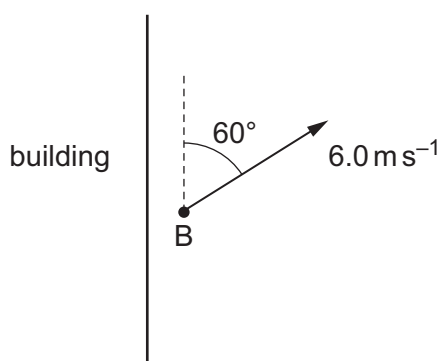
(b) In an experiment, two objects A and B are released from the side of a building, as shown in Fig. 1.1.**Fig. 1.1** (not to scale)

Object A is released from rest at a height of 10.0 m above horizontal ground.

Object B is released with an initial upward velocity of 3.0 m s^{-1} at a height h above the ground.

Both objects take the same time to reach the ground and they do not collide with each other.

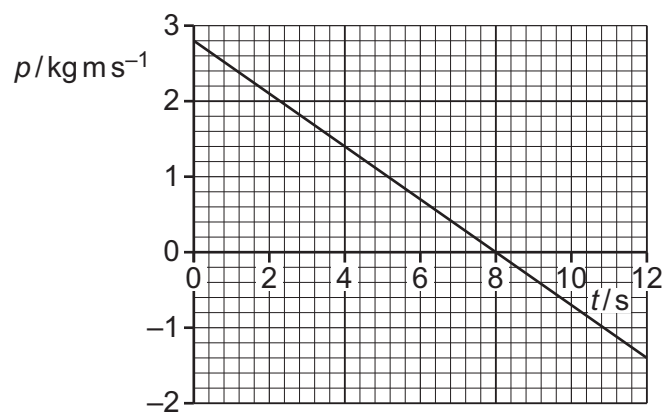
Air resistance is negligible.

Calculate h . $h =$ m [3]**(c)** In a second experiment, object B is released from the same height as in **(b)** but with a speed of 6.0 m s^{-1} at an angle of 60° to the vertical, as shown in Fig. 1.2.**Fig. 1.2****(i)** State and explain whether the time taken for object B to reach the ground is less than, the same as, or greater than the time taken in the first experiment.

[2]

9. M/J 25/P24/Q2/f

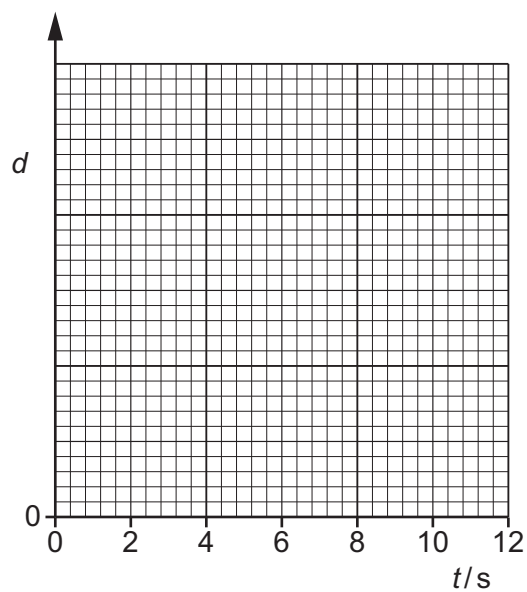
- 2 An object of constant mass moves in a straight line. The variation with time t of the momentum p of the object is shown in Fig. 2.1.

**Fig. 2.1**

- (f) At time $t = 0$ the displacement d of the object is zero.

On Fig. 2.2, sketch the variation of d with time t from $t = 0$ to $t = 12$ s.

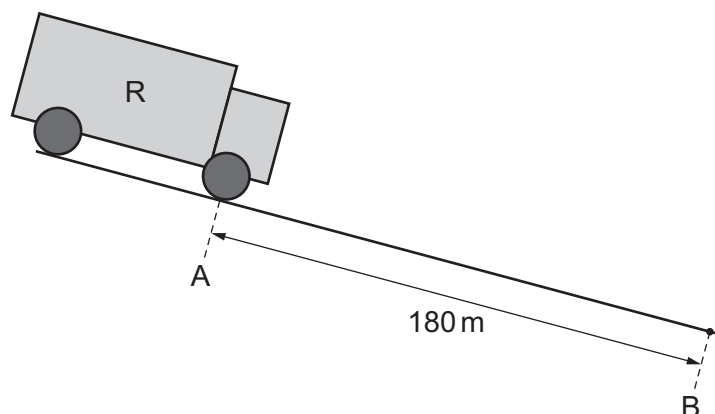
Numerical values of d are not required.

**Fig. 2.2**

[3]

10. MAR 25/P22/Q3/a(i)

- 3 (a) A truck R of mass 9400 kg moves with constant acceleration in a straight line down a slope, as illustrated in Fig. 3.1.

**Fig. 3.1**

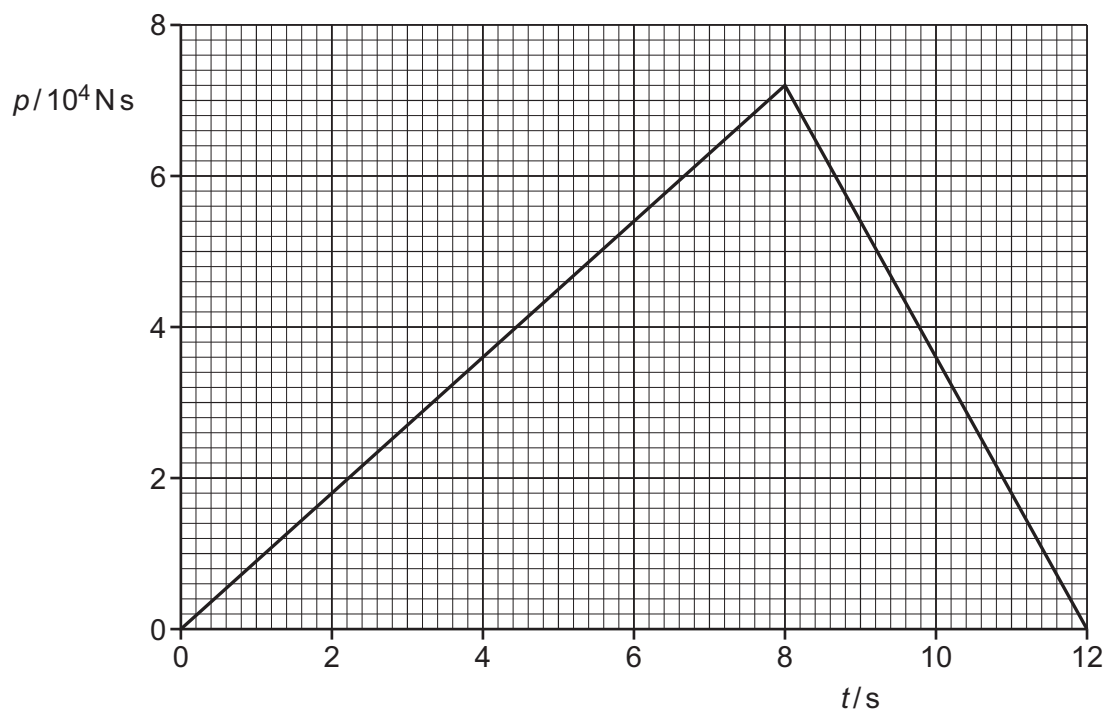
At point A the speed of the truck is 13 ms^{-1} and at point B the speed of the truck is 22 ms^{-1} . A and B are a distance of 180 m apart.

- (i) Calculate the acceleration of the truck between A and B.

acceleration = ms^{-2} [2]

11. O/N 24/P21/Q2/b(iv)

- (b) A car of mass 1800 kg is moving in a straight line. Fig. 2.1 shows the variation with time t of the momentum p of the car.

**Fig. 2.1**

- (iv) Determine the distance travelled by the car between times $t = 0$ and $t = 12.0 \text{ s}$.

distance = m [2]

12. O/N 24/P22/Q2/c

- (b) A child stands on a scooter on horizontal ground. The combined mass of the child and the scooter is 16 kg.

The child starts from rest and pushes once on the ground with her foot which causes her to accelerate. The push lasts for a time of 1.1 s. The speed of the child and the scooter after the push is 0.60 ms^{-1} .

Determine the average resultant force acting horizontally on the child and the scooter during the push.

average force = _____ N [2]

- (c) Later, the child in (b) travels down a slope at a constant angle to the horizontal, as shown in Fig. 2.1.

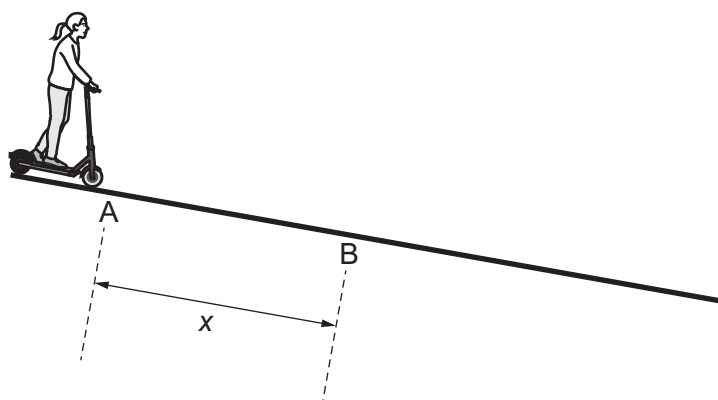


Fig. 2.1 (not to scale)

At point A her speed is 0.60 ms^{-1} . She has a constant acceleration of 0.85 ms^{-2} parallel to the slope. After a time of 3.7 s, she reaches point B.

Calculate the distance x travelled by the child along the slope from A to B.

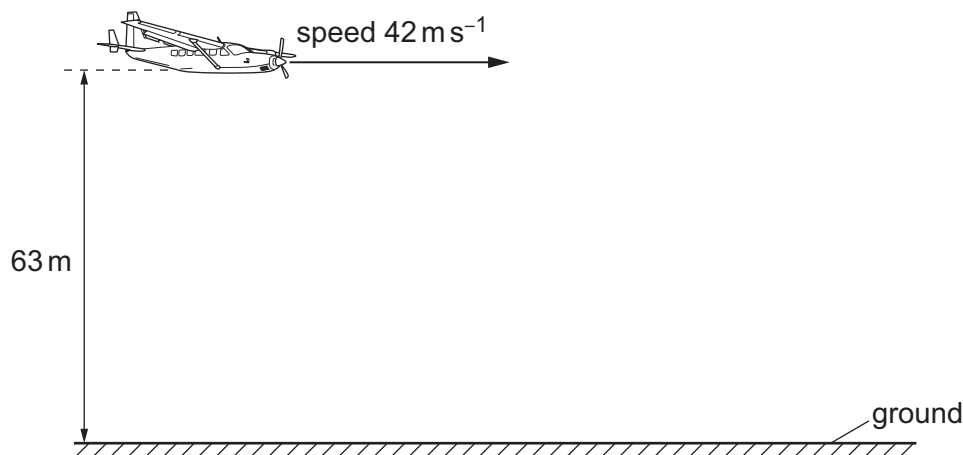
$x =$ _____ m [2]

13. O/N 24/P23/Q1/a,b

- 1** (a) Define acceleration.

[1]

- (b) A small aircraft is flying horizontally at a speed of 42 m s^{-1} at a height of 63 m above horizontal ground, as shown in Fig. 1.1.

**Fig. 1.1**

The aircraft drops a small parcel. The parcel is released from the aircraft at the instant shown in Fig. 1.1. Air resistance is negligible.

- (i) On Fig. 1.1, draw a line to show the path of the parcel as it falls from the aircraft to the ground. [1]
- (ii) Calculate the time taken from the instant of release to the instant the parcel reaches the ground.

time = s [2]

- (iii) Calculate the vertical component of the velocity of the parcel immediately before it reaches the ground.

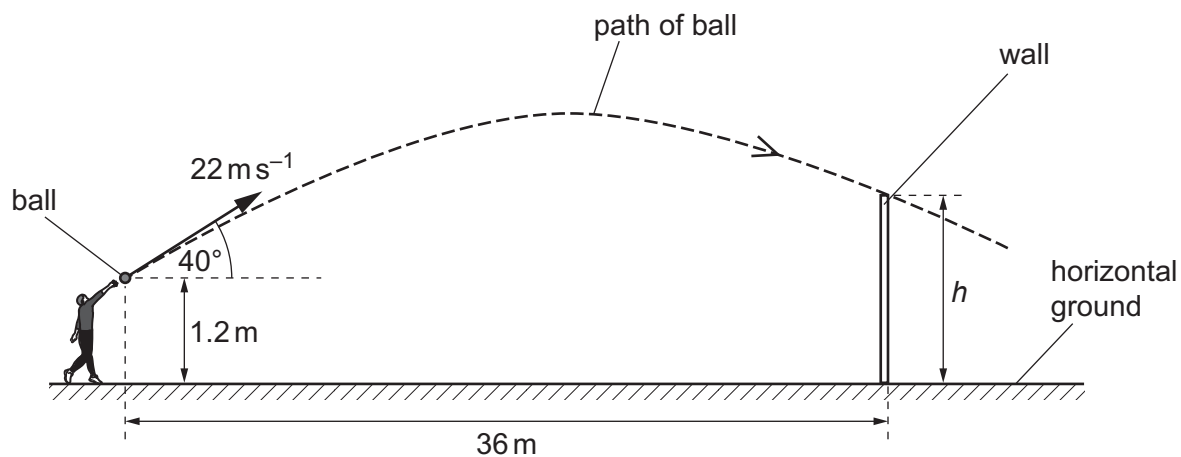
vertical component of velocity = ms^{-1} [1]

- (iv) Determine the speed at which the parcel reaches the ground.

speed = ms^{-1} [2]

14. M/J 24/P21/Q2/a,b(i,iii)**2 (a)** Define velocity.

[1]

(b) A student throws a ball over a vertical wall of height h , as shown in Fig. 2.1.**Fig. 2.1** (not to scale)

The ball leaves the hand of the student at a height of 1.2 m above the horizontal ground. The ball has an initial velocity of 22 m s^{-1} at an angle of 40° to the horizontal. The wall is a horizontal distance of 36 m from where the student releases the ball.

Air resistance is negligible.

(i) Determine the time taken for the ball to reach the wall.

time taken =

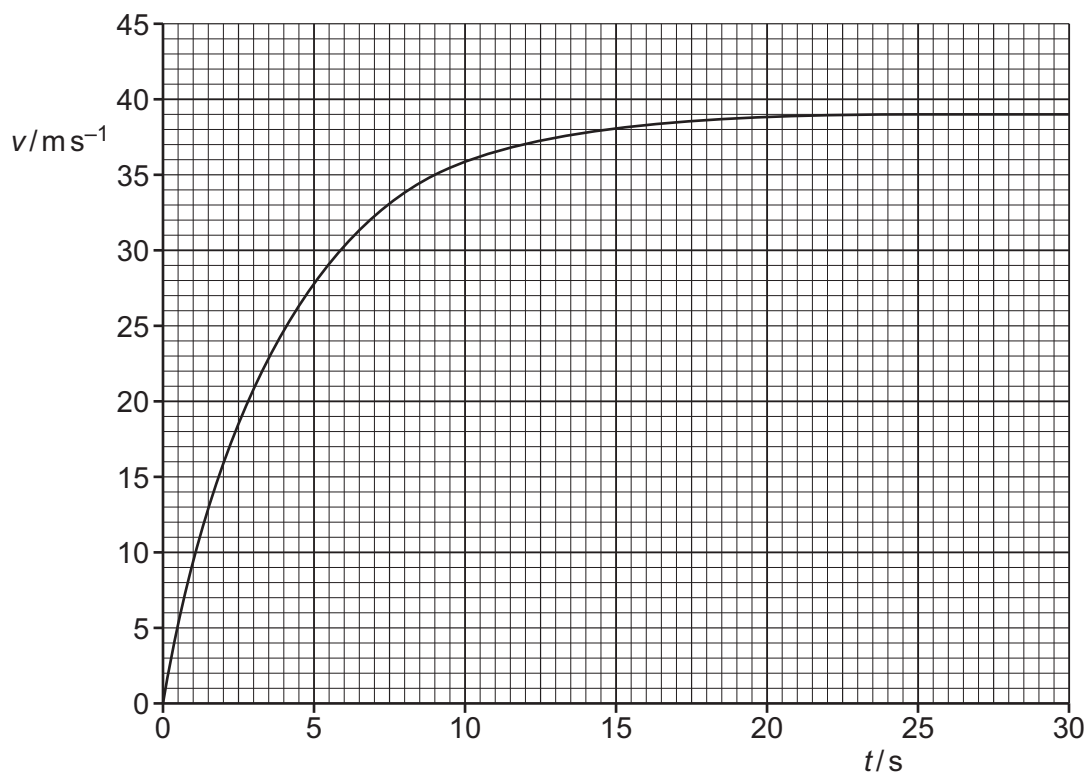
s [2]

(iii) The ball just goes over the wall.Calculate the height h of the wall. $h =$

m [3]

15. M/J 24/P22/Q2/a(ii)

- 2** A skydiver jumps from an aircraft at time $t = 0$ and falls vertically downwards. The variation with t of her velocity v is shown in Fig. 2.1.

**Fig. 2.1**

(a)

- (ii)** By drawing a suitable line on Fig. 2.1, determine the acceleration of the skydiver at time $t = 9.0 \text{ s}$.

acceleration = ms^{-2} [2]

16. M/J 24/P23/Q2/a,b(i,ii,iii,iv)

- 2 (a) Define displacement from a point.

[1]

- (b) An object is projected horizontally at a speed of 6.0 ms^{-1} from a slope, as shown in Fig. 2.1.

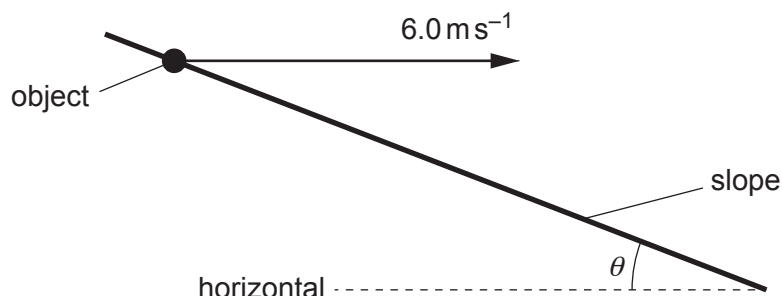


Fig. 2.1 (not to scale)

The slope is at an angle θ to the horizontal. Air resistance is negligible.

The object lands on the slope a time of 0.71 s later and stops without rolling or bouncing.

- (i) Determine the horizontal distance travelled by the object.

distance = m [1]

- (ii) Determine the vertical distance travelled by the object.

distance = m [2]

- (iii) Use your answers in (b)(i) and (b)(ii) to calculate θ .

$\theta = ^\circ$ [2]

- (iv) Determine the magnitude of the displacement of the object from its original position.

displacement = m [2]

17. MAR 24/P22/Q2/a

- 2 (a) Define acceleration.

[1]

18. O/N 23/P21/Q3/a,b(i),c

- 3 A trolley A moves along a horizontal surface at a constant velocity towards another trolley B which is moving at a lower constant speed in the same direction. Fig. 3.1 shows the trolleys at time $t = 0$.

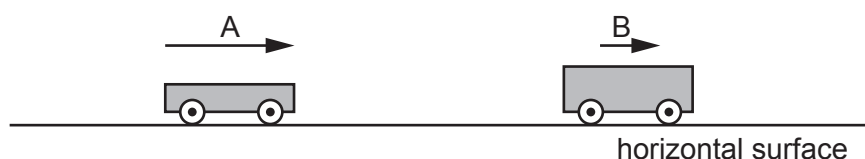


Fig. 3.1

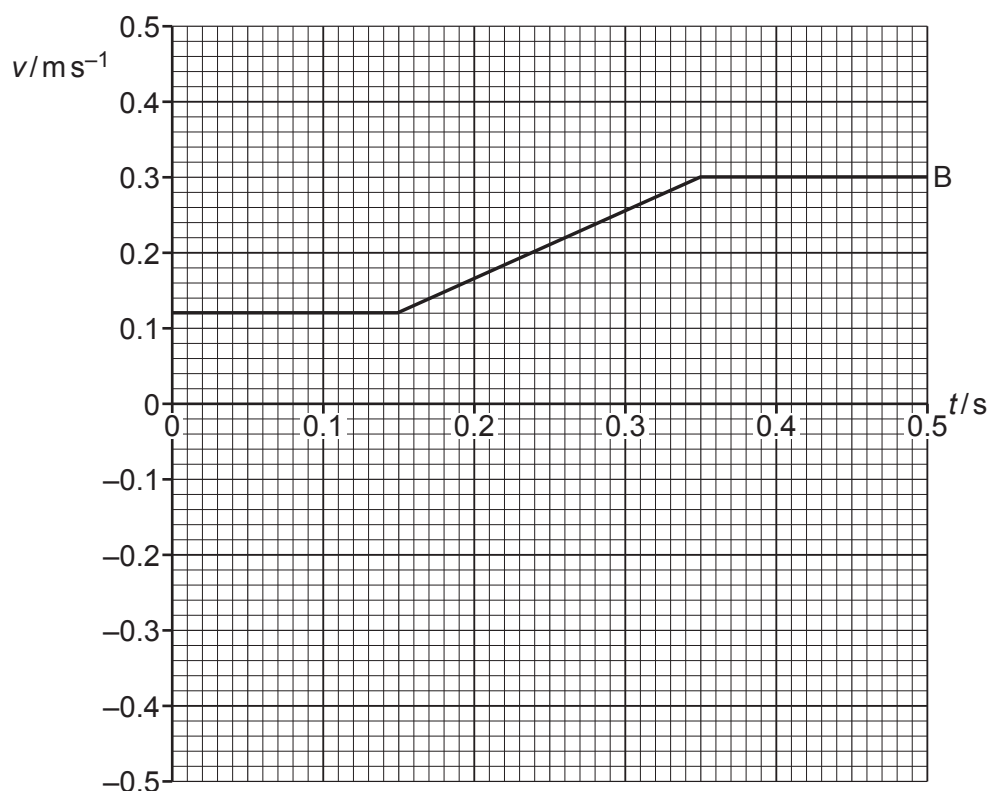
Table 3.1 shows data for the trolleys.

Table 3.1

trolley	mass/kg	initial speed/ m s^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 3.2 shows the variation with time t of the velocity v for trolley B.

**Fig. 3.2**

(a) State what is represented by the area under a velocity–time graph.

[1]

(b) Use Table 3.1 and Fig. 3.2 to determine:

(i) the acceleration of trolley B during the collision

acceleration of B = ms^{-2} [2]

(c) On Fig. 3.2, sketch the variation of the velocity of trolley A with time t from $t = 0$ to $t = 0.50$ s.

[3]

19. O/N 23/P22/Q1/b,c

(b) Under certain conditions, the distance s moved in a straight line by an object in time t is given by

$$s = \frac{1}{2}at^2$$

where a is the acceleration of the object.

State **two** conditions under which the above expression applies to the motion of the object.

1

2

[2]

- (c) The variation with time t of the velocity v of a car that is moving in a straight line is shown in Fig. 1.1.

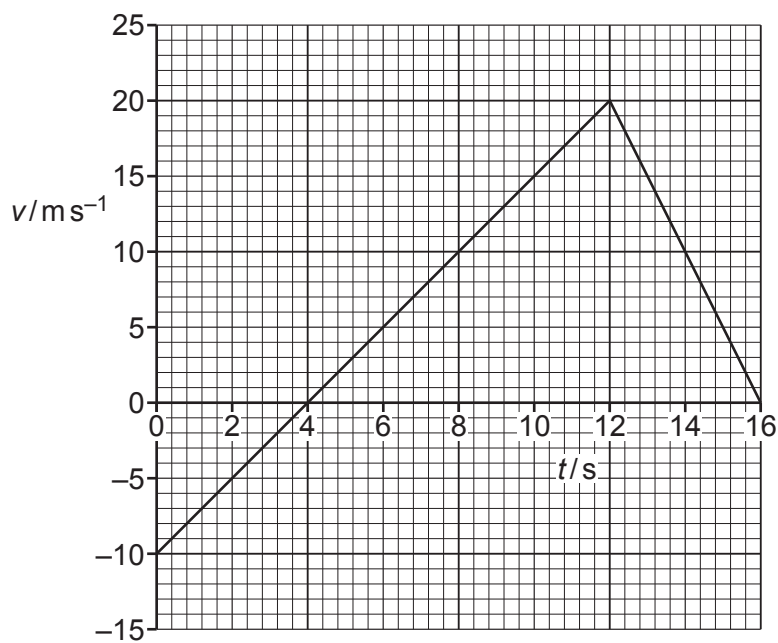


Fig. 1.1

- (i) Compare, qualitatively, the acceleration of the car at time $t = 8.0$ s and at time $t = 14.0$ s in terms of:

- magnitude
- direction.

[2]

- (ii) Determine the magnitude of the acceleration of the car at time $t = 4.0$ s.

acceleration = ms^{-2} [2]

- (iii) The car is at point X at time $t = 0$.

Determine the magnitude of the displacement of the car from X at time $t = 12.0$ s.

displacement = m [2]

20. O/N 23/P23/Q2/a(i,iii)

2 A ball on horizontal ground is kicked towards a vertical wall. Fig. 2.1 shows the path of the ball.

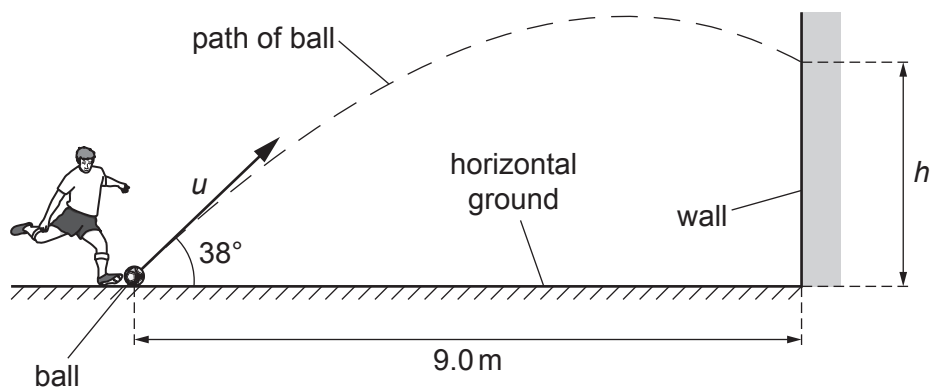


Fig. 2.1 (not to scale)

The ball has an initial velocity u at an angle of 38° to the ground. The ball travels a horizontal distance of 9.0 m before striking the wall at a height h above the ground. The horizontal component u_H of the initial velocity of the ball is 9.5 m s^{-1} .

Air resistance is negligible.

(a) (i) Show that the time t for the ball to reach the wall is 0.95 s .

[1]

(iii) Determine h .

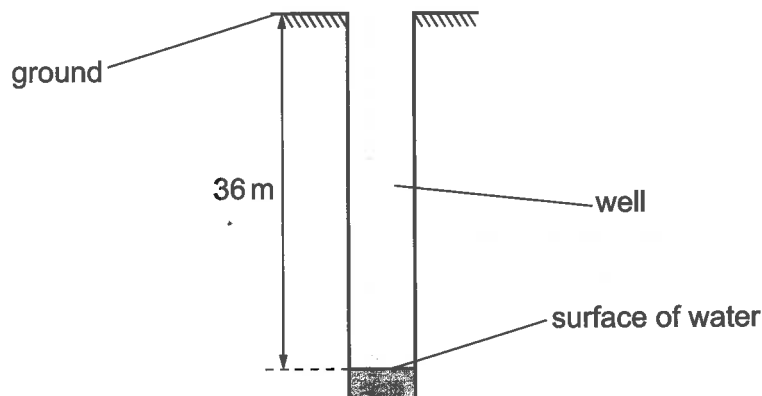
$h =$

m [2]

21. M/J 23/P23/Q1

booklet (scan + OCR)

A well has a depth of 36 m from ground level to the surface of the water in the well, as shown in Fig. 1.1.

**Fig. 1.1** (not to scale)

A student wishes to find the depth of the well. The student plans to drop a stone down the well and record the time taken from releasing the stone to hearing the splash made by the stone as it enters the water.

- (a) Assume that air resistance is negligible and that the stone is released from rest.
Calculate the time taken for the stone to fall from ground level to the surface of the water.

time = s [2]

- (b) The time recorded by the student using a stop-watch is not equal to the time in (a).
Suggest **three** possible reasons, other than the effect of air resistance, for this difference.

1

2

3 [3]

- (c) The student repeats the experiment three times and uses the results to calculate the depth of the well. The values are shown in Table 1.1.

Table 1.1

	1st experiment	2nd experiment	3rd experiment
depth/m	54.4	53.9	54.1

The true depth of the well is 36.0 m. Explain why these results may be described as precise but not accurate.

.....

..... [2]

22. MAR 23/P22/Q2

booklet (scan + OCR)

A motor uses a wire to raise a block, as illustrated in Fig. 2.1.

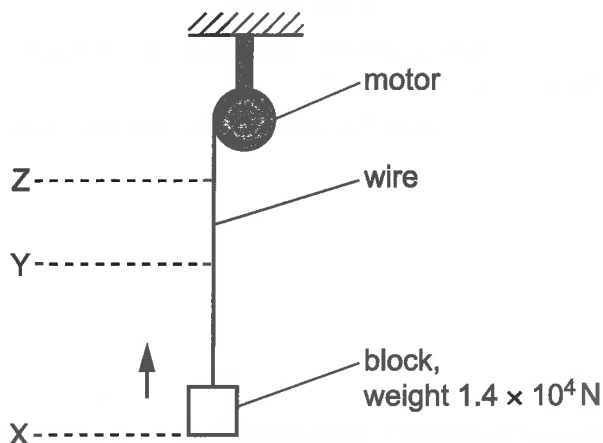


Fig. 2.1 (not to scale)

The base of the block takes a time of 0.49 s to move vertically upwards from level X to level Y at a constant speed of 0.64 m s^{-1} . During this time the wire has a strain of 0.0012. The wire is made of metal of Young modulus $2.2 \times 10^{11} \text{ Pa}$ and has a uniform cross-section.

The block has a weight of $1.4 \times 10^4 \text{ N}$. Assume that the weight of the wire is negligible.

(a) Calculate:

- (i)** the cross-sectional area A of the wire

$$A = \dots\dots\dots \text{ m}^2 \quad [2]$$

- (ii)** the increase in the gravitational potential energy of the block for the movement of its base from X to Y.

$$\text{increase in gravitational potential energy} = \dots\dots\dots \text{ J} \quad [3]$$

- (b) The motor has an efficiency of 56%.

Calculate the input power to the motor as the base of the block moves from X to Y.

input power = W [3]

- (c) The base of the block now has a uniform deceleration of magnitude 1.3 m s^{-2} from level Y until the base of the block stops at level Z.

Calculate the tension T in the wire as the base of the block moves from Y to Z.

$T = \dots\dots\dots$ N [3]

- (d) The base of the block is at levels X, Y and Z at times t_X , t_Y and t_Z respectively.

On Fig. 2.2, sketch a graph to show the variation with time t of the distance d of the base of the block from level X. Numerical values of d and t are not required.

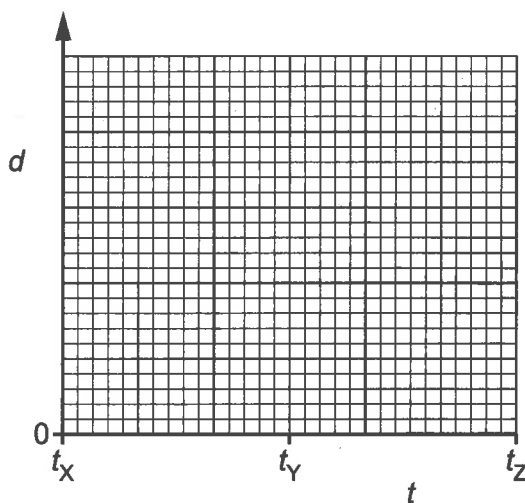


Fig. 2.2

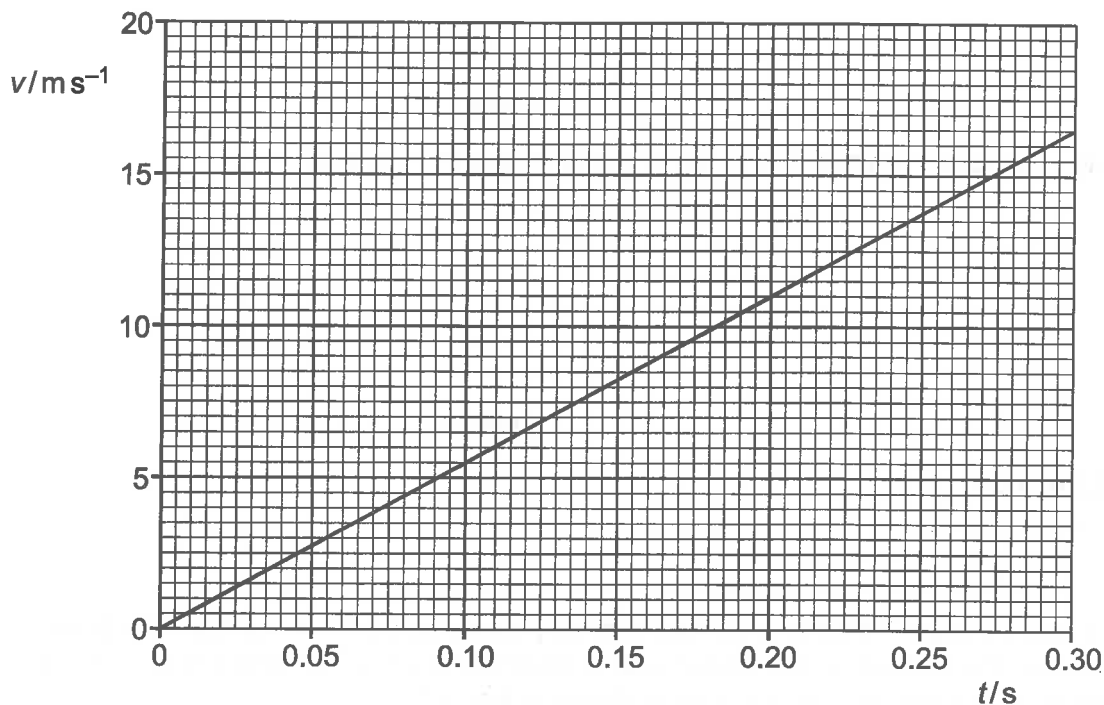
[2]

23. O/N 22/P23/Q2

booklet (scan + OCR)

The engine of a toy rocket pushes gases vertically downwards and this results in the rocket accelerating vertically upwards from the ground.

The rocket starts to move from rest at time $t = 0$. The variation with time t of the vertical velocity v of the rocket for the first 0.30 s of the flight is shown in Fig. 2.1.

**Fig. 2.1**

As the rocket moves, the thrust force T provided by the rocket engine is 16 N.

Assume that the mass of the rocket is constant for this part of its flight.

Assume that air resistance is negligible.

(a) For this part of the rocket's flight:

(i) show that the acceleration of the rocket is 55 m s^{-2}

[1]

(ii) state an expression for the resultant force F experienced by the rocket in terms of the thrust force T and the weight W of the rocket

[1]

(iii) calculate the mass of the rocket.

mass = kg [2]

- (b) At time $t = 0.30$ s, a small piece of metal separates from the rocket.

Calculate:

- (i) the height of the rocket above the ground at $t = 0.30$ s

height = m [2]

- (ii) the speed at which the piece of metal strikes the ground.

speed = ms^{-1} [3]

24. M/J 22/P21/Q1

booklet (scan + OCR)

- (a) Define velocity.

..... [1]

- (b) A rock of mass 7.5 kg is projected vertically upwards from the surface of a planet. The rock leaves the surface of the planet with a speed of 4.0 ms^{-1} at time $t = 0$. The variation with time t of the velocity v of the rock is shown in Fig. 1.1.

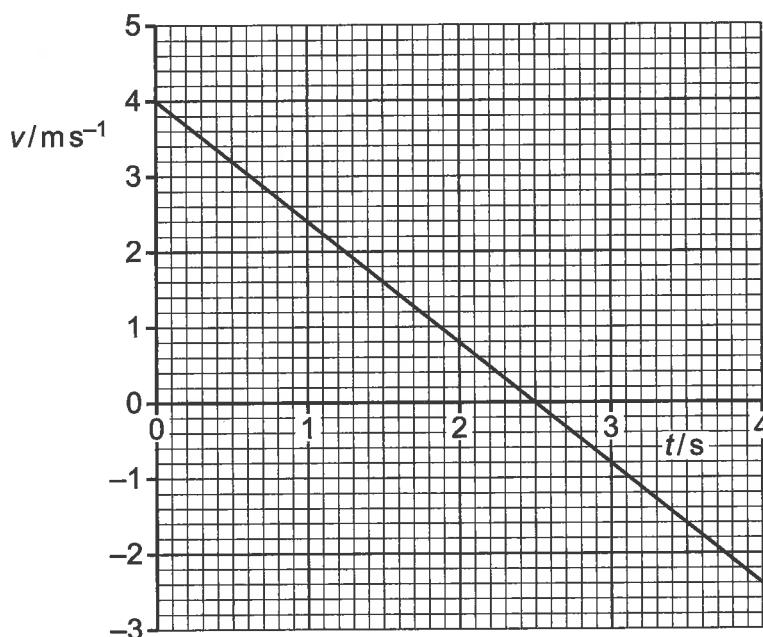


Fig. 1.1

Assume that the planet does not have an atmosphere and that the viscous force acting on the rock is always zero.

- (i) Determine the height of the rock above the surface of the planet at time $t = 4.0$ s.

height = m [3]

- (ii) Determine the change in the momentum of the rock from time $t = 0$ to time $t = 4.0$ s.

change in momentum = N s [2]

- (iii) Determine the weight W of the rock on this planet.

$W =$ N [2]

- (c) In practice, the planet in (b) does have an atmosphere that causes a viscous force to act on the moving rock.

State and explain the variation, if any, in the resultant force acting on the rock as it moves vertically upwards.

.....
..... [2]

25. M/J 22/P23/Q3

booklet (scan + OCR)

(a) Define velocity.

..... [1]

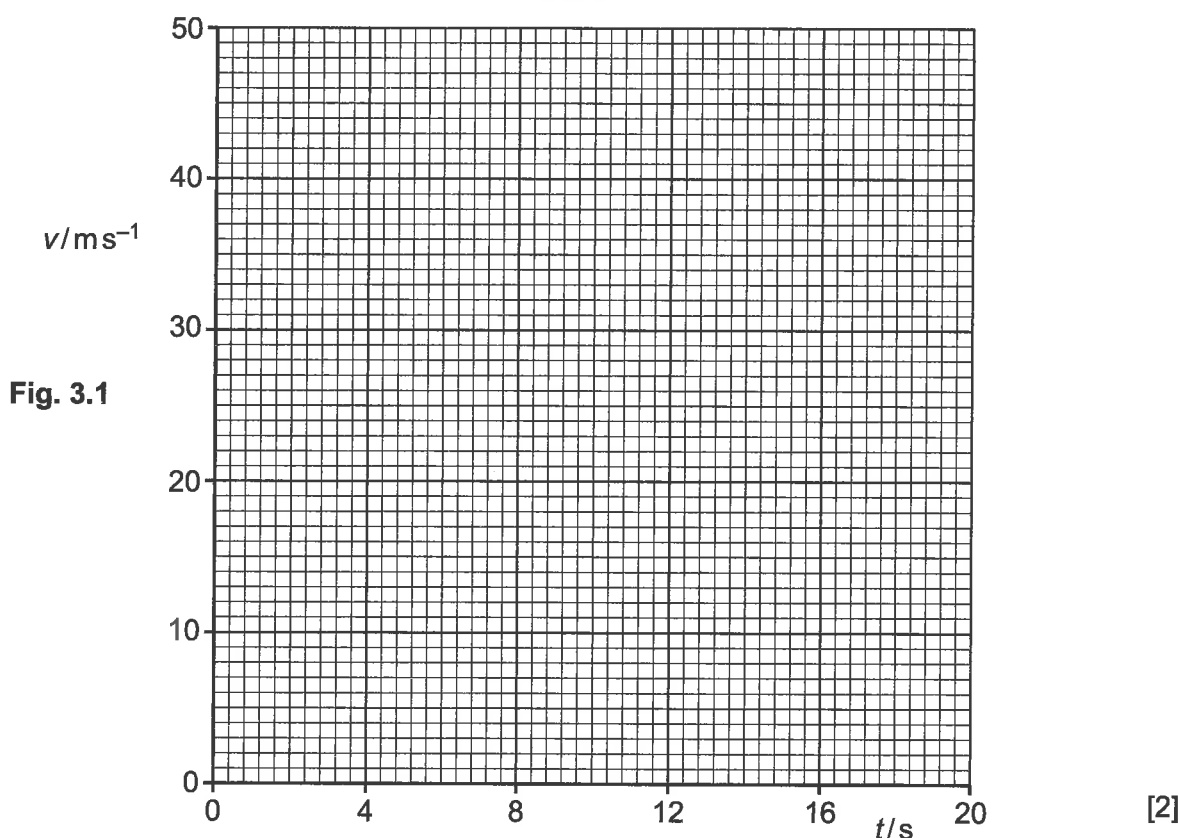
- (b)** A constant driving force of 2400 N acts on a car of mass 1200 kg. The car accelerates from rest in a straight line along a horizontal road.

Assume that the resistive forces acting on the car are negligible.

- (i)** Calculate the acceleration of the car.

acceleration = ms^{-2} [1]

- (ii)** On Fig. 3.1, sketch a graph showing the variation with time t of the velocity v of the car for the first 20 seconds of its motion. Label this line A.



- (c)** In reality, a resistive force due to air resistance acts on the car in **(b)**. This resistive force increases with speed until it becomes equal in magnitude to the driving force at time $t = 12$ s.

- (i)** On Fig. 3.1, sketch a second line to show the variation with time t of the velocity v of the car for the first 20 seconds of its motion. Label this line B. [3]

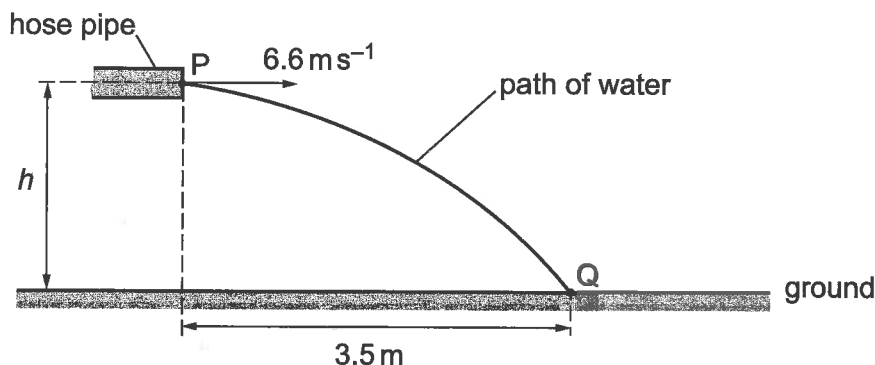
- (ii)** At time $t = 20$ s, the driving force is increased to 3000 N and remains constant at this value. Describe how the velocity of the car changes due to this increase in the driving force.

..... [2]

26. MAR 22/P22/Q2

booklet (scan + OCR)

Water leaves the end of a hose pipe at point P with a horizontal velocity of 6.6 m s^{-1} , as shown in Fig. 2.1.

**Fig. 2.1** (not to scale)

Point P is at height h above the ground. The water hits the ground at point Q. The horizontal distance from P to Q is 3.5 m .

Air resistance is negligible. Assume that the water between P and Q consists of non-interacting droplets of water and that the only force acting on each droplet is its weight.

- (a) Explain, briefly, why the horizontal component of the velocity of a droplet of water remains constant as it moves from P to Q.

..... [1]

- (b) Show that the time taken for a droplet of water to move from P to Q is 0.53 s .

[1]

- (c) Calculate height h .

$h = \dots\dots\dots \text{ m}$ [2]

- (d) For the movement of a droplet of water from P to Q, state and explain whether the displacement of the droplet is less than, more than or the same as the distance along its path.

..... [1]

- (e) Calculate the magnitude of the displacement of a droplet of water that moves from P to Q.

displacement = m [2]

27. M/J 21/P23/Q2

booklet (scan + OCR)

- (a) Define *acceleration*.

..... [1]

- (b) A stone falls vertically from the top of a cliff. Fig. 2.1 shows the variation with time t of the velocity v of the stone.

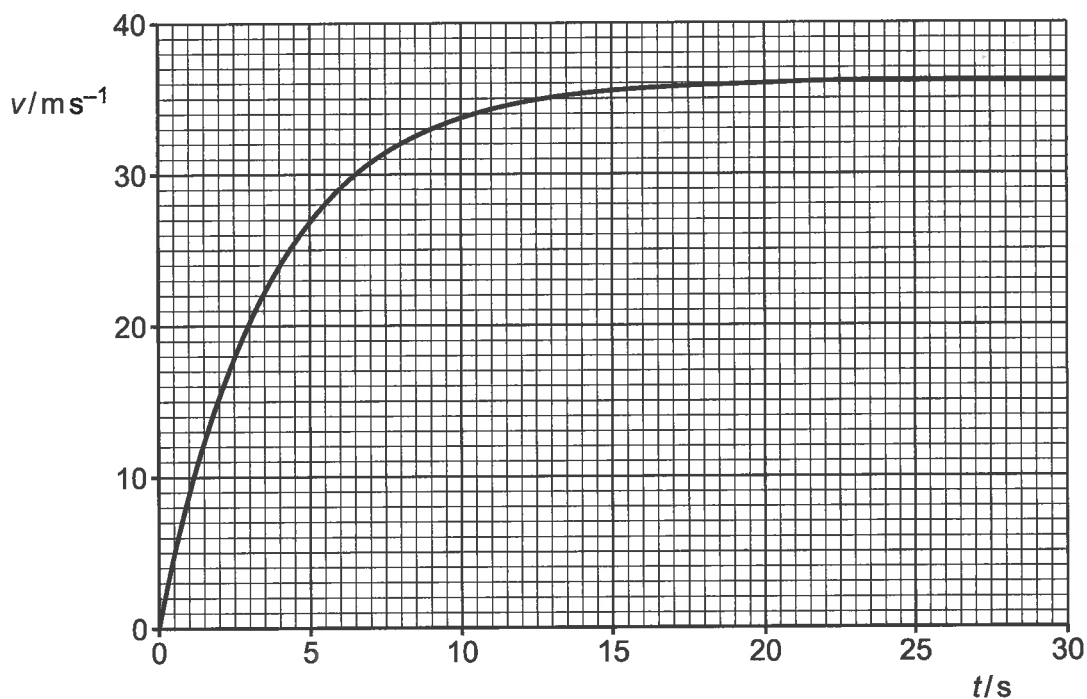


Fig. 2.1

- (i) Explain, with reference to forces acting on the stone, the shape of the curve in Fig. 2.1.

.....

 [3]

- (ii) Use Fig. 2.1 to determine the speed of the stone when the resultant force on it is zero.

speed = ms^{-1} [1]

- (iii) Use Fig. 2.1 to calculate the approximate height through which the stone falls between $t = 0$ and $t = 30 \text{ s}$.

height = m [3]

- (iv) On Fig. 2.2, sketch the variation with t of the acceleration a of the stone between $t = 0$ and $t = 30$ s.

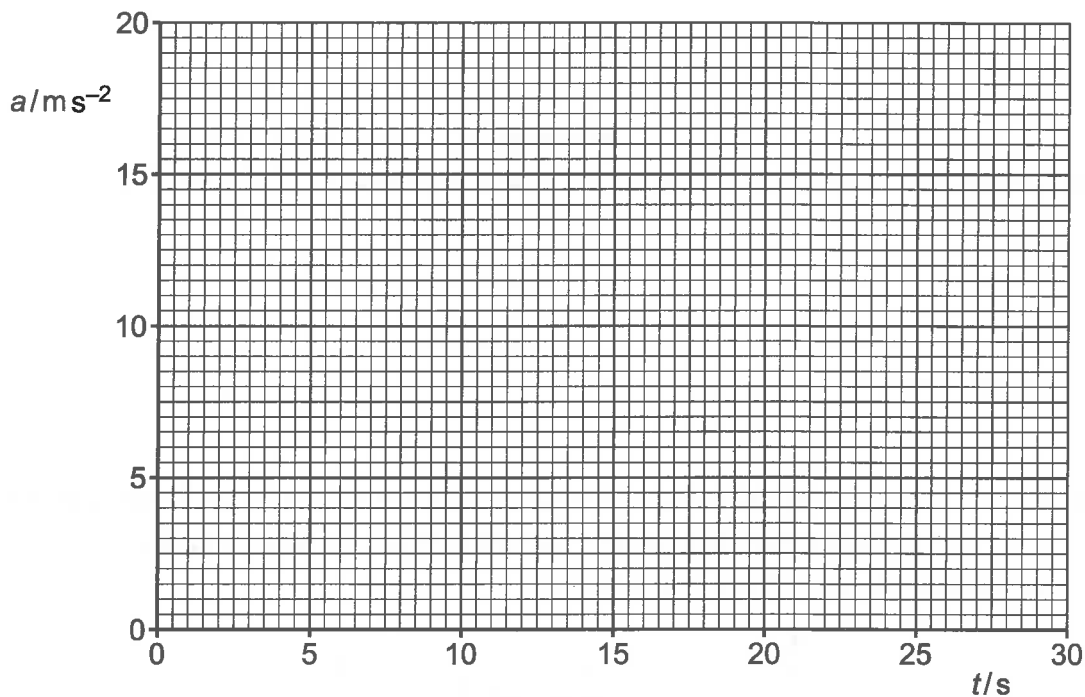


Fig. 2.2

[3]

28. MAR 21/P22/Q1

booklet (scan + OCR)

- (a) Complete Table 1.1 by stating whether each of the quantities is a vector or a scalar.

Table 1.1

quantity	vector or scalar
acceleration	
power	
work	

[2]

- (b) The variation with time t of the velocity v of an object is shown in Fig. 1.1.

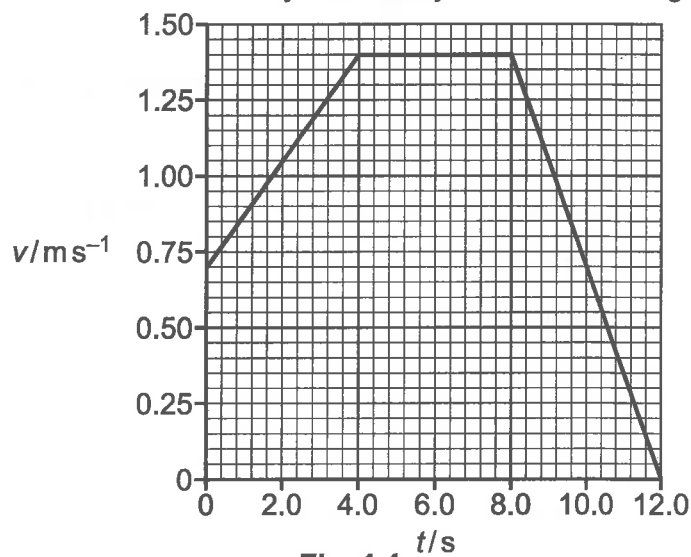


Fig. 1.1

- (i) Determine the acceleration of the object from time $t = 0$ to time $t = 4.0$ s.

acceleration = ms^{-2} [2]

- (ii) Determine the distance moved by the object from time $t = 0$ to time $t = 4.0$ s.

distance = m [2]

- (c) (i) Define *force*.

..... [1]

- (ii) The motion represented in Fig. 1.1 is caused by a resultant force F acting on the object. On Fig. 1.2, sketch the variation of F with time t from $t = 0$ to $t = 12.0$ s. Numerical values of F are not required.

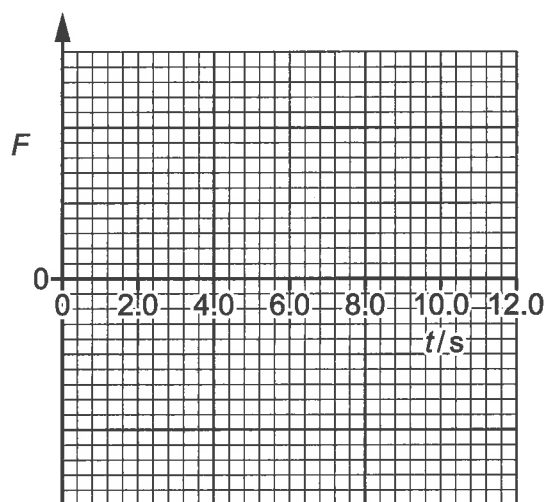


Fig. 1.2

[3]

29. O/N 20/P22/Q1

booklet (scan + OCR)

- (a) Complete Table 1.1 by putting a tick (✓) in the appropriate column to indicate whether the listed quantities are scalars or vectors.

Table 1.1

quantity	scalar	vector
acceleration		
density		
temperature		
momentum		

[2]

- (b)** A toy train moves along a straight section of track. Fig. 1.1 shows the variation with time t of the distance d moved by the train.

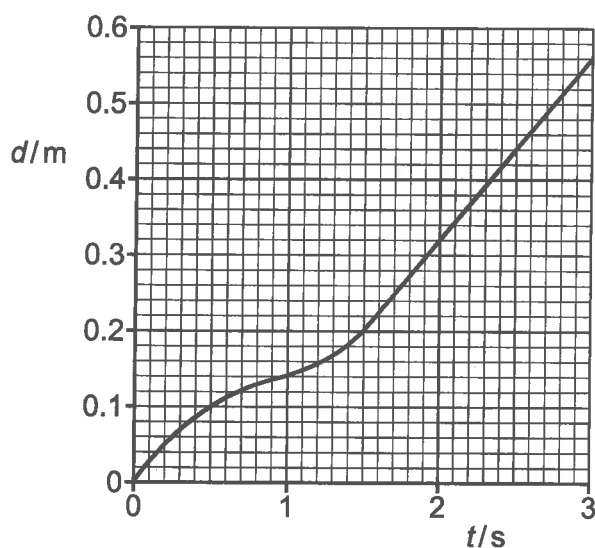


Fig. 1.1

- (i)** Describe qualitatively the motion of the train between time $t = 0$ and time $t = 1.0$ s.

..... [1]

- (ii)** Determine the speed of the train at time $t = 2.0$ s.

speed = ms^{-1} [2]

- (c)** The straight section of track in **(b)** is part of the loop of track shown in Fig. 1.2.

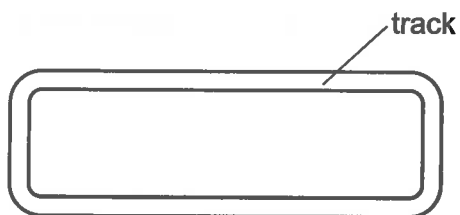


Fig. 1.2

The train completes exactly one lap of the loop.

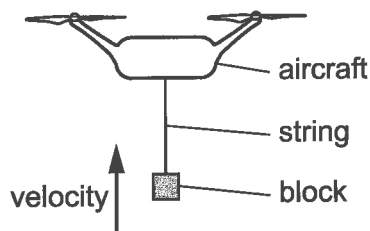
State and explain the average velocity of the train over the one complete lap.

..... [1]

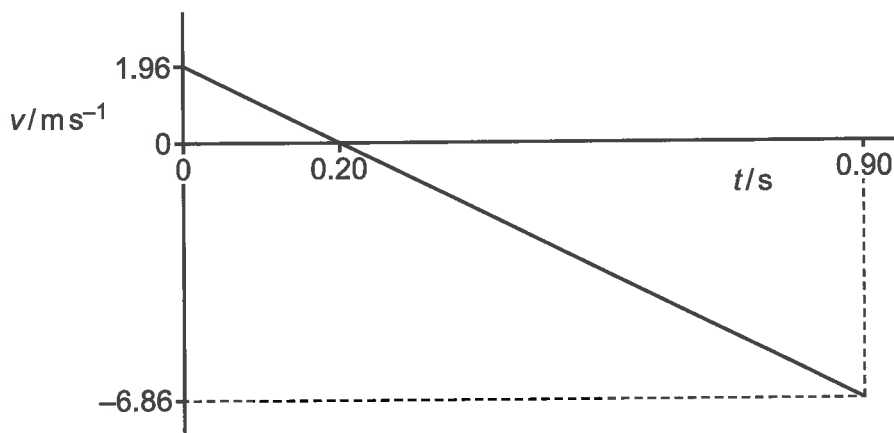
30. O/N 20/P21/Q2

booklet (scan + OCR)

A small block is lifted vertically upwards by a toy aircraft, as illustrated in Fig. 2.1.

**Fig. 2.1**

As the block is moving upwards, the string breaks at time $t = 0$. The block initially continues moving upwards and then falls and hits the ground at time $t = 0.90$ s. The variation with time t of the velocity v of the block is shown in Fig. 2.2.

**Fig. 2.2**

Air resistance is negligible.

(a) State the feature of the graph in Fig. 2.2 that shows the block has a constant acceleration.

..... [1]

(b) Use Fig. 2.2 to determine the height of the block above the ground when the string breaks at time $t = 0$.

height = m [3]

(c) The block has a weight of 0.86 N.

Calculate the difference in gravitational potential energy of the block between time $t = 0$ and time $t = 0.90$ s.

difference in gravitational potential energy = J [2]

- (d) On Fig. 2.3, sketch a line to show the variation of the distance moved by the block with time t from $t = 0$ to $t = 0.20$ s. Numerical values of distance are not required.

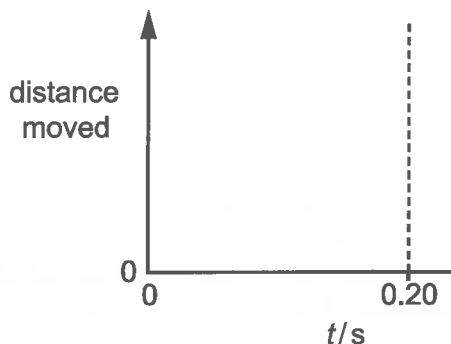


Fig. 2.3

[2]

- (e) A block of greater mass is now released from the same height with the same upward velocity. Air resistance is still negligible. State and explain the effect, if any, of the increased mass on the speed with which the block hits the ground.

[1]

31. O/N 20/P23/Q3

booklet (scan + OCR)

A ball is fired horizontally with a speed of 41.0 ms^{-1} from a stationary cannon at the top of a hill. The ball lands on horizontal ground that is a vertical distance of 57 m below the cannon, as shown in Fig. 3.1.

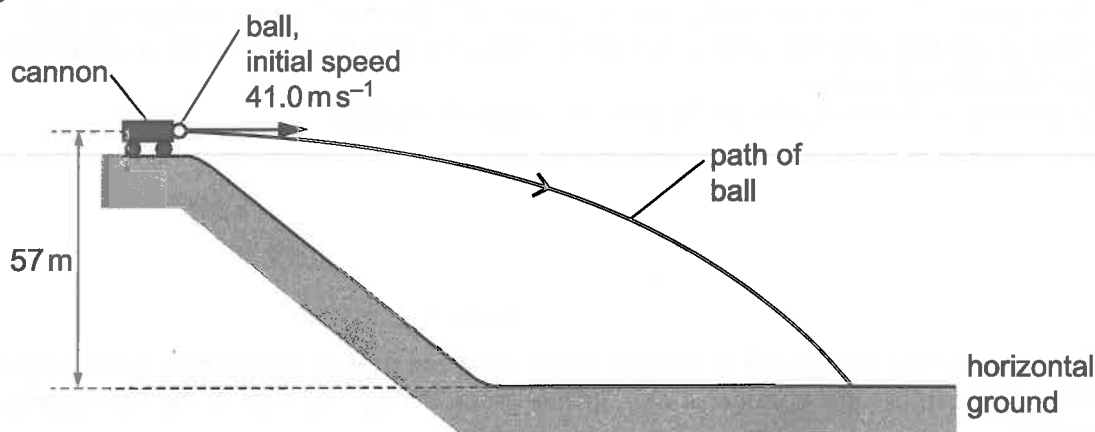


Fig. 3.1 (not to scale)

Assume air resistance is negligible.

- (a) Show that the time taken for the ball to reach the ground, after being fired, is 3.4 s .

[2]

- (b) Calculate the horizontal distance of the ball from the cannon at the point where the ball lands on the ground.

horizontal distance = m [1]

- (c) Determine the magnitude of the displacement of the ball from the cannon at the point where the ball lands on the ground.

displacement = m [2]

- (d) The ball leaves the cannon at time $t = 0$.

On Fig. 3.2, sketch a graph to show the variation of the magnitude v of the vertical component of the velocity of the ball with time t from $t = 0$ to $t = 3.4$ s.

Numerical values are not required.

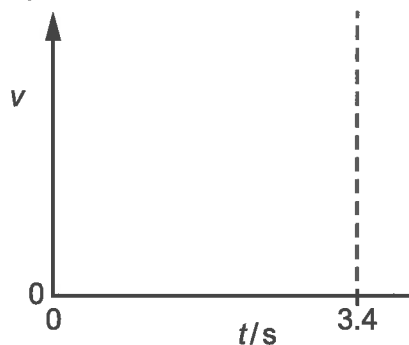


Fig. 3.2

[1]

- (e) The cannon recoils horizontally with a speed of 0.340 ms^{-1} when it fires the ball. The total mass of the ball and the cannon is 1480 kg . Assume that no external horizontal forces act on the ball-cannon system.

Determine, to three significant figures, the mass of the ball.

mass = kg [2]

- (f) The cannon now fires a ball of smaller mass. Assume that air resistance is still negligible. State and explain the change, if any, to the graph in Fig. 3.2 due to the decreased mass of the ball.

..... [2]

32. M/J 20/P22/Q2

booklet (scan + OCR)

- (a) Fig. 2.1 shows the velocity–time graph for an object moving in a straight line.

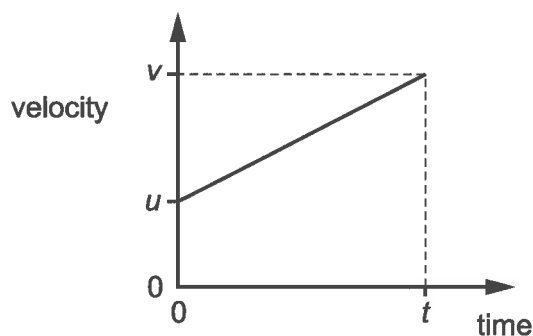


Fig. 2.1

- (i) Determine an expression, in terms of u , v and t , for the area under the graph.

area = [1]

- (ii) State the name of the quantity represented by the area under the graph.

..... [1]

- (b) A ball is kicked with a velocity of 15 ms^{-1} at an angle of 60° to horizontal ground. The ball then strikes a vertical wall at the instant when the path of the ball becomes horizontal, as shown in Fig. 2.2.

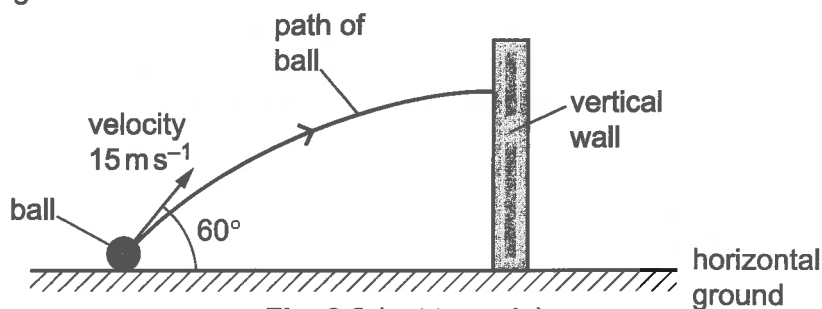


Fig. 2.2 (not to scale)

Assume that air resistance is negligible.

- (i) By considering the vertical motion of the ball, calculate the time it takes to reach the wall.

time = s [3]

- (ii) Explain why the horizontal component of the velocity of the ball remains constant as it moves to the wall.

..... [1]

- (iii) Show that the ball strikes the wall with a horizontal velocity of 7.5 ms^{-1} .

[1]

- (c) The mass of the ball in (b) is 0.40 kg . It is in contact with the wall for a time of 0.12 s and rebounds horizontally with a speed of 4.3 ms^{-1} .

- (i) Use the information from (b)(iii) to calculate the change in momentum of the ball due to the collision.

change in momentum = kg ms^{-1} [2]

- (ii) Calculate the magnitude of the average force exerted on the ball by the wall.

average force = N [1]

33. M/J 20/P21/Q2

booklet (scan + OCR)

- (a) State Newton's second law of motion.

..... [1]

- (b) A delivery company suggests using a remote-controlled aircraft to drop a parcel into the garden of a customer. When the aircraft is vertically above point P on the ground, it releases the parcel with a velocity that is horizontal and of magnitude 5.4 m s^{-1} . The path of the parcel is shown in Fig. 2.1.

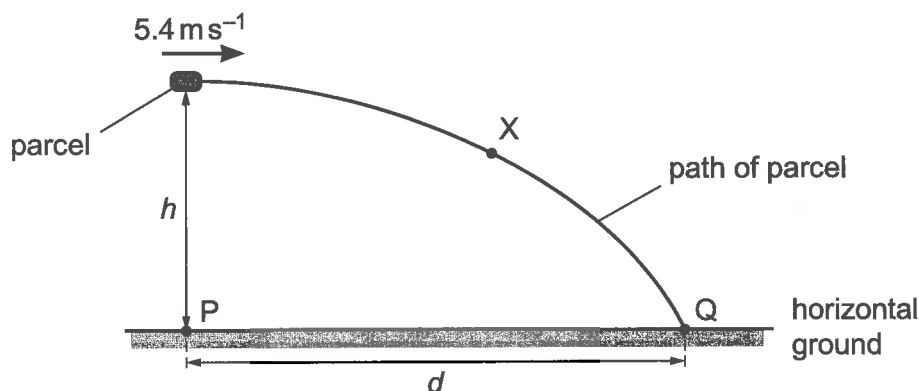


Fig. 2.1 (not to scale)

The parcel takes a time of 0.81 s after its release to reach point Q on the horizontal ground. Assume air resistance is negligible.

- (i) On Fig. 2.1, draw an arrow from point X to show the direction of the acceleration of the parcel when it is at that point. [1]
- (ii) Determine the height h of the parcel above the ground when it is released.

$h = \dots\dots\dots \text{ m}$ [2]

- (iii) Calculate the horizontal distance d between points P and Q.

$d = \dots\dots\dots \text{ m}$ [1]

- (c) Another parcel is accidentally released from rest by a different aircraft when it is hovering at a great height above the ground. Air resistance is now significant.

- (i) On Fig. 2.2, draw arrows to show the directions of the forces acting on the parcel as it falls vertically downwards. Label each arrow with the name of the force.



Fig. 2.2

[2]

- (ii) By considering the forces acting on the parcel, state and explain the variation, if any, of the acceleration of the parcel as it moves downwards before it reaches constant (terminal) speed.

.....

 [3]

- (iii) Describe the energy conversion that occurs when the parcel is falling through the air at constant (terminal) speed.

..... [1]

34. MAR 20/P22/Q2

booklet (scan + OCR)

A dolphin is swimming under water at a constant speed of 4.50 m s^{-1} .

- (a) The dolphin emits a sound as it swims directly towards a stationary submerged diver. The frequency of the sound heard by the diver is 9560 Hz . The speed of sound in the water is 1510 m s^{-1} .

Determine the frequency, to three significant figures, of the sound emitted by the dolphin.

frequency = Hz [2]

- (b) The dolphin strikes the bottom of a floating ball so that the ball rises vertically upwards from the surface of the water, as illustrated in Fig. 2.1.

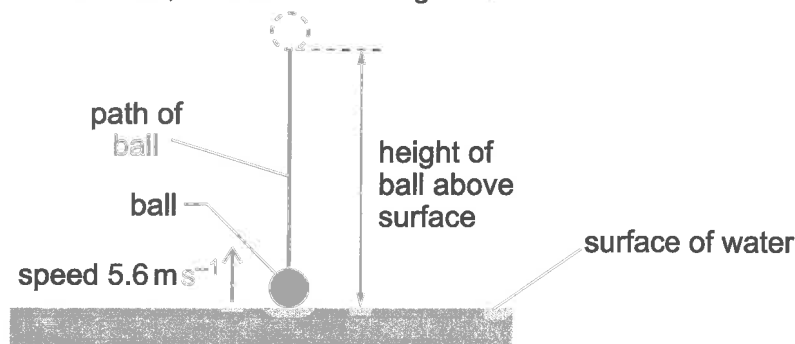


Fig. 2.1

The ball leaves the water surface with speed 5.6 m s^{-1} .

Assume that air resistance is negligible.

- (i) Calculate the maximum height reached by the ball above the surface of the water.

height = m [2]

- (ii) The ball leaves the water at time $t = 0$ and reaches its maximum height at time $t = T$.

On Fig. 2.2, sketch a graph to show the variation of the speed of the ball with time t from $t = 0$ to $t = T$. Numerical values are **not** required.

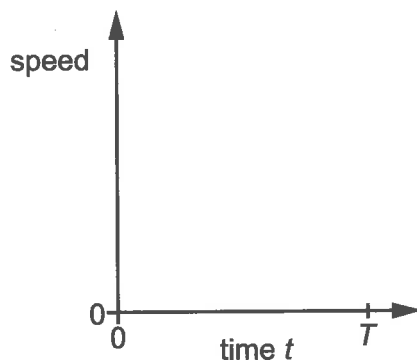


Fig. 2.2

[1]

- (iii) The mass of the ball is 0.45 kg .

Use your answer in (b)(i) to calculate the change in gravitational potential energy of the ball as it rises from the surface of the water to its maximum height.

change in gravitational potential energy = J [2]

- (iv) State and explain the variation in the magnitude of the acceleration of the ball as it falls back towards the surface of the water if air resistance is **not** negligible.

.....

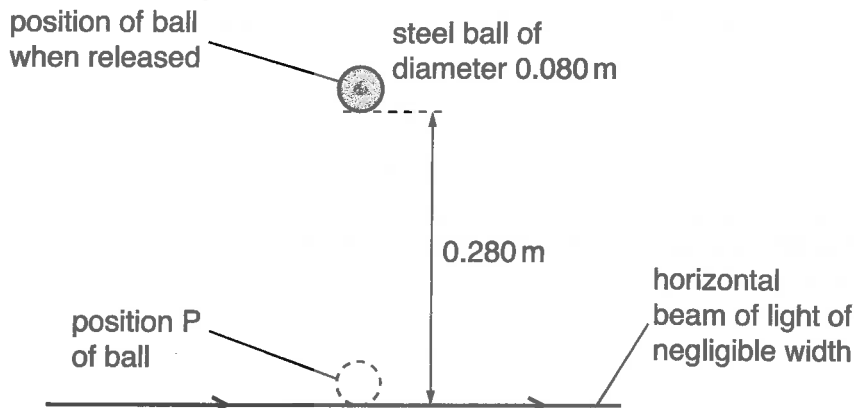
..... [2]

35. O/N 19/P22/Q2

booklet (scan + OCR)

(a) Define *acceleration*.

..... [1]

(b) A steel ball of diameter 0.080 m is released from rest and falls vertically in air, as illustrated in Fig. 2.1.**Fig. 2.1** (not to scale)

A horizontal beam of light of negligible width is a vertical distance of 0.280 m below the bottom of the ball when it is released. The ball falls through and breaks the beam of light.

(i) Explain why the force due to air resistance acting on the ball may be neglected when calculating the time taken for the ball to reach the beam of light.

..... [1]

(ii) Calculate the time taken for the ball to fall from rest to position P where the bottom of the ball touches the beam of light.

time taken = s [2]

(iii) Determine the time interval during which the beam of light is broken by the ball.

time interval = s [2]

(c) A different ball is released from the same position as the steel ball in **(b)**. This ball has the same diameter but a much lower density. For this ball, the force due to air resistance cannot be neglected as the ball falls.

State and explain the change, if any, to the time interval during which the beam of light is broken by the ball.

.....
 [2]

36. O/N 18/P22/Q1

booklet (scan + OCR)

A golfer strikes a ball so that it leaves horizontal ground with a velocity of 6.0 m s^{-1} at an angle θ to the horizontal, as illustrated in Fig. 1.1.

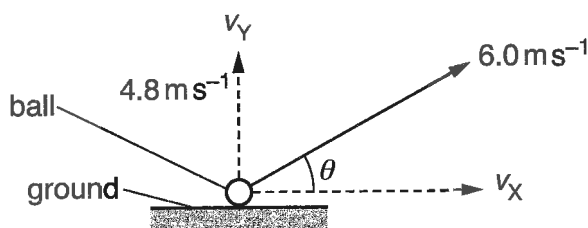


Fig. 1.1 (not to scale)

The magnitude of the initial vertical component v_y of the velocity is 4.8 m s^{-1} .
Assume that air resistance is negligible.

(a) Show that the magnitude of the initial horizontal component v_x of the velocity is 3.6 m s^{-1} .

[1]

(b) The ball leaves the ground at time $t = 0$ and reaches its maximum height at $t = 0.49 \text{ s}$.

On Fig. 1.2, sketch separate lines to show the variation with time t , until the ball returns to the ground, of

(i) the vertical component v_y of the velocity (label this line Y),

[2]

(ii) the horizontal component v_x of the velocity (label this line X).

[2]

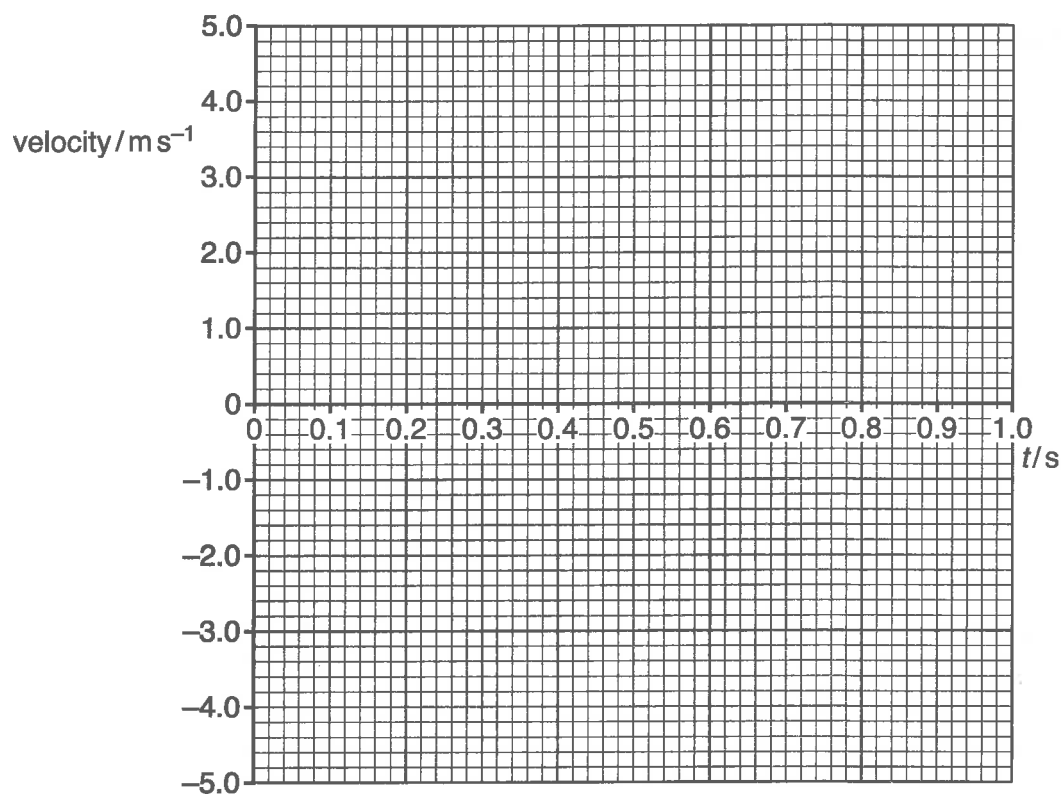


Fig. 1.2

- (c) Calculate the maximum height reached by the ball.

maximum height = m [2]

- (d) For the movement of the ball from the ground to its maximum height, determine the ratio

$$\frac{\text{kinetic energy at maximum height}}{\text{change in gravitational potential energy}}$$

ratio = [4]

- (e) In practice, significant air resistance acts on the ball. Explain why the actual time taken for the ball to reach maximum height is less than the time calculated when air resistance is assumed to be negligible.

..... [1]

37. O/N 18/P21/Q1

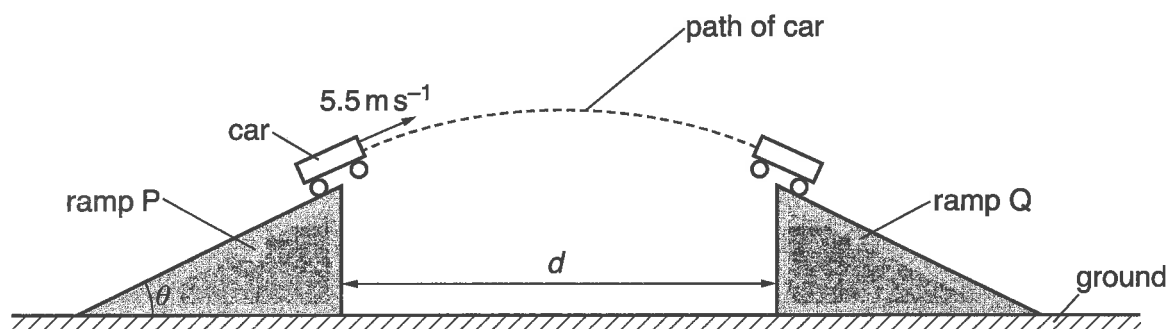
booklet (scan + OCR)

(a) Define**(i) displacement,**

..... [1]

(ii) acceleration.

..... [1]

(b) A remote-controlled toy car moves up a ramp and travels across a gap to land on another ramp, as illustrated in Fig. 1.1.**Fig. 1.1**

The car leaves ramp P with a velocity of 5.5 m s^{-1} at an angle θ to the horizontal. The horizontal component of the car's velocity as it leaves the ramp is 4.6 m s^{-1} . The car lands at the top of ramp Q. The tops of both ramps are at the same height and are distance d apart. Air resistance is negligible.

(i) Show that the car leaves ramp P with a vertical component of velocity of 3.0 m s^{-1} .

[1]

(ii) Determine the time taken for the car to travel between the ramps.

time taken = s [2]

(iii) Calculate the horizontal distance d between the tops of the ramps. $d = \dots\dots\dots \text{ m}$ [1]

(iv) Calculate the ratio

$$\frac{\text{kinetic energy of the car at its maximum height}}{\text{kinetic energy of the car as it leaves ramp P}}$$

ratio = [3]

- (c) Ramp Q is removed. The car again leaves ramp P as in (b) and now lands directly on the ground. The car leaves ramp P at time $t = 0$ and lands on the ground at time $t = T$.
On Fig. 1.2, sketch the variation with time t of the vertical component v_y of the car's velocity from $t = 0$ to $t = T$. Numerical values of v_y and t are not required.

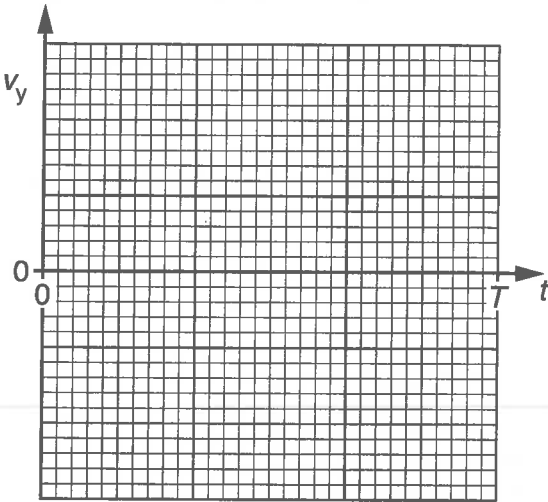


Fig. 1.2

[2]

38. MAR 18/P22/Q1
booklet (scan + OCR)

- (a) Complete Fig. 1.1 to indicate whether each of the quantities is a vector or a scalar.

quantity	vector or scalar
acceleration	
speed	
power	

Fig. 1.1

[2]

- (b) A ball is projected with a horizontal velocity of 1.1 ms^{-1} from point A at the edge of a table, as shown in Fig. 1.2.

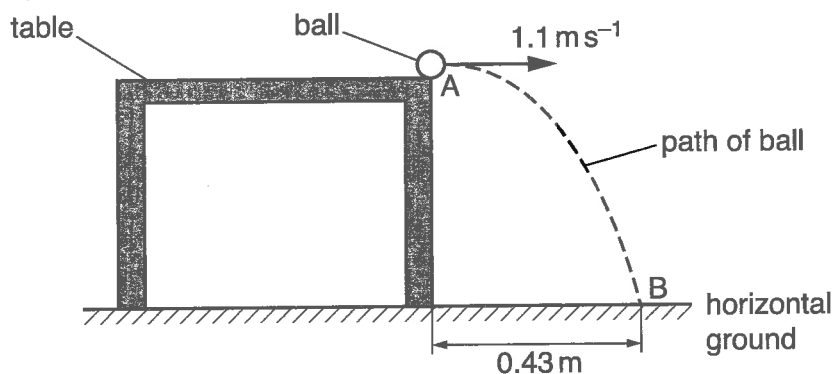


Fig. 1.2

The ball lands on horizontal ground at point B which is a distance of 0.43 m from the base of the table. Air resistance is negligible.

- (i) Calculate the time taken for the ball to fall from A to B.

time = s [1]

- (ii) Use your answer in (b)(i) to determine the height of the table.

height = m [2]

- (iii) The ball leaves the table at time $t = 0$.

For the motion of the ball between A and B, sketch graphs on Fig. 1.3 to show the variation with time t of

- the acceleration a of the ball,
- the vertical component s_v of the displacement of the ball from A.

Numerical values are not required.

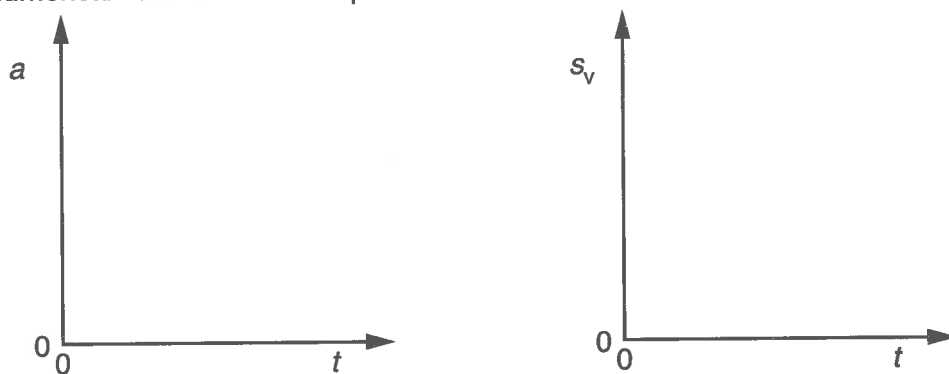


Fig. 1.3

[2]

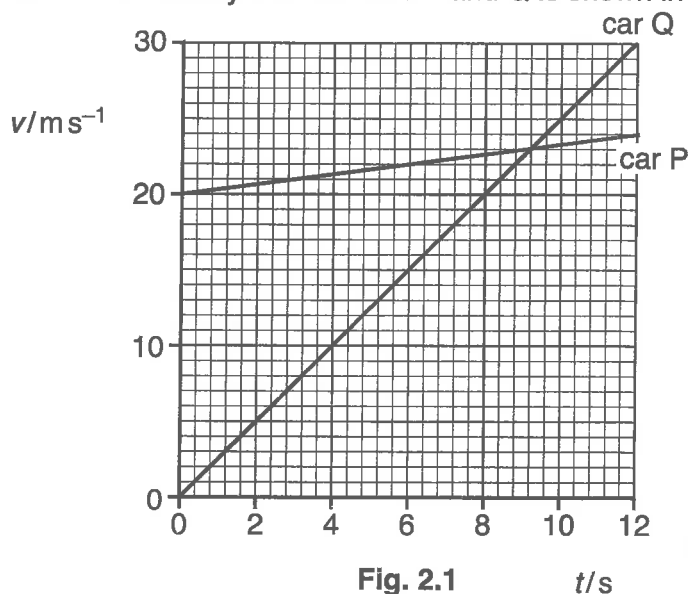
- (c) A ball of greater mass is projected from the table with the same velocity as the ball in (b). Air resistance is still negligible. State and explain the effect, if any, of the increased mass on the time taken for the ball to fall to the ground.

..... [1]

39. O/N 17/P21/Q2

booklet (scan + OCR)

The variation with time t of the velocity v of two cars P and Q is shown in Fig. 2.1.

**Fig. 2.1** t/s

The cars travel in the same direction along a straight road.

Car P passes car Q at time $t = 0$.

- (a) The speed limit for cars on the road is 100 km h^{-1} . State and explain whether car Q exceeds the speed limit.

..... [1]

- (b) Calculate the acceleration of car P.

acceleration = ms^{-2} [2]

- (c) Determine the distance between the two cars at time $t = 12 \text{ s}$.

distance = m [3]

- (d) From time $t = 12 \text{ s}$, the velocity of each car remains constant at its value at $t = 12 \text{ s}$.

Determine the time t at which car Q passes car P.

$t =$ s [2]

40. M/J 17/P22/Q2

booklet (scan + OCR)

- (a) Define *velocity*.

..... [1]

- (b)** A ball of mass 0.45 kg leaves the edge of a table with a horizontal velocity v , as shown in Fig. 2.1.

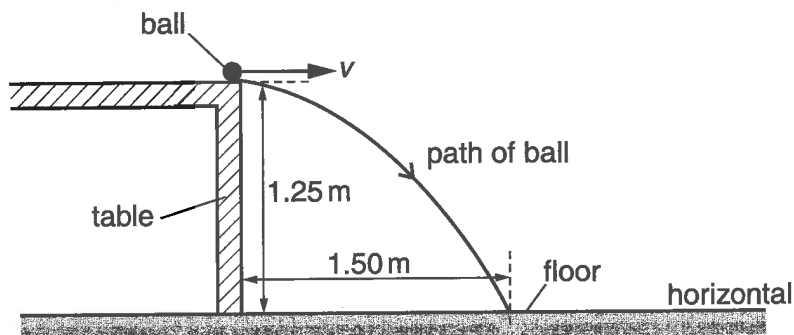


Fig. 2.1

The height of the table is 1.25 m. The ball travels a distance of 1.50 m horizontally before hitting the floor.

Air resistance is negligible.

Calculate, for the ball,

- (i)** the horizontal velocity v as it leaves the table,

$$v = \dots\dots\dots \text{ms}^{-1} \quad [3]$$

- (ii)** the velocity just as it hits the floor,

$$\text{magnitude of velocity} = \dots\dots\dots \text{ms}^{-1}$$

$$\text{angle to the horizontal} = \dots\dots\dots^\circ \quad [4]$$

- (iii)** the kinetic energy just as it hits the floor,

$$\text{kinetic energy} = \dots\dots\dots \text{J} \quad [2]$$

- (iv)** the loss in gravitational potential energy as it falls from the table to the floor.

$$\text{loss in potential energy} = \dots\dots\dots \text{J} \quad [2]$$

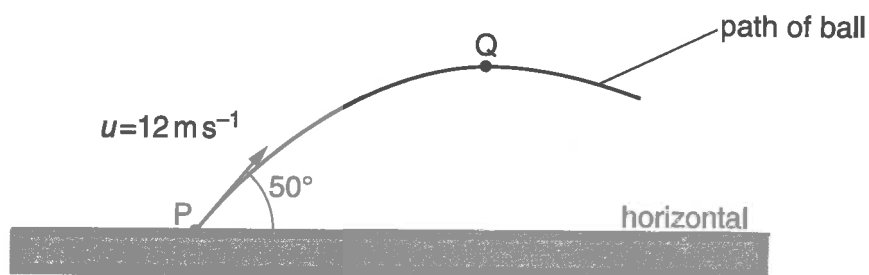
- (c)** Explain why the kinetic energy of the ball in **(b)(iii)** does not equal the loss of gravitational potential energy in **(b)(iv)**.

..... [1]

41. M/J 16/P21/Q2

booklet (scan + OCR)

A ball is thrown from a point P with an initial velocity u of 12 ms^{-1} at 50° to the horizontal, as illustrated in Fig. 2.1.

**Fig. 2.1**

The ball reaches maximum height at Q.

Air resistance is negligible.

(a) Calculate

(i) the horizontal component of u ,

horizontal component = ms^{-1} [1]

(ii) the vertical component of u .

vertical component = ms^{-1} [1]

(b) Show that the maximum height reached by the ball is 4.3 m.

[2]

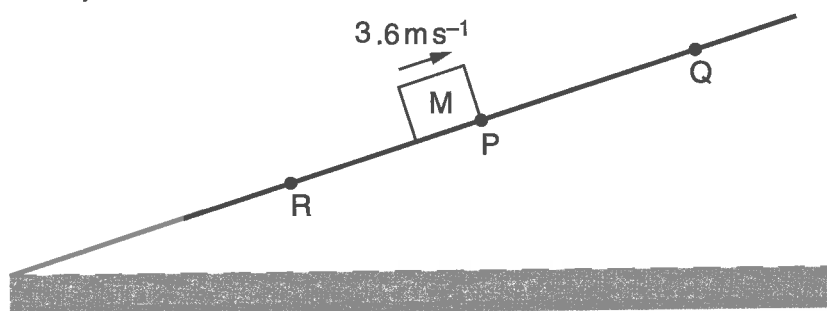
(c) Determine the magnitude of the displacement PQ.

displacement = m [4]

42. O/N 15/P22/Q2

booklet (scan + OCR)

Fig. 2.1 shows an object M on a slope.

**Fig. 2.1**

M moves up the slope, comes to rest at point Q and then moves back down the slope to point R. M has a constant acceleration of 3.0 m s^{-2} down the slope at all times. At time $t = 0$, M is at point P and has a velocity of 3.6 m s^{-1} up the slope. The total distance from P to Q and then to R is 6.0 m .

(a) Calculate, for the motion of M from P to Q,

(i) the time taken,

time = s [2]

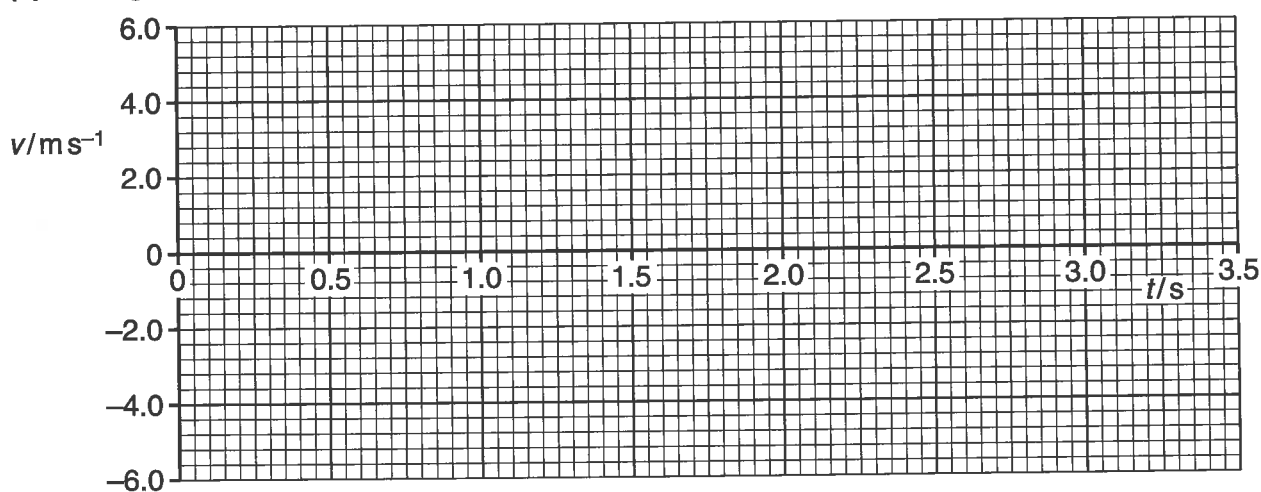
(ii) the distance travelled.

distance = m [1]

(b) Show that the speed of M at R is 4.8 m s^{-1} .

[2]

(c) On Fig. 2.2, draw the variation with time t of the velocity v of M for the motion P to Q to R.

**Fig. 2.2**

[3]

- (d) The mass of M is 450 g.
Calculate the difference in the kinetic energy of M at P and at R .

difference in kinetic energy = J [2]

43. O/N 15/P21/Q3

booklet selection; question from the official paper

Question from the official paper: printed pages 96–97 of the scanned booklet are missing

- 3 (a) An object is moved from point P to point R either by a direct path or by the path P to Q to R , as shown in Fig. 3.1.

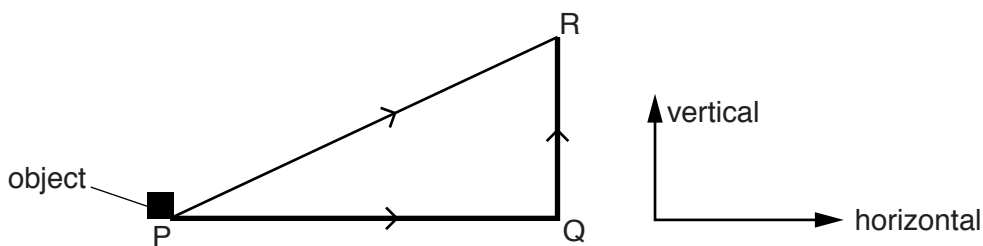


Fig. 3.1

P and Q are on the same horizontal level. R is vertically above Q .

Explain whether the work done moving the object against the gravitational field is the same or different along paths PR and PQR .

[2]

- (b) A ball is thrown with an initial velocity V at an angle θ to the horizontal, as shown in Fig. 3.2.

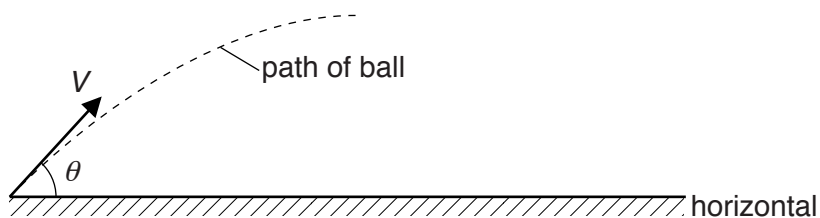


Fig. 3.2 (not to scale)

The variation with time t of the height h of the ball is shown in Fig. 3.3.

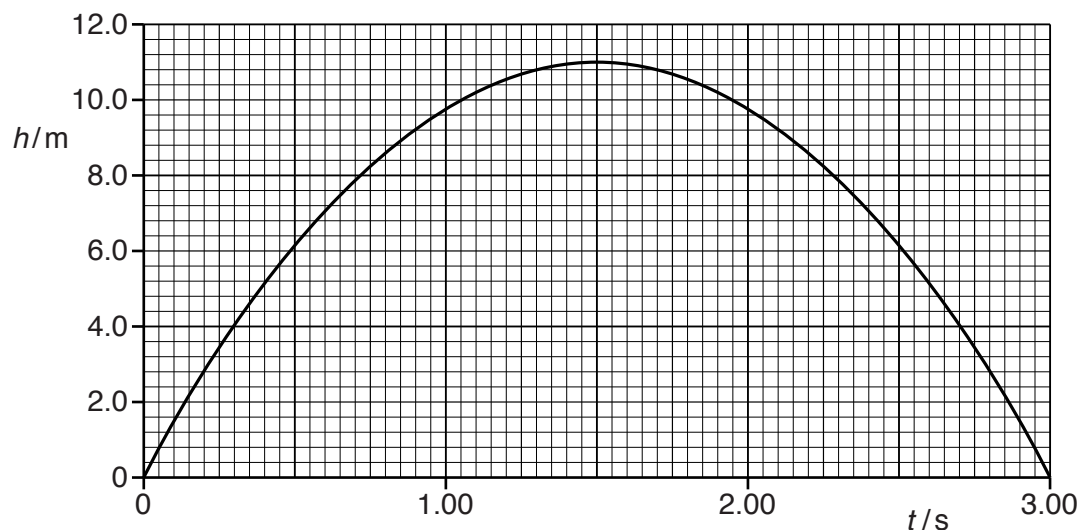


Fig. 3.3

Air resistance is negligible.

- (i) Use the time to reach maximum height to determine the vertical component V_v of the velocity of the ball for time $t = 0$.

$$V_v = \quad \text{ms}^{-1} \quad [2]$$

- (ii) The horizontal displacement of the ball at $t = 3.00$ s is 25.5 m.
On Fig. 3.4, draw the variation with t of the horizontal displacement x of the ball.

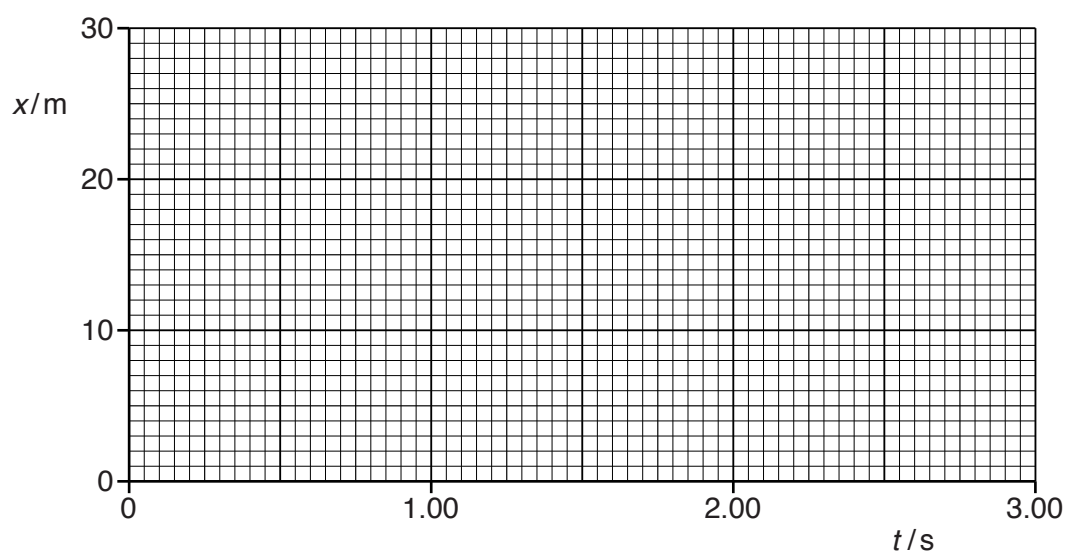


Fig. 3.4

[1]

- (iii) For the ball at maximum height, calculate the ratio

$$\frac{\text{potential energy of the ball}}{\text{kinetic energy of the ball}}.$$

$$\text{ratio} = \quad [3]$$

- (iv) In practice, air resistance is not negligible. State and explain the effect of air resistance on the time taken for the ball to reach maximum height.

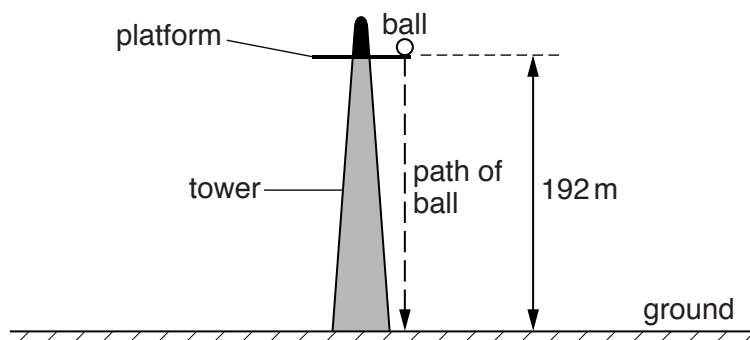
[2]

44. O/N 15/P23/Q3

booklet selection; question and answer from the official paper

The scanned booklet lost this question (missing pages); only its answer survives

- 3** A steel ball falls from a platform on a tower to the ground below, as shown in Fig. 3.1.

**Fig. 3.1**

The ball falls from rest through a vertical distance of 192 m. The mass of the ball is 270 g.

- (a)** Assume air resistance is negligible.

- (i)** Calculate

1. the time taken for the ball to fall to the ground,

time taken = _____ s [2]

2. the maximum kinetic energy of the ball.

maximum kinetic energy = _____ J [2]

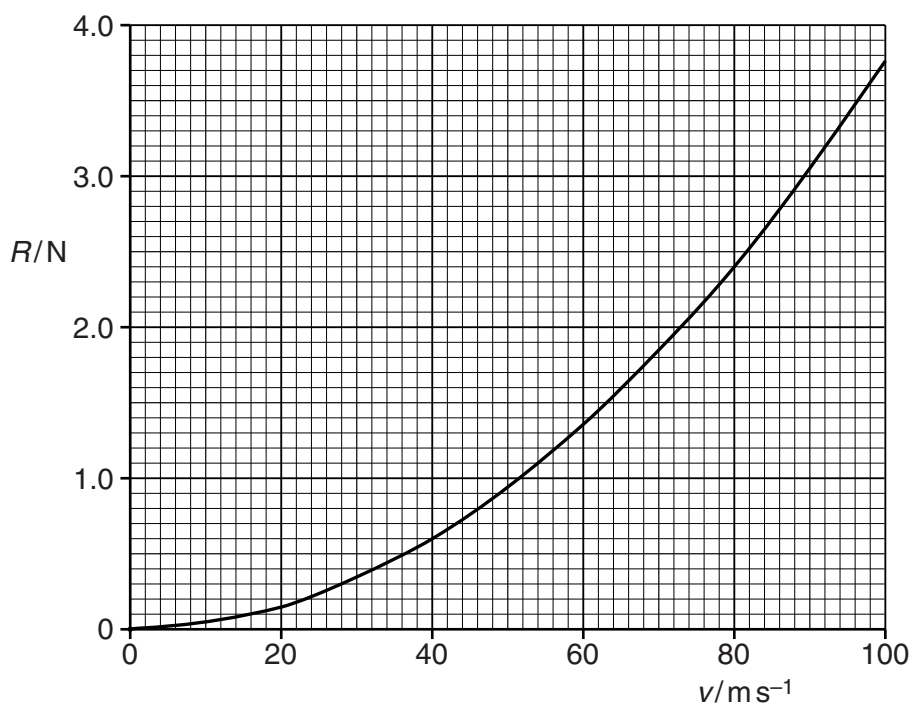
- (ii)** State and explain the variation of the velocity of the ball with time as the ball falls to the ground.

[1]

- (iii)** Show that the velocity of the ball on reaching the ground is approximately 60 ms^{-1} .

[1]

- (b)** In practice, air resistance is not negligible. The variation of the air resistance R with the velocity v of the ball is shown in Fig. 3.2.

**Fig. 3.2**

- (i) Use Fig. 3.2 to state and explain qualitatively the variation of the acceleration of the ball with the distance fallen by the ball.

[3]

- (ii) The speed of the ball reaches 40 ms^{-1} . Calculate its acceleration at this speed.

acceleration = ms^{-2} [2]

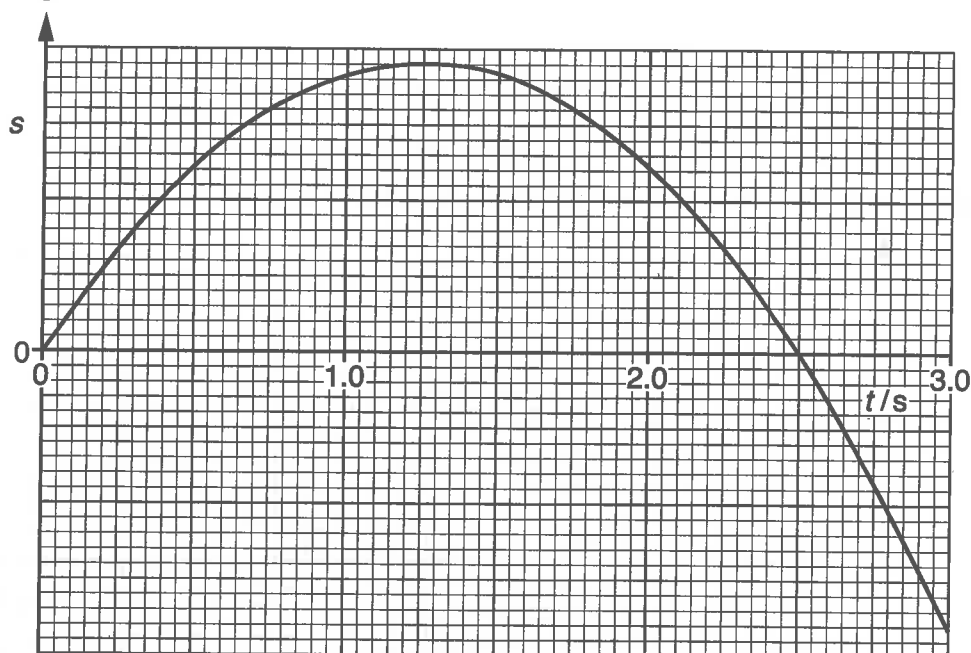
- (iii) Use information from (a)(iii) and Fig. 3.2 to state and explain whether the ball reaches terminal velocity.

[2]

45. M/J 15/P22/Q2

booklet (scan + OCR)

A stone is thrown vertically upwards. The variation with time t of the displacement s of the stone is shown in Fig. 2.1.

**Fig. 2.1**

- (a) Use Fig. 2.1 to describe, without calculation, the speed of the stone from $t = 0$ to $t = 3.0$ s.

.....

..... [2]

- (b) Assume air resistance is negligible and therefore the stone has constant acceleration. Calculate, for the stone,

(i) the speed at 3.0 s,

speed = ms^{-1} [3]

(ii) the distance travelled from $t = 0$ to $t = 3.0$ s,

distance = m [3]

(iii) the displacement from $t = 0$ to $t = 3.0$ s.

displacement = m

direction [2]

(c) On Fig. 2.2, draw the variation with time t of the velocity v of the stone from $t = 0$ to $t = 3.0$ s.

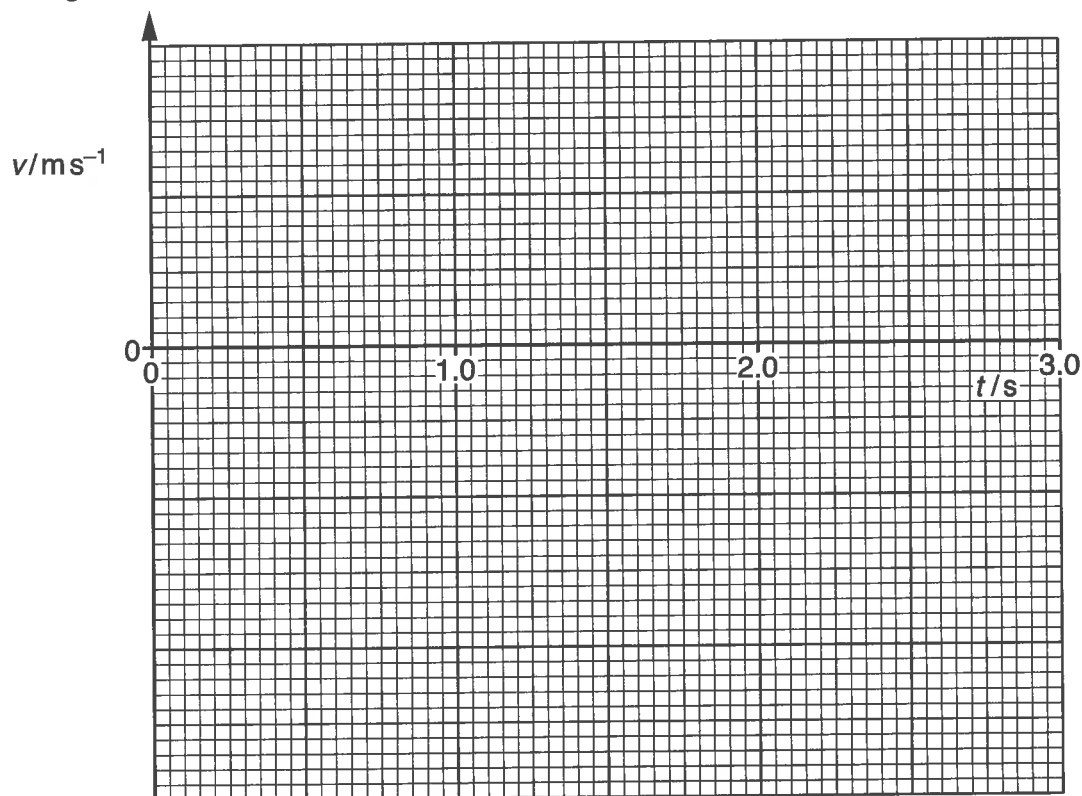


Fig. 2.2

[3]

46. M/J 15/P21/Q2

booklet (scan + OCR)

(a) Define *speed* and *velocity* and use these definitions to explain why one of these quantities is a scalar and the other is a vector.

speed:

velocity: [2]

(b) A ball is released from rest and falls vertically. The ball hits the ground and rebounds vertically, as shown in Fig. 2.1.

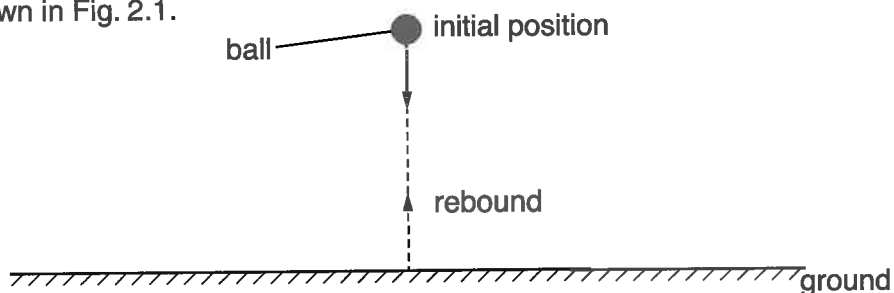


Fig. 2.1

The variation with time t of the velocity v of the ball is shown in Fig. 2.2.

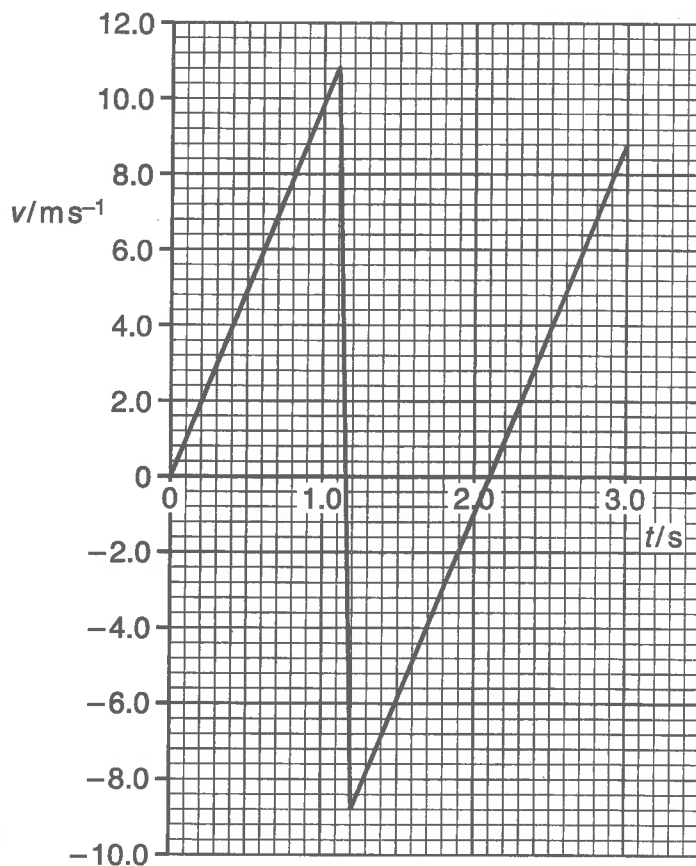


Fig. 2.2

Air resistance is negligible.

- (i) Without calculation, use Fig. 2.2 to describe the variation with time t of the velocity of the ball from $t = 0$ to $t = 2.1$ s.

.....

.....

..... [3]

- (ii) Calculate the acceleration of the ball after it rebounds from the ground. Show your working.

acceleration = ms⁻² [3]

(iii) Calculate, for the ball, from $t = 0$ to $t = 2.1$ s,

1. the distance moved,

distance = m [3]

2. the displacement from the initial position.

displacement = m [2]

(iv) On Fig. 2.3, sketch the variation with t of the speed of the ball.

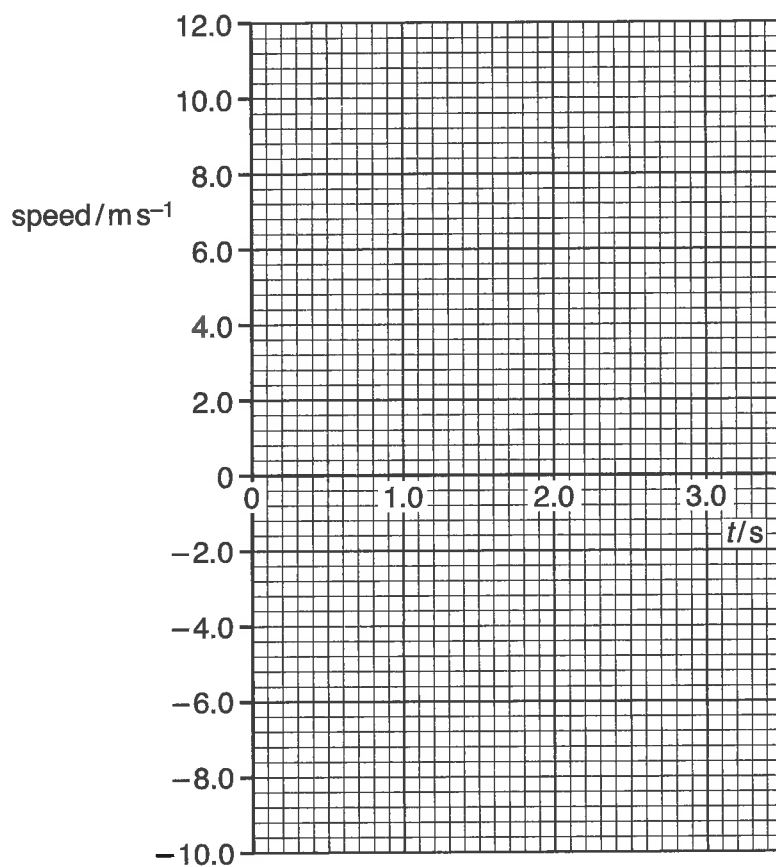


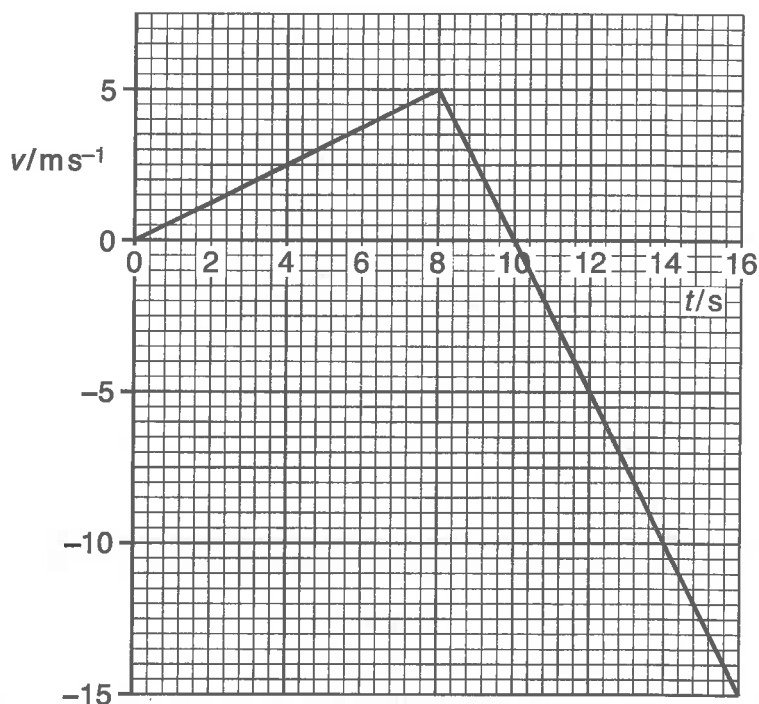
Fig. 2.3

[2]

47. M/J 15/P23/Q2

booklet (scan + OCR)

The variation with time t of the velocity v of a ball is shown in Fig. 2.1.

**Fig. 2.1**

The ball moves in a straight line from a point P at $t = 0$. The mass of the ball is 400g.

(a) Use Fig. 2.1 to describe, without calculation, the velocity of the ball from $t = 0$ to $t = 16$ s.

[2]

(b) Use Fig. 2.1 to calculate, for the ball,

(i) the displacement from P at $t = 10$ s,

displacement = m [2]

(ii) the acceleration at $t = 10$ s,

acceleration = m s^{-2} [2]

(iii) the maximum kinetic energy.

kinetic energy = J [2]

(c) Use your answers in (b)(i) and (b)(ii) to determine the time from $t = 0$ for the ball to return to P.

time = s [2]

48. O/N 14/P22/Q2

booklet (scan + OCR)

A ball is thrown from A to B as shown in Fig. 2.1.

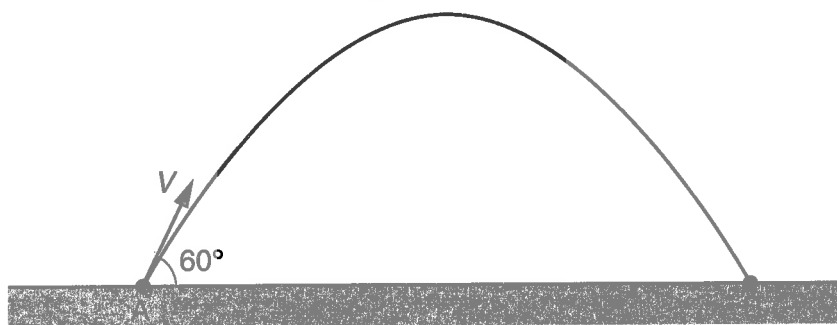


Fig. 2.1

The ball is thrown with an initial velocity V at 60° to the horizontal.

The variation with time t of the vertical component V_v of the velocity of the ball from $t = 0$ to $t = 0.60$ s is shown in Fig. 2.2.

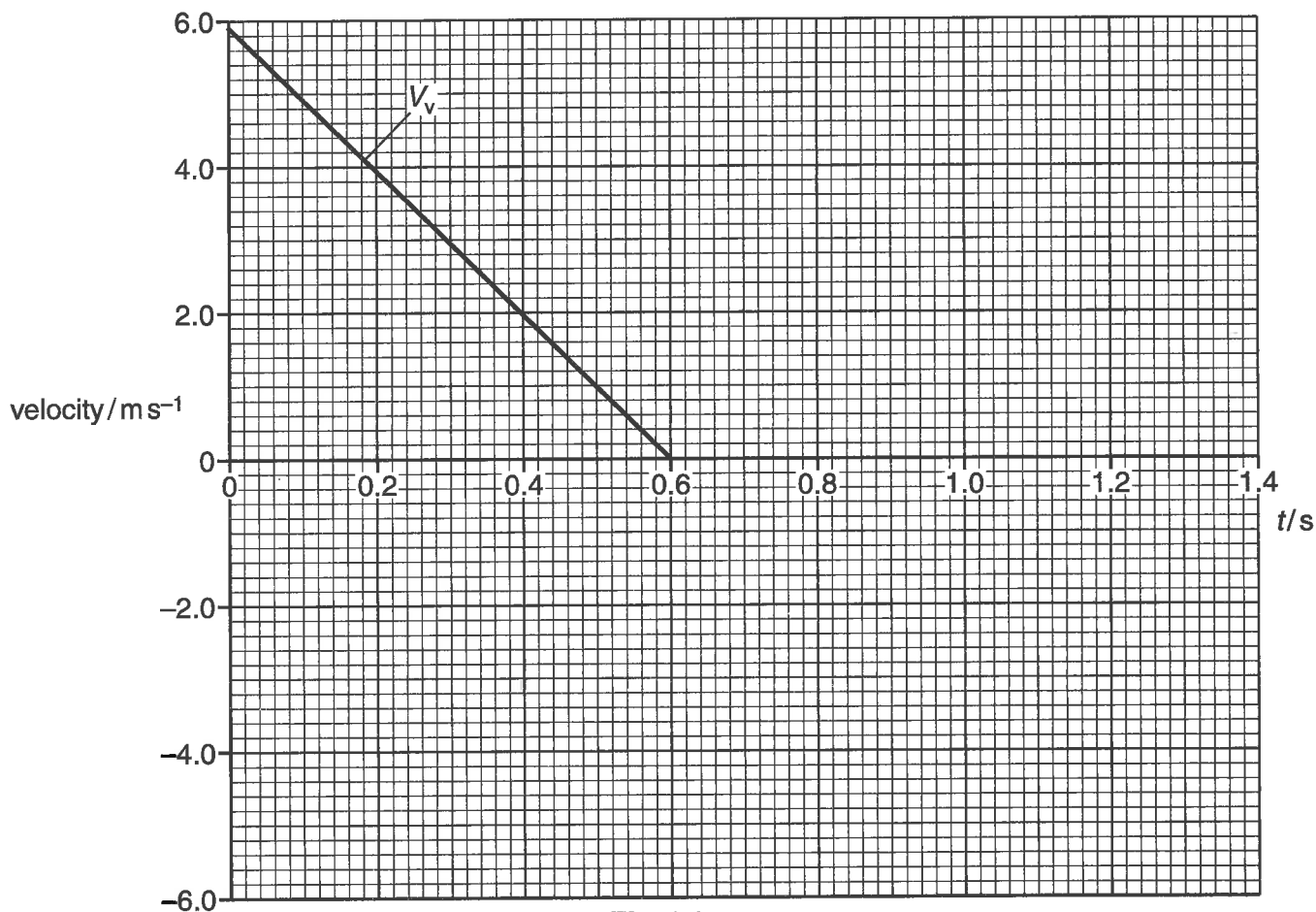


Fig. 2.2

Assume air resistance is negligible.

- (a) (i) Complete Fig. 2.2 for the time until the ball reaches B. [2]
 (ii) Calculate the maximum height reached by the ball.

height =m [2]

- (iii) Calculate the horizontal component V_h of the velocity of the ball at time $t = 0$.

$V_h = \dots\dots\dots \text{ms}^{-1}$ [2]

- (iv) On Fig. 2.2, sketch the variation with t of V_h . **Label** this sketch V_h . [1]

- (b) The ball has mass 0.65 kg.
 Calculate, for the ball,
 (i) the maximum kinetic energy,

maximum kinetic energy =J [3]

- (ii) the maximum potential energy above the ground.

maximum potential energy =J [2]

49. M/J 14/P21/Q1

booklet (scan + OCR)

- (a) (i) Define *velocity*.

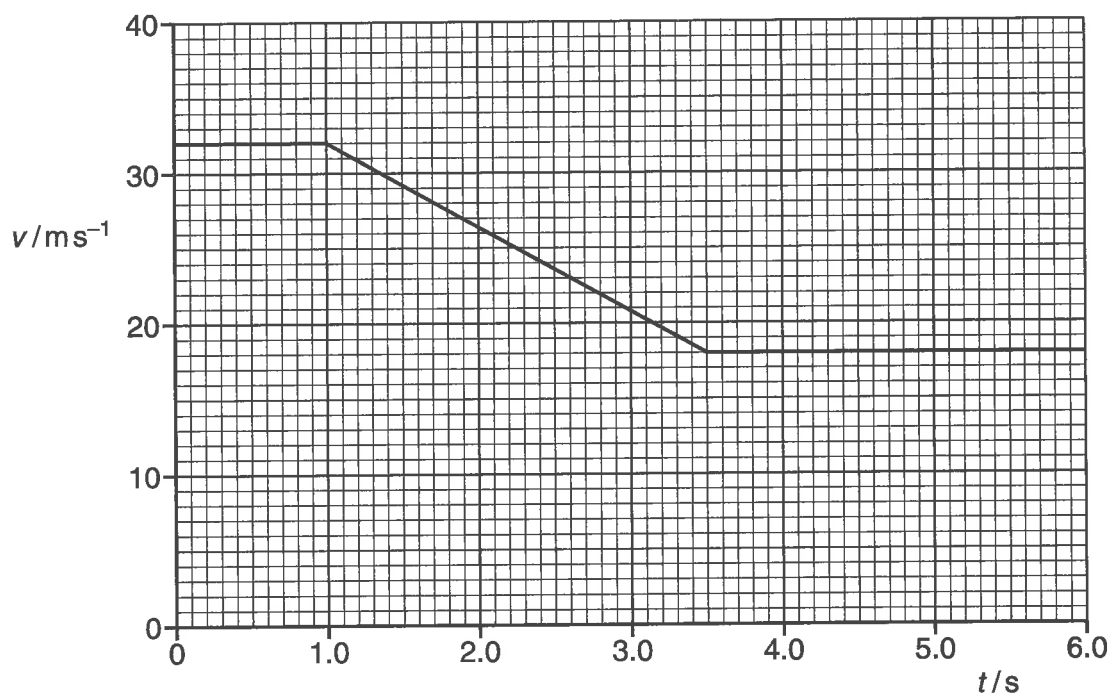
.....

- (ii) Distinguish between *speed* and *velocity*.

.....

..... [2]

- (b) A car of mass 1500 kg moves along a straight, horizontal road. The variation with time t of the velocity v for the car is shown in Fig. 1.1.

**Fig. 1.1**

The brakes of the car are applied from $t = 1.0\text{ s}$ to $t = 3.5\text{ s}$.

For the time when the brakes are applied,

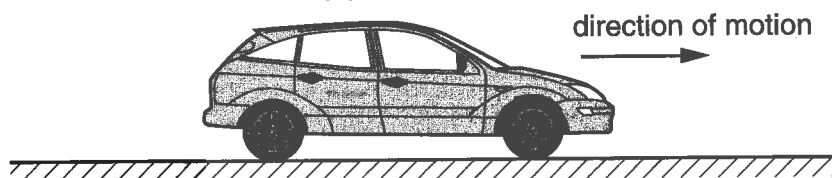
(i) calculate the distance moved by the car,

distance = m [3]

(ii) calculate the magnitude of the resultant force on the car.

resultant force = N [3]

(c) The direction of motion of the car in (b) at time $t = 2.0\text{ s}$ is shown in Fig. 1.2.

**Fig. 1.2**

On Fig. 1.2, show with arrows the directions of the acceleration (label this arrow A) and the resultant force (label this arrow F). [1]

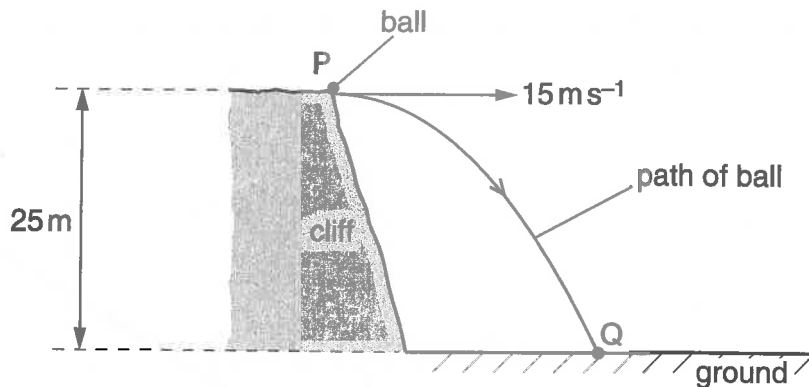
50. M/J 14/P23/Q2

booklet (scan + OCR)

(a) Explain what is meant by a *scalar* quantity and by a *vector* quantity.

scalar:

vector: [2]

(b) A ball leaves point P at the top of a cliff with a horizontal velocity of 15 ms^{-1} , as shown in Fig. 2.1.**Fig. 2.1**

The height of the cliff is 25 m. The ball hits the ground at point Q.

Air resistance is negligible.

(i) Calculate the vertical velocity of the ball just before it makes impact with the ground at Q.vertical velocity = ms^{-1} [2]**(ii)** Show that the time taken for the ball to fall to the ground is 2.3 s.

[1]

(iii) Calculate the magnitude of the displacement of the ball at point Q from point P.

displacement = m [4]

(iv) Explain why the distance travelled by the ball is different from the magnitude of the displacement of the ball.

.....

..... [2]

51. O/N 13/P23/Q3

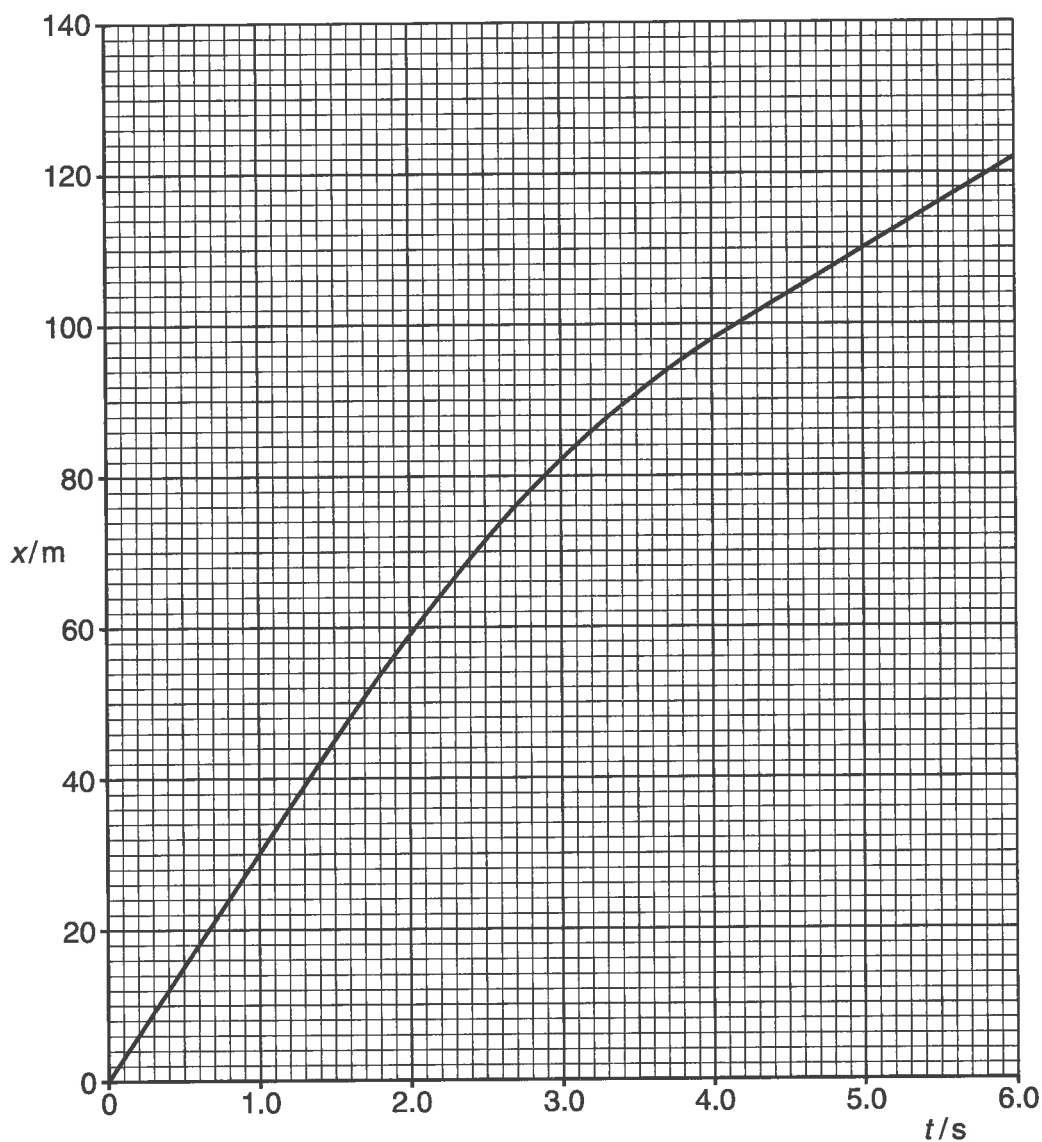
booklet (scan + OCR)

(a) Define**(i) velocity,**

..... [1]

(ii) acceleration.

..... [1]

(b) A car of mass 1500 kg travels along a straight horizontal road.The variation with time t of the displacement x of the car is shown in Fig. 3.1.**Fig. 3.1**

- (i) Use Fig. 3.1 to describe qualitatively the velocity of the car during the first six seconds of the motion shown. Give reasons for your answers.**

.....

.....

.....

[3]

(ii) Calculate the average velocity during the time interval $t = 0$ to $t = 1.5$ s.

average velocity = ms^{-1} [1]

(iii) Show that the average acceleration between $t = 1.5$ s and $t = 4.0$ s is -7.2 ms^{-2} .

(iv) Calculate the average force acting on the car between $t = 1.5$ s and $t = 4.0$ s.

[2]

force = N [2]

52. M/J 13/P23/Q2

booklet (scan + OCR)

(a) A student walks from A to B along the path shown in Fig. 2.1.

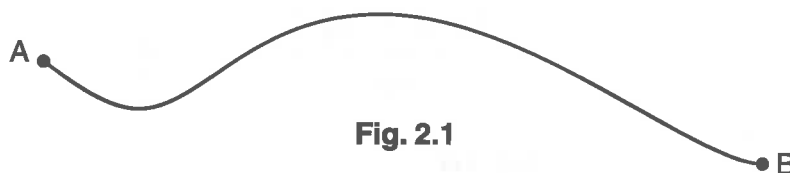


Fig. 2.1

The student takes time t to walk from A to B.

(i) State the quantity, apart from t , that must be measured in order to determine the average value of

1. speed,

..... [1]

2. velocity.

..... [1]

(ii) Define *acceleration*.

..... [1]

(b) A girl falls vertically onto a trampoline, as shown in Fig. 2.2.

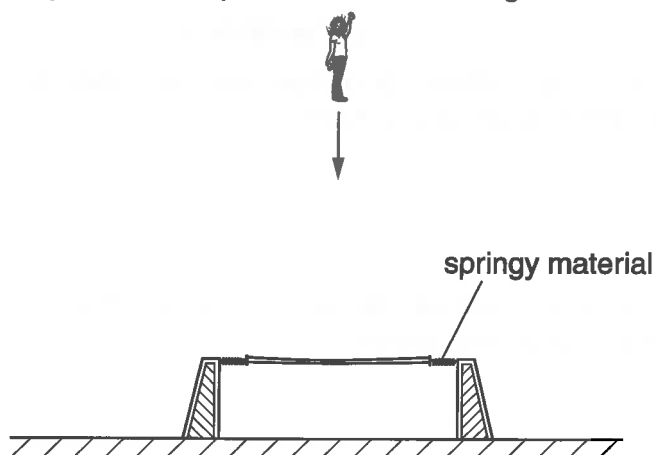


Fig. 2.2

The trampoline consists of a central section supported by springy material. At time $t = 0$ the girl starts to fall. The girl hits the trampoline and rebounds vertically. The variation with time t of velocity v of the girl is illustrated in Fig. 2.3.

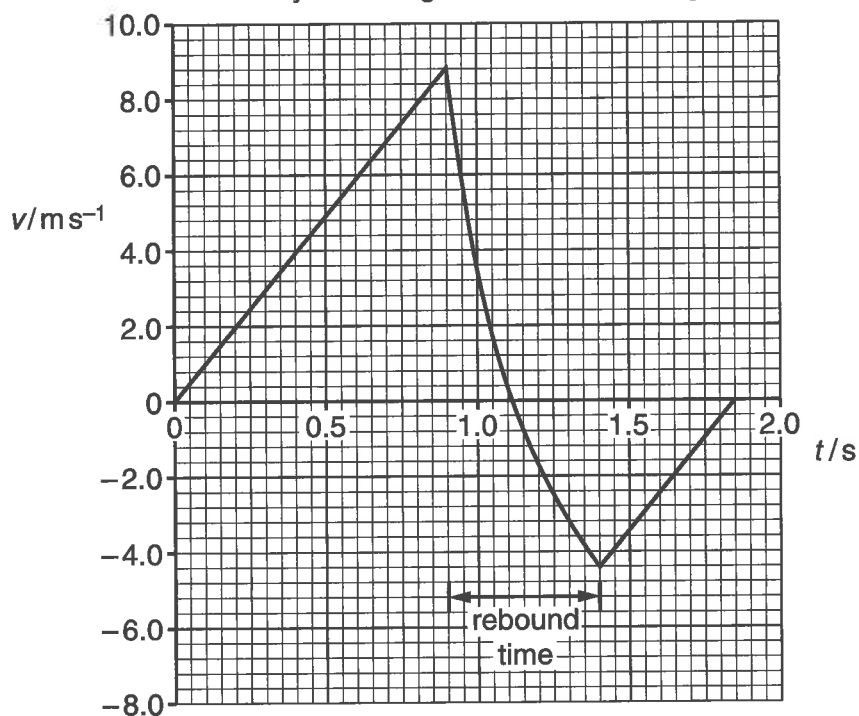


Fig. 2.3

For the motion of the girl, calculate

- (i) the distance fallen between time $t = 0$ and when she hits the trampoline,

distance = m [2]

- (ii) the average acceleration during the rebound.

acceleration = ms^{-2} [2]

- (c) (i) Use Fig. 2.3 to compare, without calculation, the accelerations of the girl before and after the rebound. Explain your answer.

.....
 [2]

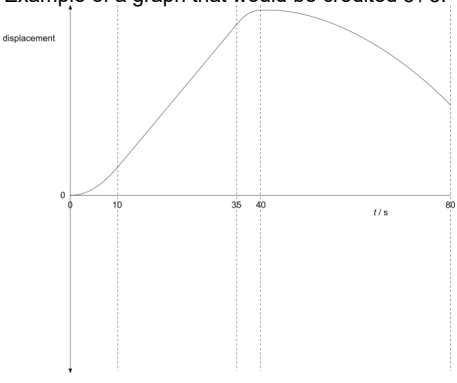
- (ii) Use Fig. 2.3 to compare, without calculation, the potential energy of the girl at $t = 0$ and $t = 1.85$ s. Explain your answer.

.....
 [2]

Answers Section

1. M/J 26/P22/Q1/b

1(b)(i)	area (under graph line)	B1	
1(b)(ii)	distance = 390 m	A1	
1(b)(iii)	$a = (v - u) / t$ or $a = \Delta v / (\Delta)t$ or gradient (of graph) e.g. = $(0 - 12) / 5$	C1	Allow $a = (u - v) / t$ or $a = (\Delta)v / (\Delta)t$ or $a = -\text{gradient}$ Allow $a = (y_2 - y_1) / (x_2 - x_1)$ Allow a numerical expression equivalent to one of the three non-zero gradients on the graph, e.g. $a = 12 / 5$ or $a = 12 / 10$ or $a = 10 / 40$ No ECF of incorrect velocity from 1(b)(ii)
	magnitude of acceleration = 2.4 m s^{-2}	A1	Allow -2.4 m s^{-2}
1(b)(iv)	$v^2 = 2 \times -0.25 \times -150$ $= 75$	C1	No mark for symbol formulae. For each method, the C1 mark is given for the full substitution of data into the formula. Allow $v^2 = 2 \times 0.25 \times 150$
	speed = 8.7 m s^{-1}	A1	Allow speed = -8.7 m s^{-1} for all methods
	<u>Alternative methods:</u> $(-150 = \frac{1}{2} \times -0.25 \times t^2 \text{ so } t = 34.6)$ $v = -0.25 \times 34.6$ $= -8.7$	(C1)	Allow $v = 0.25 \times 34.6$ Allow use of $t = 35$ to imply the C1 mark, but do not credit the A1 mark if leads to incorrect speed (8.8 m s^{-1})
	speed = 8.7 m s^{-1}	(A1)	
1(b)(iv)	OR $-150 = \frac{1}{2} \times v \times 34.6$ $v = -8.7$	(C1)	Allow $150 = \frac{1}{2} \times v \times 34.6$ Allow use of $t = 35$ to imply the C1 mark, but do not credit the A1 mark if leads to incorrect speed (8.6 m s^{-1})
	speed = 8.7 m s^{-1}	(A1)	
	OR $-150 = (v \times 34.6) - (\frac{1}{2} \times -0.25 \times 34.6^2)$ $v = -8.7$	(C1)	Allow $150 = (v \times 34.6) - (\frac{1}{2} \times 0.25 \times 34.6^2)$
	speed = 8.7 m s^{-1}	(A1)	
	OR $t = 34.6 + 40$ $= 74.6$ At $t = 74.6$, $v = -8.7$ (from graph)	(C1)	No tolerance (for consistency with other methods)
	speed = 8.7 m s^{-1}	(A1)	

1(b)(v)	A line from 0,0 to $t = 80$ s with increasing displacement then decreasing displacement that does not cross the t -axis	B1	For all marking points ignore line after $t = 80$ s Allow line to have discontinuous gradient at $t = 10, 35, 40$ Do not allow step change in displacement Allow graph to end up to 5 mm before $t = 80$ s Allow line to end at the t -axis Do not allow displacement that increases, decreases and then increases again
	A line with a maximum displacement at $t = 40$ s	B1	Allow centre of peak ± 3 mm horizontally from $t = 40$ s
	A line with - increasing gradient from $t = 0$ to $t = 10$ s - constant positive gradient between $t = 10$ s and $t = 35$ s - decreasing gradient between $t = 35$ s and $t = 40$ s - decreasing gradient between $t = 40$ s and $t = 80$ s	B1	Allow line to have discontinuous gradient at $t = 10, 35, 40$ Do not allow step change in displacement If unclear whether a section of the line is straight or curved give BOD Allow graph to end up to 5 mm before $t = 80$ s Example of a graph that would be credited 3 / 3: 

2. M/J 26/P23/Q1/b(i,iii),c(ii)

1(b)(i)	$a = \Delta v / (\Delta t)$ or $= (v - u) / (\Delta t)$ or = gradient of slope	C1	
	acceleration = $(-0.8 - 1.6) / (0.12 - 0.09)$ $= (-80 \text{ m s}^{-2})$	A1	No tolerance.
1(b)(iii)	$\frac{1}{2} \times 1.60 \times 0.020$ or $1.6^2 / (2 \times 80)$ or $1.6 \times 0.02 - \frac{1}{2} \times 80 \times 0.02^2$ or $\frac{1}{2} \times 80 \times 0.02^2$ $= 0.016$	C1	ECF from (b)(i) for v read-offs ECF a from (b)(i) .
	OR $\frac{1}{2} \times 0.80 \times 0.010$ or $0.8^2 / (2 \times 8)$ or $-0.8 \times 0.01 + \frac{1}{2} \times 80 \times 0.01^2$ or $\frac{1}{2} \times 80 \times 0.01^2$ $= 0.004$	(C1)	
	distance = $0.016 + 0.004$ $= 0.020 \text{ m}$	A1	Allow 0.02 (1SF) treat finding displacement as XP e.g. $(-0.8)^2 - 1.6^2 / 2 \times 80 = 0.12 \text{ m}$
1(c)(ii)	a straight horizontal line from (0, 0) to (0.09, 0) or a straight horizontal line from (0.12, 0.8) to (0.20, 0.8)	B1	tolerance $\pm \frac{1}{2}$ small square
	Both horizontal lines correct <u>and</u> straight diagonal line from (0.09, 0) to (0.12, 0.8)	B1	Allow horizontal line from (0, 0) to (0.09, 0) to be missing if diagonal line starts at (0.09, 0)

3. M/J 26/P24/Q1/b(ii),c(ii)

1(b)(ii)	$0 = 15^2 + (2 \times a \times 35)$	C1	Allow any subject.
	$a = -3.2 \text{ m s}^{-2}$	A1	
	OR $(35 = \frac{1}{2} \times 15 \times t \text{ so } t = 4.7 \text{ so})$ $0 = 15 + (a \times 4.7) \text{ or}$ $35 = -\frac{1}{2} \times a \times 4.7^2 \text{ or}$ $35 = 15 \times 4.7 + (\frac{1}{2} \times a \times 4.7^2)$	(C1)	
	$a = -3.2 \text{ m s}^{-2}$	(A1)	
1(b)(iii)	$F = ma$ $= 1400 \times 3.2$	C1	ecf a from 1(b)(ii)
	$= 4500 \text{ N}$	A1	
	OR $F = W/s$ $= 1.6 \times 10^5 / 35$	(C1)	Allow $E/E_K/KE$ for W . Allow d or x for s
	$= 4600 \text{ N}$	(A1)	Allow $\pm 4600 \text{ N}$ ecf E_K from 1(b)(i) Allow 4500 or 4600 for either method.
1(c)(ii)	The (initial) kinetic energy is four times greater	B1	Allow kinetic energy is proportional to speed squared
	(So for the same applied force) the distance travelled will be four times greater	B1	Allow distance = 140 m
	OR acceleration is the same and using $s = (-)u^2 / 2a$	(B1)	Not $v^2 = u^2 + 2as$ unless $v = 0$ also given Allow acceleration the same and distance is proportional to initial speed squared
	The distance will be four times greater	(B1)	Allow distance = 140 m

4. MAR 26/P22/Q4/b

4(b)(i)	$s = ut + \frac{1}{2} at^2$ $1.6 = \frac{1}{2} \times 9.81 \times t^2$	C1
	$t = ((2 \times 1.6) / 9.81)^{0.5}$ $= 0.57 \text{ s}$	A1
	OR $v = (2 \times 9.81 \times 1.6)^{0.5} = 5.6$ $t = (v - u) / a$ $= (5.6 - 0) / 9.81$	(C1)
	$= 0.57 \text{ s}$	(A1)
	OR $t = 2s / v$ $= 2 \times 1.6 / 5.6$	(C1)
	$= 0.57 \text{ s}$	(A1)
4(b)(ii)	a curved line from the origin	B1
	a line from the origin with a continuously decreasing positive gradient	B1

5. O/N 25/P21/Q1/a,b

1(a)	rate of change of velocity	B1
1(b)(i)	acceleration = $320 / 20$ = 16 m s^{-2}	A1
1(b)(ii)	$s = \frac{1}{2} ((u) + v)t$ or distance = area under graph	C1
	(height) = $\frac{1}{2} \times 20 \times 320 = 3200 \text{ m}$ or 3.2 km	A1

6. O/N 25/P24/Q1/a(ii,iii,iv)

1(a)(ii)	time = $16 / 9.81 = 1.6 \text{ s}$	A1
1(a)(iii)	horizontal straight line at $v = 23 \text{ m s}^{-1}$ from $t = 0$ to $t = 3.2 \text{ s}$	B1
1(a)(iv)	straight diagonal line starting at a positive velocity from $t = 0$ to $t = 3.2 \text{ s}$, crossing the time axis	B1
	line starting at $v = 16 \text{ m s}^{-1}$ and ending at $v = -16 \text{ m s}^{-1}$	B1
	line passing through $v = 0$ at $t = 1.6 \text{ s}$	B1

7. M/J 25/P21/Q1/a,b,c(i)

1(a)	rate of change of velocity	B1
1(b)(i)	$s = ut + \frac{1}{2}at^2$ and $u = 0$ or $s = \frac{1}{2}at^2$ $t = \sqrt{(2 \times 14 / 9.81)}$	C1
	$t = 1.7 \text{ s}$	A1
	OR $v = \sqrt{(0^2 + 2 \times 9.81 \times 14)}$ $v = 17 \text{ (m s}^{-1}\text{)}$ $17 = (0 + 9.81) \times t$ or $17 = 9.81 \times t$ or $14 = \frac{1}{2} \times (0 + 17) \times t$ or $14 = \frac{1}{2} \times 17 \times t$	(C1)
	$t = 1.7 \text{ s}$	(A1)
1(b)(ii)	$s = ut + \frac{1}{2}at^2$ $u = (3.6 - \frac{1}{2} \times 9.81 \times 1.7^2) / 1.7$	C1
	$u = (-) 6.2 \text{ m s}^{-1}$	A1
	OR $v = (3.6 + \frac{1}{2} \times 9.81 \times 1.7^2) / 1.7$ $v = 10 \text{ (m s}^{-1}\text{)}$ $10 = u + 9.81 \times 1.7$ or $10^2 = u^2 + (2 \times 9.81 \times 3.6)$ or $3.6 = \frac{1}{2} \times (u + 10) \times 1.7$	(C1)
	$u = (-) 6.2 \text{ m s}^{-1}$	(A1)
1(c)(i)	Less time (as it reaches a lower height)	B1
	(because) the initial <u>vertical</u> (component of the) velocity is smaller (than in part (b))	B1

8. M/J 25/P23/Q1/a,b,c(i)

1(a)	Rate of change of displacement	B1
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1(b)	For object A: $s = ut + \frac{1}{2}at^2$ so $t = \sqrt{\frac{2 \times 10}{9.81}}$ $= 1.4$	C1
	Then for object B: $s = ut + \frac{1}{2}at^2$ $h = -(3 \times 1.4) + (0.5 \times 9.81 \times 1.4^2)$ $= 5.7 \text{ m}$	C1
	OR $h = -(3 \times 1.4) + 10$ $= 5.7 \text{ m}$	(C1)
		(A1)
1(c)(i)	time taken (to reach the ground is) same	B1
	The <u>initial vertical</u> (component of the) velocity is the same (as in part (1b))	B1

9. M/J 25/P24/Q2/f

2(f)	line from origin with decreasing positive gradient	B1
	gradient changes from positive to negative at 8.0 s	B1
	after $t = 8.0 \text{ s}$ the line has a negative gradient of increasing magnitude <u>and</u> a positive value of d at $t = 12 \text{ s}$	B1

10. MAR 25/P22/Q3/a(i)

3(a)(i)	$a = (v^2 - u^2) / 2s$ $= (22^2 - 13^2) / (2 \times 180)$ $= 0.88 \text{ m s}^{-2}$	C1
		A1
	OR [$t = (180 \times 2) / (22 + 13) = 10.3$] $180 = 13 \times 10.3 + \frac{1}{2} a \times 10.3^2$ or $180 = 22 \times 10.3 - \frac{1}{2} a \times 10.3^2$ or $22 = 13 + a \times 10.3$	(C1)
	$a = 0.88 \text{ m s}^{-2}$	(A1)

11. O/N 24/P21/Q2/b(iv)

2(b)(iv)	distance = $\frac{1}{2} \times 40 \times (8 + 4)$ or distance = $\frac{1}{2} \times 72\,000 \times (8 + 4) / 1800$ or distance = $[40^2 / (2 \times 5)] + [40^2 / (2 \times 10)]$ or distance = $\frac{1}{2} \times 5 \times 8^2 + (40 \times 4 - \frac{1}{2} \times 10 \times 4^2)$	C1
	distance = 240 m	A1

12. O/N 24/P22/Q2/c

2(c)	$x = ut + \frac{1}{2}at^2$ $x = 0.60 \times 3.7 + \frac{1}{2} \times 0.85 \times 3.7^2$	C1
	or	
	$v = 0.60 + 0.85 \times 3.7 (= 3.75 \text{ m s}^{-1})$ $x = 3.75 \times 3.7 - 0.5 \times 0.85 \times 3.7^2$ or $x = \frac{1}{2} \times (0.60 + 3.75) \times 3.7$ or $x = (3.75^2 - 0.60^2) / (2 \times 0.85)$	(C1)
	$x = 8.0 \text{ m}$	A1

13. O/N 24/P23/Q1/a,b

1(a)	rate of change of velocity	B1
1(b)(i)	curved path from aircraft to ground, starting horizontal at aircraft and then with increasing negative gradient as it moves towards the ground	B1
1(b)(ii)	$s = ut + \frac{1}{2}at^2$ $63 = \frac{1}{2} \times 9.81 \times t^2$	C1
	time = 3.6 s	A1
1(b)(iii)	$v^2 = 2 \times 9.81 \times 63$ or $v = 0 + (9.81 \times 3.6)$ or $63 = (v \times 3.6) - (\frac{1}{2} \times 9.81 \times 3.6^2)$ or $63 = \frac{1}{2} \times (0 + v) \times 3.6$ $v = 35 \text{ m s}^{-1}$	A1
1(b)(iv)	speed ² = 35 ² + 42 ²	C1
	speed = 55 m s ⁻¹	A1

14. M/J 24/P21/Q2/a,b(i,iii)

2(a)	change in displacement / time (taken)	B1
2(b)(i)	horizontal velocity = $22 \times \cos 40^\circ$	C1
	time taken = $36 / (22 \times \cos 40^\circ)$ = 2.1 s	A1
2(b)(iii)	$s = ut + \frac{1}{2}at^2$ = $(14 \times 2.1) + (\frac{1}{2} \times -9.81 \times 2.1^2)$	C1
	= 7.8 (m)	C1
	(therefore) height of wall = 7.8 + 1.2 = 9.0 m	A1

2(b)(iii)	or	
	<i>other methods possible e.g.</i> time to maximum height = $(0 - 14) / -9.81$ (= 1.43 s) time from maximum height to wall = $2.1 - 1.43$ (= 0.67 s) maximum height above release = $(14 \times 1.43) + (\frac{1}{2} \times -9.81 \times 1.43^2)$ = 9.99 (m)	(C1)
	height from maximum to wall = $0.5 \times 9.81 \times 0.67^2$ (= 2.2 m) height above release = $9.99 - 2.2$ = 7.8 (m)	(C1)
	height of wall = $1.2 + 7.8$ = 9.0 m	(A1)

15. M/J 24/P22/Q2/a(ii)

2(a)(ii)	tangent line to curve drawn on Fig. 2.1	C1
	$a = \text{gradient of tangent line} = \Delta v / \Delta t$	A1
	e.g. = $(44 - 26) / (18 - 0)$	
	$0.9 \leq a \leq 1.1 \text{ ms}^{-2}$	

16. M/J 24/P23/Q2/a,b(i,ii,iii,iv)

2(a)	distance (from the point) in a straight line in a given direction	B1
2(b)(i)	distance = speed $\times$ time = 6.0×0.71 = 4.3 m	A1
2(b)(ii)	$s = ut + \frac{1}{2} at^2$ = $\frac{1}{2} \times 9.81 \times 0.71^2$	C1
	= 2.5 m	A1
2(b)(iii)	$\tan \theta = 2.5 / 4.3$ or hypotenuse = $\sqrt{(4.3^2 + 2.5^2)}$ (= 4.97 m) $\cos \theta = 4.3 / 4.97$ or $\sin \theta = 2.5 / 4.97$	C1
	$\theta = 30^\circ$	A1
2(b)(iv)	displacement = $\sqrt{(4.3^2 + 2.5^2)}$ = 4.9 m or 5.0 m	C1 A1
	or	
	displacement = $2.5 / \sin 30^\circ$ or displacement = $4.3 / \cos 30^\circ$	(C1)
	= 5.0 m	(A1)

17. MAR 24/P22/Q2/a

2(a)	rate of change of velocity	B1
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18. O/N 23/P21/Q3/a,b(i),c

3(a)	displacement	A1
3(b)(i)	$a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ or $a = (v - u) / t$	C1
	e.g. $a = (0.30 - 0.12) / (0.35 - 0.15)$	A1
	$a = 0.90 \text{ m s}^{-2}$	
3(c)	sketch: horizontal line from (0, 0.48) to (0.15, 0.48)	B1
	horizontal line from (0.35, -0.06) to (0.5, -0.06)	B1
	straight line between (0.15, 0.48) and (0.35, -0.06)	B1

19. O/N 23/P22/Q1/b,c

1(b)	initial speed / velocity is zero	B1
	(non-zero magnitude of) acceleration is constant / uniform (and in a straight line)	B1
1(c)(i)	• magnitude of acceleration at $t = 8.0 \text{ s}$ is less than that at $t = 14.0 \text{ s}$	B1
	• direction of acceleration at $t = 8.0 \text{ s}$ is opposite to that at $t = 14.0 \text{ s}$	B1
1(c)(ii)	$a = \text{gradient}$ or $a = (v - u) / t$ or $a = \Delta v / (\Delta)t$	C1
	$a = \text{e.g. } (20 + 10) / 12$ or $(0 + 10) / 4$ or $(20 - 0) / (12 - 4)$	A1
	$a = 2.5 \text{ m s}^{-2}$	
1(c)(iii)	displacement = $[\frac{1}{2} \times (12 - 4) \times 20] - [\frac{1}{2} \times 4 \times 10]$ or displacement = $(-10 \times 12) + (\frac{1}{2} \times 2.5 \times 12^2)$ or displacement = $(20 \times 12) - (\frac{1}{2} \times 2.5 \times 12^2)$ or displacement = $\frac{1}{2} \times (20 - 10) \times 12$	C1
	displacement = 60 m	A1

20. O/N 23/P23/Q2/a(i,iii)

2(a)(i)	(time / $t =$) $9(.0) / 9.5 = 0.95 \text{ (s)}$	A1
2(a)(iii)	$s = ut + \frac{1}{2}at^2$ $(h =) 7.4 \times 0.95 - \frac{1}{2} \times 9.81 \times 0.95^2$	C1
	$h = 2.6 \text{ m}$	A1

21. M/J 23/P23/Q1

(a) $t = \sqrt{(2s/g)}$
 $= \sqrt{[(2 \times 36)/9.81]}$
 $= 2.7 \text{ s}$ [2]

- (b) • reaction time between hearing the splash and stopping the stop-watch
 • the sound (of the splash) takes time to reach the student **or** the stone hits the water at a different time to the sound being heard **or** the sound (of the splash) has to travel to the student
 • the student might not let go of the stone from ground level
 • the student might not let go of the stone and start the stop-watch at the same time
 • stop-watch may not be properly calibrated / has a zero error
 • (local value of) g is not (exactly) $9.81 \text{ (m s}^{-2}\text{)}$
 • stone given initial velocity / initial velocity not zero
 • stone does not fall (exactly) vertically / in a straight line

Any three points, 1 mark each [3]

- (c) precise: results are close together / have little scatter
 not accurate: the values are not close to / 50% different / (very) different from the true value [2]

22. MAR 23/P22/Q2

(a) (i) $E = \sigma / \epsilon$ **or** $E = F / A\epsilon$
 $A = 1.4 \times 10^4 / (2.2 \times 10^{11} \times 0.0012)$
 $= 5.3 \times 10^{-5} \text{ m}^2$ [2]

(ii) $(\Delta)h = 0.64 \times 0.49 (= 0.3136)$
 $(\Delta)E = mg(\Delta)h$ **or** $W(\Delta)h$
 $= 1.4 \times 10^4 \times 0.64 \times 0.49$
 $= 4.4 \times 10^3 \text{ J}$ [3]

(b) $P = Fv$ **or** W/t
 $= (1.4 \times 10^4 \times 0.64) / 0.56$ **or** $(4.4 \times 10^3 / 0.49) / 0.56$
 $= 1.6 \times 10^4 \text{ W}$ [3]

(c) $m = 1.4 \times 10^4 / 9.81$
 $(= 1427 \text{ kg})$
 (resultant) $F = (1.4 \times 10^4 / 9.81) \times 1.3$
 $(= 1855 \text{ N})$
 $T = 1.4 \times 10^4 - 1855$ **or** $(1.4 \times 10^4 / 9.81) \times (9.81 - 1.3)$
 $= 1.2 \times 10^4 \text{ N}$ [3]

- (d) upward sloping straight line from $(t_x, 0)$ to t_y
 from t_y to t_z : an upward sloping curve with decreasing magnitude of gradient (that is horizontal at t_z) [2]

23. O/N 22/P23/Q2

(a) (i) (e.g. $a =$) $[16.5 - 0] / [0.30 - 0] = 55 \text{ (m s}^{-2}\text{)}$ [1]

(ii) $(F =) T - W$ [1]

(iii) $ma = T - mg$
 $m = 16 / (55 + 9.81)$
 $m = 0.25 \text{ kg}$ [2]

(b) (i) $s = ut + \frac{1}{2}at^2$ or $v^2 = u^2 + 2as$ or $s = vt - \frac{1}{2}at^2$ or $s = \frac{1}{2}(u + v)t$
 or $s = \text{area under graph}$
 $s = \frac{1}{2} \times 55 \times 0.30^2$
 or
 $s = 16.5^2 / (2 \times 55)$
 or
 $s = 16.5 \times 0.30 - \frac{1}{2} \times 55 \times 0.30^2$
 or
 $s = \frac{1}{2} \times 16.5 \times 0.30$
 $s = 2.5 \text{ m}$ [2]

(ii) $u = 16.5 \text{ (m s}^{-1}\text{)}$
 $v^2 = u^2 + 2as$
 $v^2 = 16.5^2 + 2 \times 9.81 \times 2.5$
 $v = 18 \text{ m s}^{-1}$ [3]

24. M/J 22/P21/Q1

(a) change in displacement / time (taken) [1]

(b) (i) (displacement =) area under graph
 (at $t = 4.0 \text{ s}$) $v = (-) 2.4$
 $\text{height} = \frac{1}{2} \times 2.5 \times 4.0 - \frac{1}{2} \times 1.5 \times 2.4$
 $= 3.2 \text{ m}$ [3]

(ii) change in momentum = $7.5 (-4.0 - 2.4)$
 $= (-) 48 \text{ N s}$ [2]

(iii) $W = \Delta p / (\Delta)t$ or $\Delta mv / (\Delta)t$
 $= 48 / 4.0$
 $= 12 \text{ N}$
 or
 $W = ma$ or mg or $m(v - u) / t$
 $= 7.5 \times 1.6$ or $7.5 \times (4 + 2.4) / 4.0$
 $= 12 \text{ N}$ [2]

(c) speed/velocity decreases so viscous force decreases
 viscous force decreases (and weight constant) so resultant force decreases [2]

25. M/J 22/P23/Q3

(a) (velocity =) change in displacement / time [1]

(b) (i) $a = 2400 / 1200$
 $= 2.0 \text{ m s}^{-2}$ [1]

(ii) straight line from the origin with positive gradient (labelled A)
 ending at (20, 40) [2]

- (c) (i) line starting at origin (with the same gradient as A) and beneath A at all points
gradient decreasing to zero
straight horizontal line from $t = 12$ s and ending at $t = 20$ s (and labelled B) [3]
- (ii) the velocity/speed will increase
to a new terminal/constant/maximum velocity/speed [2]
or
the car has an acceleration
to a new (higher) terminal/constant/maximum velocity/speed [2]

26. MAR 22/P22/Q2

- (a) force (on droplet of water) in horizontal direction is zero. [1]
- (b) (time taken =) $3.5 / 6.6 = 0.53$ (s) [1]
- (c) $s = ut + \frac{1}{2}at^2$
 $s = \frac{1}{2} \times 9.81 \times 0.53^2$
 $h = 1.4$ m [2]
- (d) displacement is straight-line distance (from P to Q) so less (than distance along path)
or
displacement is the shortest distance (from P to Q). [1]
- (e) (displacement) $^2 = 3.5^2 + 1.4^2$
displacement = 3.8 m [2]

27. M/J 21/P23/Q2

- (a) change in velocity / time (taken) [1]
- (b) (i) air resistance increases (with speed/with time)
resultant force decreases (as speed increases/with time) so acceleration decreases
(as speed increases/with time)
when air resistance equals the weight the speed/velocity/v becomes constant [3]
- (ii) speed = 36 m s^{-1} [1]
- (iii) height given by area under the curve
height = 950 m
Round to two significant figures and award 2 marks for a value in the range 920–980 m and 1 mark for a value in the range 900–910 m or 990–1000 m. [3]
- (iv) line starting at (0, 9.8)
curve with negative gradient between $t = 0$ and $t = 20$ s
line showing zero acceleration between $t = 20$ s and $t = 30$ s [3]

28. MAR 21/P22/Q1

- (a) acceleration: vector
work: scalar
power: scalar
Three correct scores 2 marks. Two correct scores 1 mark. [2]
- (b) (i) $a = (v - u) / t$ or $a = \text{gradient}$ or $a = \Delta v / (\Delta t)$
e.g. $a = (1.40 - 0.70) / 4.0$
 $= 0.18 \text{ m s}^{-2}$ [2]
- (ii) distance = $0.5 \times (0.70 + 1.40) \times 4.0$ or $(0.70 \times 4.0) + (0.5 \times 0.70 \times 4.0)$
 $= 4.2$ m [2]

- (c) (i) (force equal to) rate of change of momentum [1]
 (ii) horizontal line starting from $t = 0$ and ending at $t = 4.0$ s at a positive value of F
 horizontal line starting from $t = 4.0$ s and ending at $t = 8.0$ s at $F = 0$
 horizontal line starting from $t = 8.0$ s and ending at $t = 12.0$ s at a negative value of F
and the magnitude of F is larger than from $t = 0$ to 4.0 s [3]

29. O/N 20/P22/Q1

- (a) density and temperature indicated as scalars
 acceleration and momentum indicated as vectors [2]
 (b) (i) decelerates **or** speed/velocity decreases [1]
 (ii) speed = $(\Delta)d / (\Delta)t$ **or** gradient
 = e.g. $(0.56 - 0.20) / 1.5$
 = 0.24 m s^{-1} [2]
 (c) displacement is zero (so) average velocity is zero [1]

30. O/N 20/P21/Q2

- (a) constant gradient [1]
 (b) (displacement until 0.20 s =) $\frac{1}{2} \times 1.96 \times 0.20$ (= 0.196 m) **or**
 (displacement after 0.20 s =) $\frac{1}{2} \times 6.86 \times 0.70$ (= 2.401 m)
 height = $2.401 - 0.196 = 2.2$ m
(alternative methods are possible using equations of uniformly accelerated motion) [3]
 (c) $(\Delta)E = mg(\Delta)h$ **or** $W(\Delta)h$
 $(\Delta)E = 0.86 \times 2.2 = 1.9$ J [2]
 (d) curved line from the origin gradient of curved line decreases and is zero at $t = 0.20$ s only [2]
 (e) acceleration (of free fall) is unchanged/is not dependent on mass and (so) no effect [1]

31. O/N 20/P23/Q3

- (a) $s = \frac{1}{2}at^2$
 $57 = \frac{1}{2} \times 9.81 \times t^2$ **and** $t = 3.4$ (s) [2]
 (b) horizontal distance = 41×3.4
 = 140 m [1]
 (c) (displacement) $^2 = 57^2 + 140^2$
 displacement = $(57^2 + 140^2)^{0.5}$
 = 150 m [2]
 (d) straight line from the origin with positive gradient [1]
 (e) $(1480 - m) \times 0.340 = m \times 41.0$
 $m = 12.2$ kg **or** $m_c 0.34 = m_b 41$ **and** $m_c + m_b = 1480$
 $m_c = (41 / 0.34)m_b$
 $(41 / 0.34)m_b + m_b = 1480$
 $m_b = 12.2$ kg [2]
 (f) acceleration (of free fall) is unchanged/is not dependent on mass
 (so) no change (to the graph) [2]

32. M/J 20/P22/Q2

- (a) (i) area = $ut + \frac{1}{2}(v - u)t$ **or**
 area = $vt - \frac{1}{2}(v - u)t$ **or**
 area = $\frac{1}{2}(u + v)t$ [1]
 (ii) displacement [1]

- (b) (i) $u = 15 \sin 60^\circ (= 13 \text{ m s}^{-1})$
 $t = 15 \sin 60^\circ / 9.81$
 $= 1.3 \text{ s}$ [3]
- (ii) the force in the horizontal direction is zero [1]
- (iii) (velocity $\Rightarrow$) $15 \cos 60^\circ = 7.5 \text{ (m s}^{-1}\text{)}$ or
 (velocity $\Rightarrow$) $15 \sin 30^\circ = 7.5 \text{ (m s}^{-1}\text{)}$ [1]
- (c) (i) $p = mv$ or 0.40×7.5 or 0.40×4.3
 $\Delta p = 0.40 (7.5 + 4.3)$
 $= 4.7 \text{ kg m s}^{-1}$ [2]
- (ii) force $= 4.7 / 0.12$ or $0.40 \times [(7.5 + 4.3) / 0.12]$
 $= 39 \text{ N}$ [1]

33. M/J 20/P21/Q2

- (a) (resultant) force proportional to rate of change of momentum [1]
- (b) (i) arrow drawn vertically downwards from point X [1]
- (ii) $s = ut + \frac{1}{2}at^2$
 $h = \frac{1}{2} \times 9.81 \times 0.81^2$
 $= 3.2 \text{ m}$ [2]
- (iii) $d = 5.4 \times 0.81$
 $= 4.4 \text{ m}$ [1]
- (c) (i) downward pointing arrow labelled weight
 upward pointing arrow labelled air resistance [2]
- (ii) air resistance increases weight constant or resultant force decreases
 (so) acceleration decreases [3]
- (iii) gravitational potential energy to thermal/internal energy [1]

34. MAR 20/P22/Q2

- (a) $f_0 = f_s v / (v - v_s)$
 $9560 = f \times 1510 / (1510 - 4.50)$
 $f = 9530 \text{ Hz}$ [2]
- (b) (i) $v^2 = u^2 + 2as$
 height $= 5.6^2 / (2 \times 9.81)$
 $= 1.6 \text{ m}$ [2]
- (ii) downward sloping straight line starting from a point on the speed axis and ending at point (T, 0) [1]
- (iii) $(\Delta)E = mg(\Delta)h$
 $= 0.45 \times 9.81 \times 1.6$
 $= 7.1 \text{ J}$ [2]
- (iv) air resistance increases (and weight constant)
 (resultant force decreases so) acceleration decreases [2]

35. O/N 19/P22/Q2

- (a) change in velocity / time (taken) [1]
- (b) (i) weight $\gg$ (force due to) air resistance or
 (force due to) air resistance is negligible compared to weight [1]
- (ii) $s = ut + \frac{1}{2}at^2$
 $0.280 = \frac{1}{2} \times 9.81 \times t^2$
 $t = 0.24 \text{ s}$ [2]

(iii) total distance fallen = $0.280 + 0.080 = 0.360$

$$0.360 = \frac{1}{2} \times 9.81 \times t^2$$

$$t = 0.27 \text{ s}$$

$$\text{time taken} = 0.27 - 0.24$$

$$= 0.03 \text{ s} \quad \text{or}$$

$$v = 9.81 \times 0.239 \quad \text{or} \quad (2 \times 9.81 \times 0.280)^{0.5} \quad \text{or} \quad (2 \times 0.280) / 0.239$$

$$= 2.34 \text{ (ms}^{-1}\text{)}$$

$$0.080 = 2.34t + \frac{1}{2} \times 9.81 \times t^2$$

$$\text{solving quadratic equation gives } t = 0.03 \text{ s}$$

allow any correct method using equations of uniform accelerated motion

[2]

- (c) (average) resultant force/acceleration/speed/velocity (of low-density ball) is less
(so) time interval is longer

[2]

36. O/N 18/P22/Q1

(a) $v_x = (6.0^2 - 4.8^2)^{1/2} = 3.6 \text{ (ms}^{-1}\text{)} \quad \text{or}$

$$6.0 \sin \theta = 4.8 \text{ (so } \theta = 53.1^\circ \text{) and } v_x = 6.0 \cos 53.1^\circ = 3.6 \text{ (ms}^{-1}\text{)}$$

[1]

- (b) (i) straight line from (0, 4.8) to (0.49, 0)

straight line continues with same slope to (0.98, -4.8) (labelled Y)

[2]

- (ii) a horizontal line from (0, 3.6) to (0.98, 3.6) (labelled X)

[2]

(c) $s = ut + \frac{1}{2}at^2$

$$= (4.8 \times 0.49) + (\frac{1}{2} \times -9.81 \times 0.49^2) \quad \text{or}$$

$$s = \frac{1}{2}(u + v)t \quad \text{or area under graph}$$

$$= \frac{1}{2} \times (4.8 + 0) \times 0.49 \quad \text{or}$$

$$s = vt - \frac{1}{2}at^2$$

$$= \frac{1}{2} \times 9.81 \times 0.49^2 \quad \text{or}$$

$$v^2 = u^2 + 2as$$

$$s = 4.8^2 / (2 \times 9.81)$$

$$s = 1.2 \text{ m}$$

[2]

(d) $(\Delta)E = mg(\Delta)h$

$$E = \frac{1}{2}mv^2$$

$$\text{ratio} = (\frac{1}{2} \times m \times 3.6^2) / (m \times 9.81 \times 1.2) \quad \text{or}$$

$$\text{ratio} = [(\frac{1}{2} \times m \times 6.0^2) - (m \times 9.81 \times 1.2)] / (m \times 9.81 \times 1.2) \quad \text{or}$$

$$\text{ratio} = (\frac{1}{2} \times m \times 3.6^2) / (\frac{1}{2} \times m \times 4.8^2)$$

$$\text{ratio} = 0.56$$

[4]

- (e) (force due to) air resistance acts in opposite direction to the velocity **or**
(with air resistance, average) resultant force is larger (than weight)

[1]

37. O/N 18/P21/Q1

- (a) (i) distance in a specified direction (from a point)

[1]

- (ii) change in velocity /time (taken)

[1]

- (b) (i) vertical component of velocity $= (5.5^2 - 4.6^2)^{1/2} = 3.0 \text{ (ms}^{-1}\text{)}$ **or**
 $5.5 \cos \theta = 4.6$ (so $\theta = 33.2^\circ$) **and** $5.5 \sin 33.2^\circ = 3.0 \text{ (ms}^{-1}\text{)}$ [1]
- (ii) $s = ut + \frac{1}{2}at^2$
 $0 = (3.0 \times t) - (\frac{1}{2} \times 9.81 \times t^2)$ **or**
 $v = u + at$
 $-3.0 = 3.0 - 9.81t$
 $t = 0.61 \text{ s}$ [2]
- (iii) $d = 4.6 \times 0.61$
 $= 2.8 \text{ m}$ [1]
- (iv) $E = \frac{1}{2}mv^2$
ratio $= (\frac{1}{2} \times m \times 4.6^2) / (\frac{1}{2} \times m \times 5.5^2)$ **or**
ratio $= (\frac{1}{2} \times m \times 5.5^2 - m \times 9.81 \times 0.459) / (\frac{1}{2} \times m \times 5.5^2)$
ratio $= 0.70$ [3]
- (c) straight line from positive value of v_y at $t = 0$ to negative value of v_y
straight line ends at $t = T$ and final magnitude of v_y greater than initial magnitude of v_y [2]

38. MAR 18/P22/Q1

- (a) acceleration: vector
speed: scalar
power: scalar
All three correct scores 2 marks. Only two correct scores 1 mark. [2]
- (b) (i) time $= 0.43 / 1.1$
 $= 0.39 \text{ s}$ [1]
- (ii) $s = ut + \frac{1}{2}at^2$
 $= \frac{1}{2} \times 9.81 \times 0.39^2$
 $= 0.75 \text{ m}$ [2]
- (iii) 1 horizontal line at a non-zero value of a .
2 curved line from origin with increasing gradient. [2]
- (c) acceleration (of free fall) is unchanged / not dependent on mass
and so no effect (on time taken). [1]

39. O/N 17/P21/Q2

- (a) $30 \text{ ms}^{-1} = 108 \text{ km h}^{-1}$ **or**
 $100 \text{ km h}^{-1} = 28 \text{ ms}^{-1}$
and so exceeds speed limit [1]
- (b) acceleration = gradient **or** $\Delta v / (\Delta)t$ **or** $(v - u) / t$
e.g. acceleration $= (24 - 20) / 12$ [other points on graph line may be used]
 $= 0.33 \text{ ms}^{-2}$ [2]
- (c) distance travelled by Q $= \frac{1}{2} \times 12 \times 30 (= 180 \text{ m})$
distance travelled by P $= \frac{1}{2} \times (20 + 24) \times 12 (= 264 \text{ m})$
distance between cars $= 264 - 180$
 $= 84 \text{ m}$ [3]
- (d) $30 - 24 = 6 \text{ ms}^{-1}$
'extra' time $T = 84 / 6 (= 14 \text{ s})$ **or**
 $180 + 30T = 264 + 24T$
'extra' time $T = 84 / 6 (= 14 \text{ s})$
 $t = 12 + 14 = 26 \text{ s}$ [2]

40. M/J 17/P22/Q2**(a)** rate of change of displacement **or** change in displacement/time taken [1]

(b) (i) $s = ut + \frac{1}{2}at^2$
 $t = [(2 \times 1.25) / 9.81]^{1/2} (= 0.5048 \text{ s})$ **or**
 $v^2 = u^2 + 2as$
 $v_{\text{vert}} = (2 \times 9.81 \times 1.25)^{1/2} (= 4.95)$
 $t = [2s / (u + v)] = 2 \times 1.25 / 4.95 (= 0.5048 \text{ s})$
 $v = d/t = 1.5 / 0.5048$
 $= 3.0 \text{ (2.97) ms}^{-1}$ [3]

(ii) vertical velocity $= at$
 $= 9.81 \times 0.5048 (= 4.95)$ [using $t = 0.50$ gives 4.9]
velocity $= [(v_h)^2 + (v_v)^2]^{1/2}$
 $= [(2.97)^2 + (4.95)^2]^{1/2}$
 $= 5.8 \text{ (5.79)}$ [using $t = 0.50$ leads to 5.7]
direction $(= \tan^{-1} 4.95/2.97) = 59^\circ$ [4]

(iii) kinetic energy $= \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 0.45 \times (5.8)^2$
 $= 7.6 \text{ (7.57) J}$ [using $t = 0.50$ leads to 7.3 J] [2]

(iv) potential energy $= mgh$
 $= (0.45 \times 9.81 \times 1.25)$
 $= 5.5 \text{ (5.52) J}$ [2]

(c) there is KE of the ball at the start/leaving table **or**
the ball has an initial/constant horizontal velocity **or**
the ball has velocity at start/leaving table [1]

41. M/J 16/P21/Q2**(a) (i)** horizontal component $(= 12 \cos 50^\circ) = 7.7 \text{ ms}^{-1}$ [1]**(ii)** vertical component $(= 12 \sin 50^\circ \text{ or } 7.7 \tan 50^\circ) = 9.2 \text{ ms}^{-1}$ [1]

(b) $v^2 = u^2 + 2as$ **and** $v = 0$ **or** $mgh = \frac{1}{2}mv^2$ **or** $s = v^2 \sin^2 \theta / 2g$
 $9.2^2 = 2 \times 9.81 \times h$ hence $h = 4.3 \text{ (4.31) m}$
alternative methods using time to maximum height of 0.94 s:
 $s = ut + \frac{1}{2}at^2$ and $t = 0.94 \text{ (s)}$
 $s = 9.2 \times 0.94 - \frac{1}{2} \times 9.81 \times 0.94^2$ hence $s = 4.3 \text{ m}$ **or**
 $s = vt - \frac{1}{2}at^2$ and $t = 0.94 \text{ (s)}$
 $s = \frac{1}{2} \times 9.81 \times 0.94^2$ hence $s = 4.3 \text{ m}$ **or**
 $s = \frac{1}{2}(u + v)t$ and $t = 0.94 \text{ (s)}$
 $s = \frac{1}{2} \times 9.2 \times 0.94$ hence $s = 4.3 \text{ m}$ [2]

(c) $t (= 9.2/9.81) = 0.94 \text{ (0.938) s}$
horizontal distance $= 0.938 \times 7.7 (= 7.23 \text{ m})$
displacement $= [4.3^2 + 7.23^2]^{1/2} = 8.4 \text{ m}$ [4]

42. O/N 15/P22/Q2

(a) (i) $v = u + at$
 $0 = 3.6 - 3.0t$
 $t (= 3.6/3.0) = 1.2 \text{ s}$ [2]

(ii) (distance to rest from P = $(3.6 \times 1.2)/2 = 2.2$ (2.16) m or

$$[0 - (3.6)^2]/[2 \times (-3.0)] = 2.2 \text{ (2.16) m or}$$

$$3.6 \times 1.2 - \frac{1}{2} \times 3.0 \times (1.2)^2 = 2.2 \text{ (2.16) m or}$$

$$0 + \frac{1}{2} \times 3.0 \times (1.2)^2 = 2.2 \text{ (2.16) m}$$

[1]

(b) distance = $6.0 - 2.16 = 3.84$

$$v^2 = u^2 + 2as = 2 \times 3.0 \times 3.84 (= 23.04) \text{ or}$$

$$x + 2 \times 2.16 = 6.0 \text{ gives } x = 1.68 \text{ (m)}$$

$$v^2 = 3.6^2 + 2 \times 1.68 \times 3.0 (= 23.04)$$

or correct method with intermediate time calculated ($t = 1.6$ s from Q to R)

$$v = 4.8 \text{ m s}^{-1}$$

[2]

(c) straight line from $v = 3.6 \text{ m s}^{-1}$ to $v = 0$ at $t = 1.2$ s

straight line continues with the same gradient as v changes sign

straight line from $v = 0$ intercept to $v = -4.8 \text{ m s}^{-1}$

[3]

(d) difference in KE = $\frac{1}{2}m(v^2 - u^2)$

$$= 0.5 \times 0.45 (4.8^2 - 3.6^2) [= 5.184 - 2.916]$$

$$= 2.3 \text{ (2.27) J}$$

[2]

43. O/N 15/P21/Q3

(a) work done is force $\times$ distance moved in direction of force or no work done along PQ as no displacement/distance moved in direction of force work done is same in vertical direction as same distance moved in direction of force

[2]

(b) (i) at maximum height $t = 1.5$ (s) or $s = \frac{1}{2}(u + v)t$, $s = 11$ m and $t = 1.5$ s

$$V_v = 0 + 9.81 \times 1.5$$

$$V_v = (11 \times 2) / 1.5$$

$$= 15 \text{ (14.7) m s}^{-1}$$

[2]

(ii) straight line from (0,0) to (3.00, 25.5)

[1]

(iii) at maximum height $V_h = 25.5/3 (= 8.5 \text{ m s}^{-1})$

$$\text{ratio} = mgh / \frac{1}{2}mv^2$$

$$= (2 \times 9.81 \times 11.0) / (8.5)^2$$

$$= 3.0 \text{ (2.99)}$$

[3]

(iv) deceleration is greater/resultant force (weight and friction force) is greater time is less [2]

44. O/N 15/P23/Q3

Answer from the official mark scheme

- 3 (a) (i) 1.** $s = ut + \frac{1}{2}at^2$
- $192 = \frac{1}{2} \times 9.81 \times t^2$ C1
- $t = 6.3 \text{ (6.26) s}$ A1 [2]
- 2.** $\max E_k (= mgh) = 0.27 \times 9.81 \times 192$ C1
- or**
- calculation of v ($= 61.4$) and use of $E_k (= \frac{1}{2}mv^2) = \frac{1}{2} \times 0.27 \times (61.4)^2$ (C1)
- $\max E_k = 510 \text{ (509) J}$ A1 [2]
- (ii)** velocity is proportional to time **or** velocity increases at a constant rate
as acceleration is constant or resultant force is constant B1 [1]
- (iii)** use of $v = at$ or $v^2 = 2as$ or $E = \frac{1}{2}mv^2$ to give $v = 61(.4) \text{ m s}^{-1}$ B1 [1]
- (b) (i)** R increases with velocity B1
- resultant force is $mg - R$ **or** resultant force decreases B1
- acceleration decreases B1 [3]
- (ii)** at $v = 40 \text{ m s}^{-1}$, $R = 0.6 \text{ (N)}$ C1
- $0.27 \times 9.8 - 0.6 = 0.27 \times a$
- $a = 7.6 \text{ (7.58) m s}^{-2}$ A1 [2]
- (iii)** $R = \text{weight}$ for terminal velocity B1
- either* weight requires velocity to be about 80 m s^{-1}
or at 60 m s^{-1} , R is less than weight
- so does not reach terminal velocity B1 [2]

45. M/J 15/P22/Q2

- (a) speed decreases/stone decelerates to rest/zero at 1.25 s
 speed then increases/stone accelerates in (opposite direction) [2]

(b) (i) $v = u + at$ (or $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$)
 $= 0 + (3.00 - 1.25) \times 9.81$
 $= 17.2 \text{ (17.17) m s}^{-1}$ [3]

(ii) $s = ut + \frac{1}{2}at^2$
 $s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$
 $s = \frac{1}{2} \times 9.81 \times (1.75)^2 [= 15.02]$
 (distance = 7.66 + 15.02)
 $[v = u + at = 0 + 9.81 \times (2.50 - 1.25) = 12.26 \text{ m s}^{-1}]$ or
 $s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$
 $s = 12.26 \times 0.50 + \frac{1}{2} \times 9.81 \times (3.00 - 2.50)^2 [= 7.36]$
 (distance = $2 \times 7.66 + 7.36$)
Example alternative method:
 $s = (v^2 - u^2)/2a = (12.26^2 - 0)/2 \times 9.81 [= 7.66]$
 $s = (v^2 - u^2)/2a = (17.17^2 - 12.26^2)/2 \times 9.81 [= 7.36]$
 (distance = $2 \times 7.66 + 7.36$)
 22.7 (22.69 or 23) m [3]

(iii) ($s = 15.02 - 7.66 =$) 7.4 (7.36) m (ignore sign in answer)
 down [2]

- (c) straight line from positive value of v to t axis
 same straight line crosses t axis at $t = 1.25$ s
 same straight line continues with same gradient to $t = 3.0$ s [3]

46. M/J 15/P21/Q2

- (a) speed = distance/time and velocity = displacement/time
 speed is a scalar as distance has no direction **and**
 velocity is a vector as displacement has direction [2]

- (b) (i) constant acceleration or linear/uniform increase in velocity until 1.1 s
 rebounds or bounces or changes direction
 decelerates to zero velocity at the same acceleration as initial value [3]

(ii) $a = (v - u)/t$ or use of gradient implied
 $= (8.8 + 8.8)/1.8$ or appropriate values from line or $= (8.6 + 8.6)/1.8$
 $= 9.8 \text{ (9.78) m s}^{-2}$ or $= 9.6 \text{ m s}^{-2}$ [3]

(iii) 1. distance = first area above graph + second area below graph
 $= (1.1 \times 10.8)/2 + (0.9 \times 8.8)/2 (= 5.94 + 3.96)$
 $= 9.9 \text{ m}$ [3]

2. displacement = first area above graph – second area below graph
 $= (1.1 \times 10.8)/2 - (0.9 \times 8.8)/2$
 $= 2.0 \text{ (1.98) m}$ [2]

- (iv) correct shape with straight lines and all lines above the time axis or all below
 correct times for zero speeds (0.0, 1.15 s, 2.1 s) and peak speeds
 (10.8 m s^{-1} at 1.1 s and 8.8 m s^{-1} at 1.2 s and 3.0 s) [2]

47. M/J 15/P23/Q2

- (a) constant rate of increase in velocity/acceleration from $t = 0$ to $t = 8$ s
constant deceleration from $t = 8$ s to $t = 16$ s or constant rate of increase in velocity in the opposite direction from $t = 10$ s to $t = 16$ s [2]
- (b) (i) area under lines to 10 s
 (displacement =) $(5.0 \times 8.0) / 2 + (5.0 \times 2.0) / 2 = 25$ m
 or $\frac{1}{2} (10.0 \times 5.0) = 25$ m [2]
- (ii) $a = (v - u) / t$ or gradient of line
 $= (-15.0 - 5.0) / 8.0$
 $= (-) 2.5 \text{ ms}^{-2}$ [2]
- (iii) $\text{KE} = \frac{1}{2} m v^2$
 $= 0.5 \times 0.4 \times (15.0)^2 = 45$ J [2]
- (c) (distance =) 25 (m) (= $ut + \frac{1}{2} at^2$) = $0 + \frac{1}{2} \times 2.5 \times t^2$
 ($t = 4.5$ (4.47) s therefore) time to return = 14.5 s [2]

48. O/N 14/P22/Q2

- (a) (i) straight line from $t = 0.60$ s to $t = 1.2$ s and $|V_v| = 5.9$ at $t = 1.2$ s
 $V_v = -5.9$ at $t = 1.2$ s i.e. line is for negative values of V_v [2]
- (ii) $s = 0 + \frac{1}{2} \times 9.81 \times (0.6)^2$ or area of graph = $(5.9 \times 0.6) / 2$
 $= 1.8$ (1.77) m [2]
- (iii) $V_h = V \cos 60^\circ$ and $V_v = V \sin 60^\circ$ or $V_h = 5.9 / \tan 60^\circ$ or $V_h = 5.9 \tan 30^\circ$
 $V_h = 3.4 \text{ ms}^{-1}$ [2]
- (iv) horizontal line at 3.4 from $t = 0$ to $t = 1.2$ s [to half a small square] [1]
- (b) (i) $\text{KE} = \frac{1}{2} m v^2$
 $= \frac{1}{2} \times 0.65 \times (6.81)^2$ [allow if valid method to find v]
 $= 15$ (15.1) J [3]
- (ii) $\text{PE} = 0.65 \times 9.81 \times 1.77$
 $= 11$ (11.3) J [2]

49. M/J 14/P21/Q1

- (a) (i) *either* rate of change of displacement
 or (change in) displacement / time (taken)
- (ii) speed has magnitude only
 velocity has magnitude and direction [2]
- (b) (i) idea of area under graph / use of $s = \frac{(u + v)}{2} \times t$
 $s = \frac{(18 + 32)}{2} \times 2.5$
 $= 62.5$ m [3]
- (ii) $a = (18 - 32) / 2.5 (= -5.6)$
 $F = ma$
 $F = 1500 \times (-) 5.6 = (-) 8400$ N [3]
- (c) arrow labelled A and arrow labelled F both to the left [1]

50. M/J 14/P23/Q2

- (a) scalar has magnitude only
 vector has magnitude and direction [2]

- (b) (i) $v^2 = 0 + 2 \times 9.81 \times 25$ (or using $\frac{1}{2}mv^2 = mgh$)
 $v = 22(.1) \text{ m s}^{-1}$ [2]
- (ii) $22.1 = 0 + 9.81 \times t$ (or $25 = \frac{1}{2} \times 9.81 \times t^2$)
 $t (=22.1/9.81) = 2.26 \text{ s}$ or $t [= (5.097)^{1/2}] = 2.26 \text{ s}$ [1]
- (iii) horizontal distance $= 15 \times t$
 $= 15 \times 2.257 = 33.86$ (allow $15 \times 2.3 = 34.5$)
 $(\text{displacement})^2 = (\text{horizontal distance})^2 + (\text{vertical distance})^2$
 $= (25)^2 + (33.86)^2$
displacement $= 42$ (42.08) m (allow 43 (42.6) m, allow 2 or more s.f.) [4]
- (iv) distance is the actual (curved) path followed by ball
displacement is the straight line / minimum distance P to Q [2]

51. O/N 13/P23/Q3

- (a) (i) velocity = rate of change of displacement
OR displacement change / time (taken) [1]
- (ii) acceleration = rate of change of velocity
OR change in velocity / time (taken) [1]
- (b) (i) initial constant velocity as straight line / gradient constant
middle section deceleration / speed / velocity decreases / slowing down as gradient decreases
last section lower velocity (than at start) as gradient (constant and) smaller [3]
[special case: all three stages correct descriptions but no reasons 1/3]
- (ii) velocity $= 45 / 1.5 = 30 \text{ m s}^{-1}$ [1]
- (iii) velocity at 4.0 s is $(122 - 98) / 2.0 = 12 \text{ (m s}^{-1}\text{)}$ (allow 12 to 13)
acceleration $= (12 - 30) / 2.5 = -7.2 \text{ m s}^{-2}$ (if answer not this value then comment needed to explain why, e.g. difficulty in drawing tangent) [2]
- (iv) $F = ma$
 $= (-)1500 \times 7.2 = (-)11000$ (10800) N [2]

52. M/J 13/P23/Q2

- (a) (i) 1. distance of path / along line AB [1]
2. shortest distance between AB / distance in straight line between AB or displacement from A to B [1]
- (ii) acceleration = rate of change of velocity [1]
- (b) (i) distance $= \text{area under line or } (v/2)t \text{ or } s = (8.8)^2 / (2 \times 9.81)$
 $= 8.8 / 2 \times 0.90 = 3.96 \text{ m}$ or $s = 3.95 \text{ m} = 4(.0) \text{ m}$ [2]
- (ii) acceleration $= (-4.4 - 8.8) / 0.50$
 $= (-)26(.4) \text{ m s}^{-2}$ [2]
- (c) (i) the accelerations are constant as straight lines
the accelerations are the same as same gradient or
no air resistance as acceleration is constant or
change of speed in opposite directions (one speeds up one slows down) [2]
- (ii) area under the lines represents height
or KE at trampoline equals PE at maximum height
second area is smaller / velocity after rebound smaller hence KE less
hence less height means loss in potential energy [2]